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COMPRISING SAME****Publication Classification**(51) **Int. Cl.**

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21/28 (2013.01); *H04B 1/40* (2013.01); *H04M*
1/0216 (2013.01); *H04M 1/0268* (2013.01);
H04M 1/0277 (2013.01)

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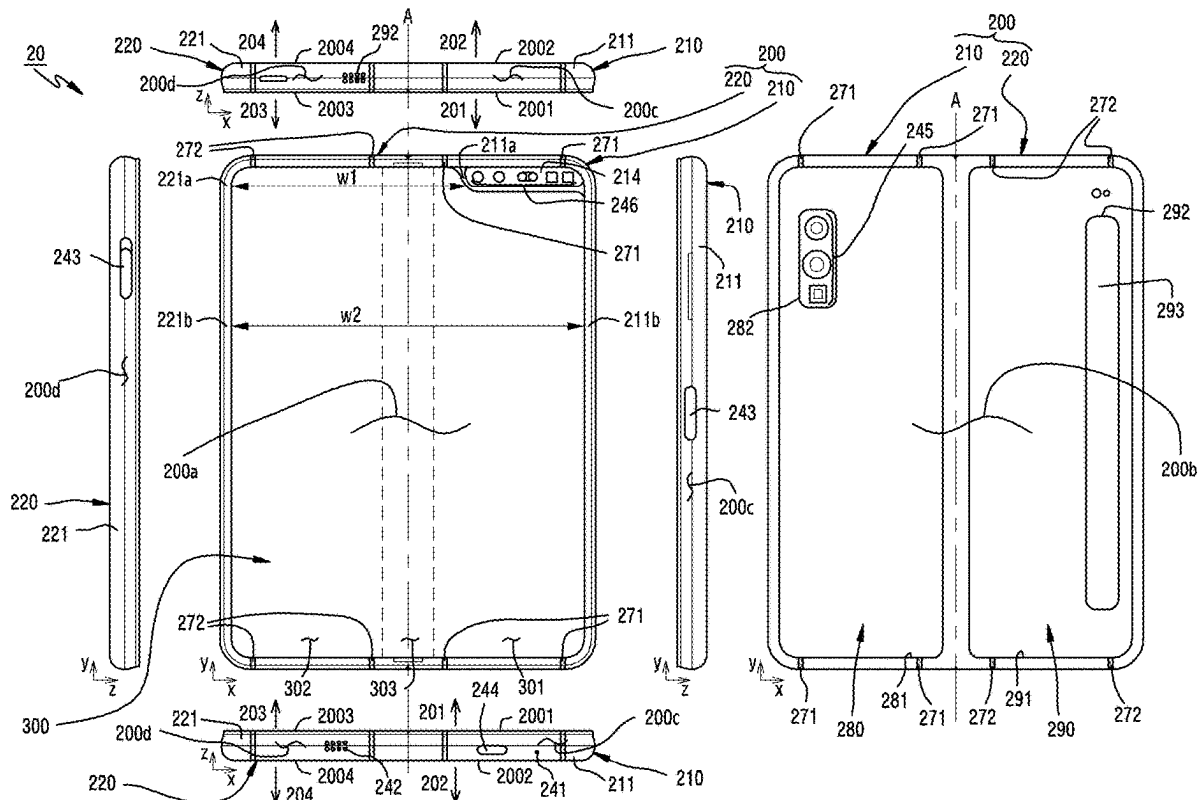
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(57)

ABSTRACT

An electronic device according to an embodiment may include a first housing structure, a second housing structure, and a foldable housing structure for connecting the first housing structure and the second housing structure. The first housing structure and the second housing structure may include a front plate for interconnecting front surfaces with a flexible display, a rear plate which is an opposite surface to the front plate, a side member which surrounds a space between the front plate and the rear plate, and includes at least in part a conductive portion and an insulating portion, a communication circuit and at least one switch electrically connected to the communication circuit. Other various embodiments are possible.



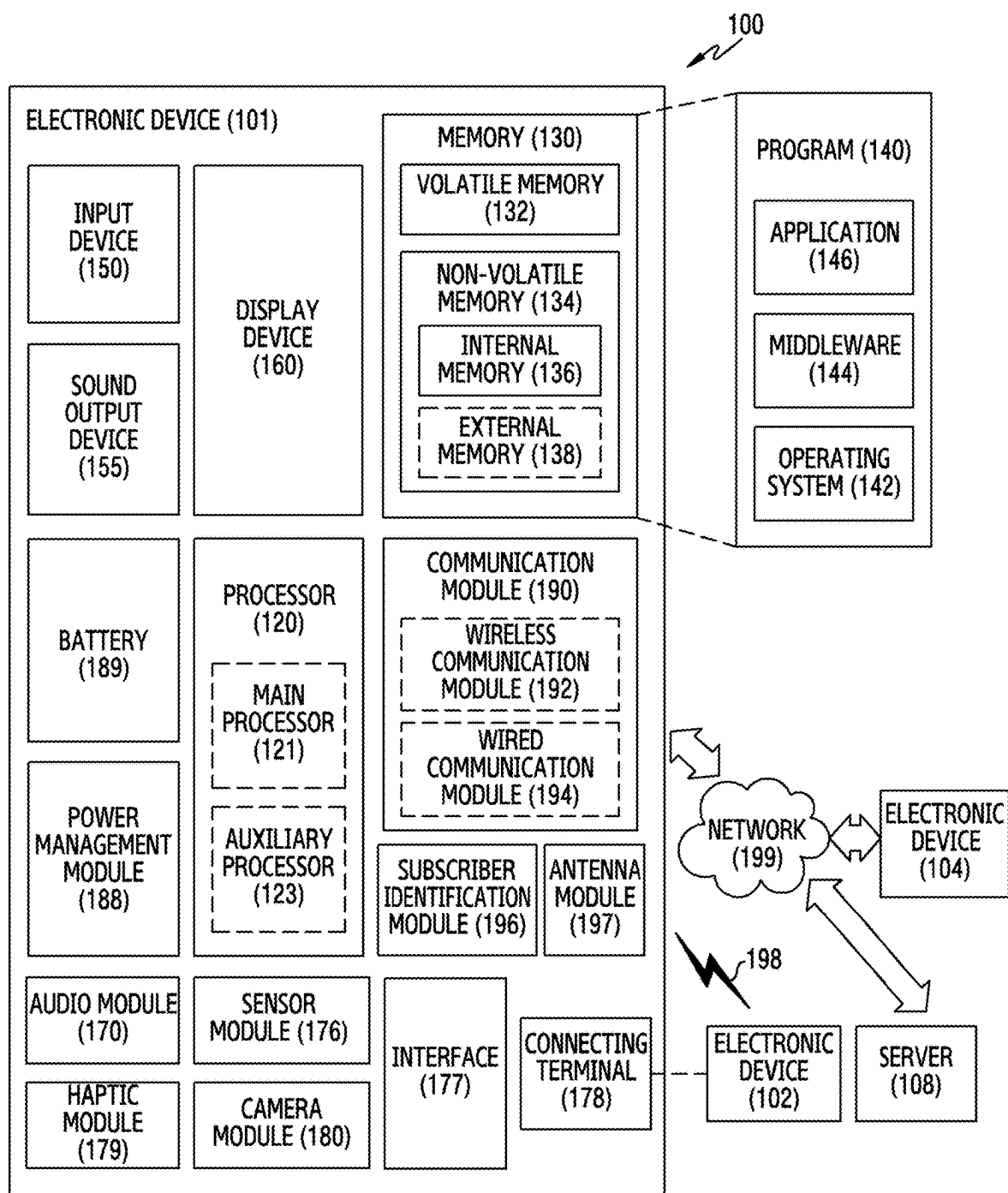


FIG. 1

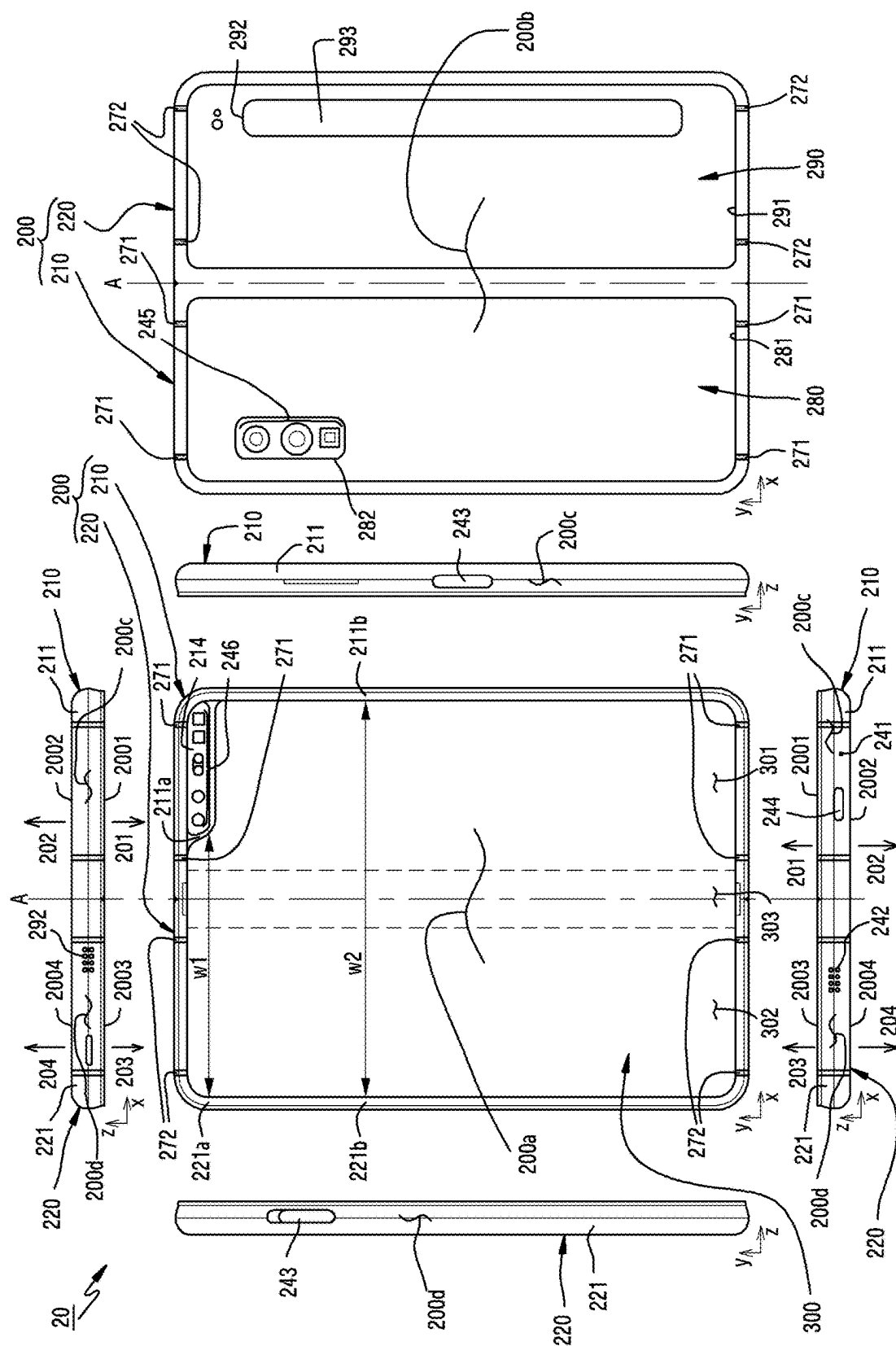


FIG. 2

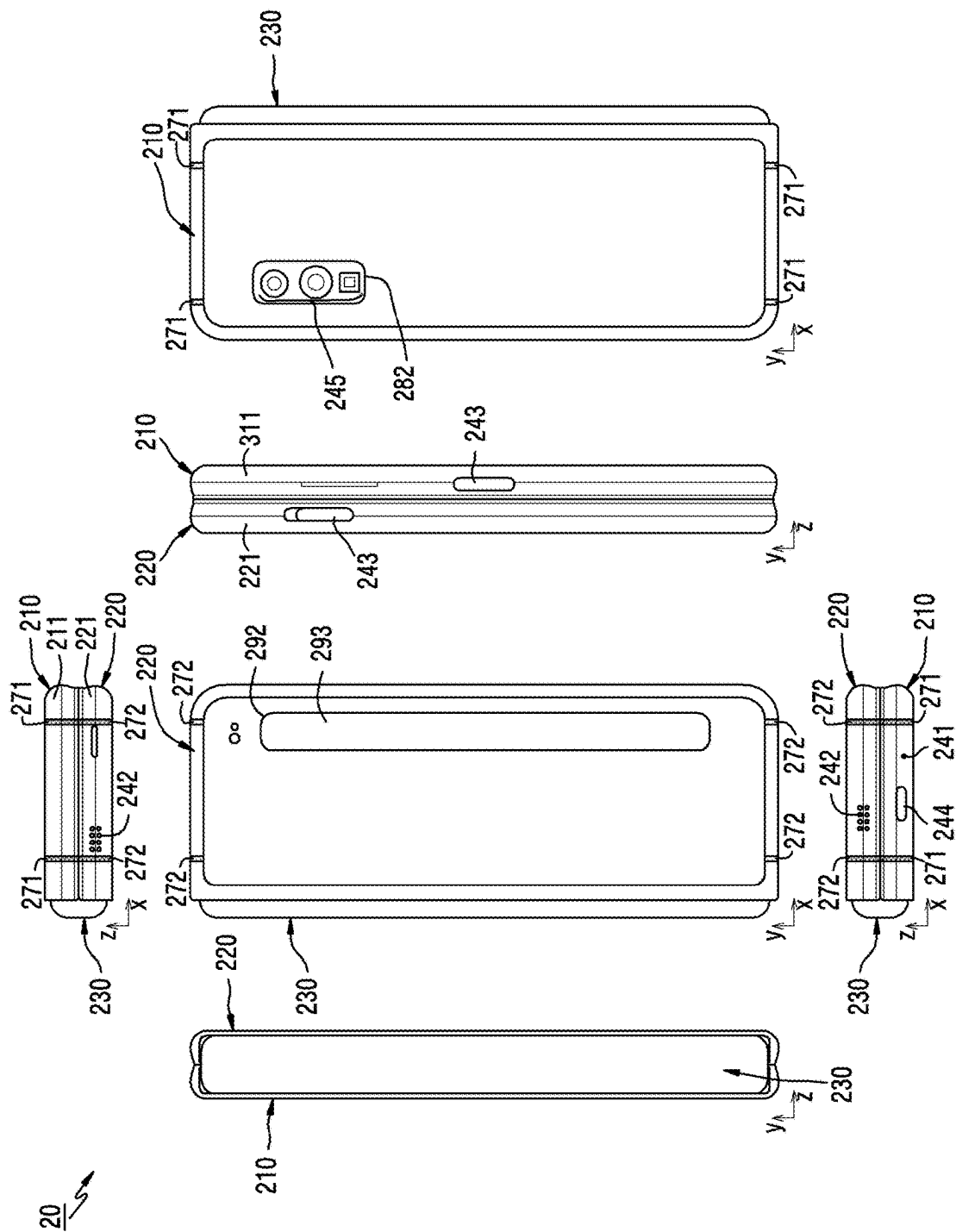


FIG. 3

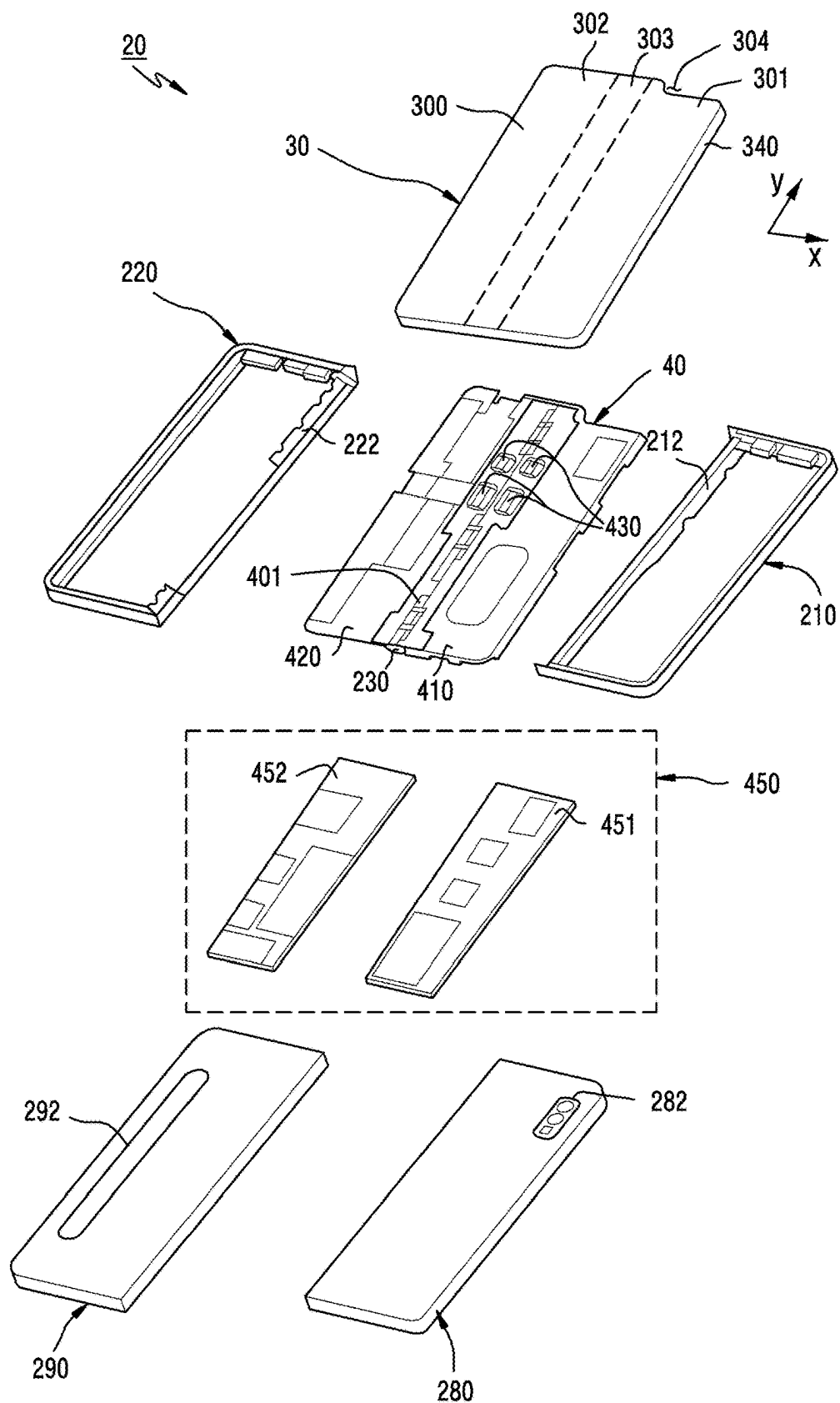


FIG. 4

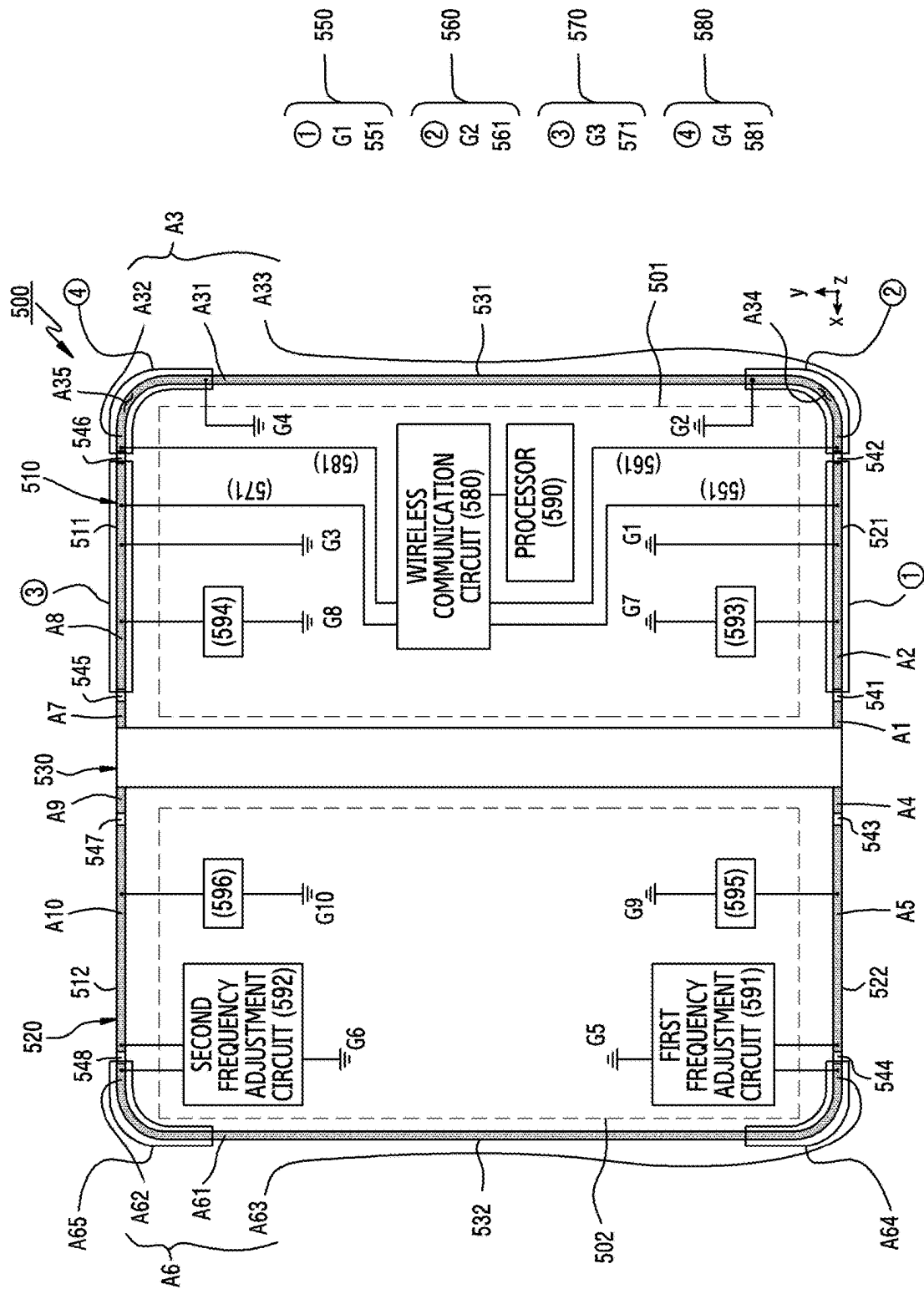


FIG. 5

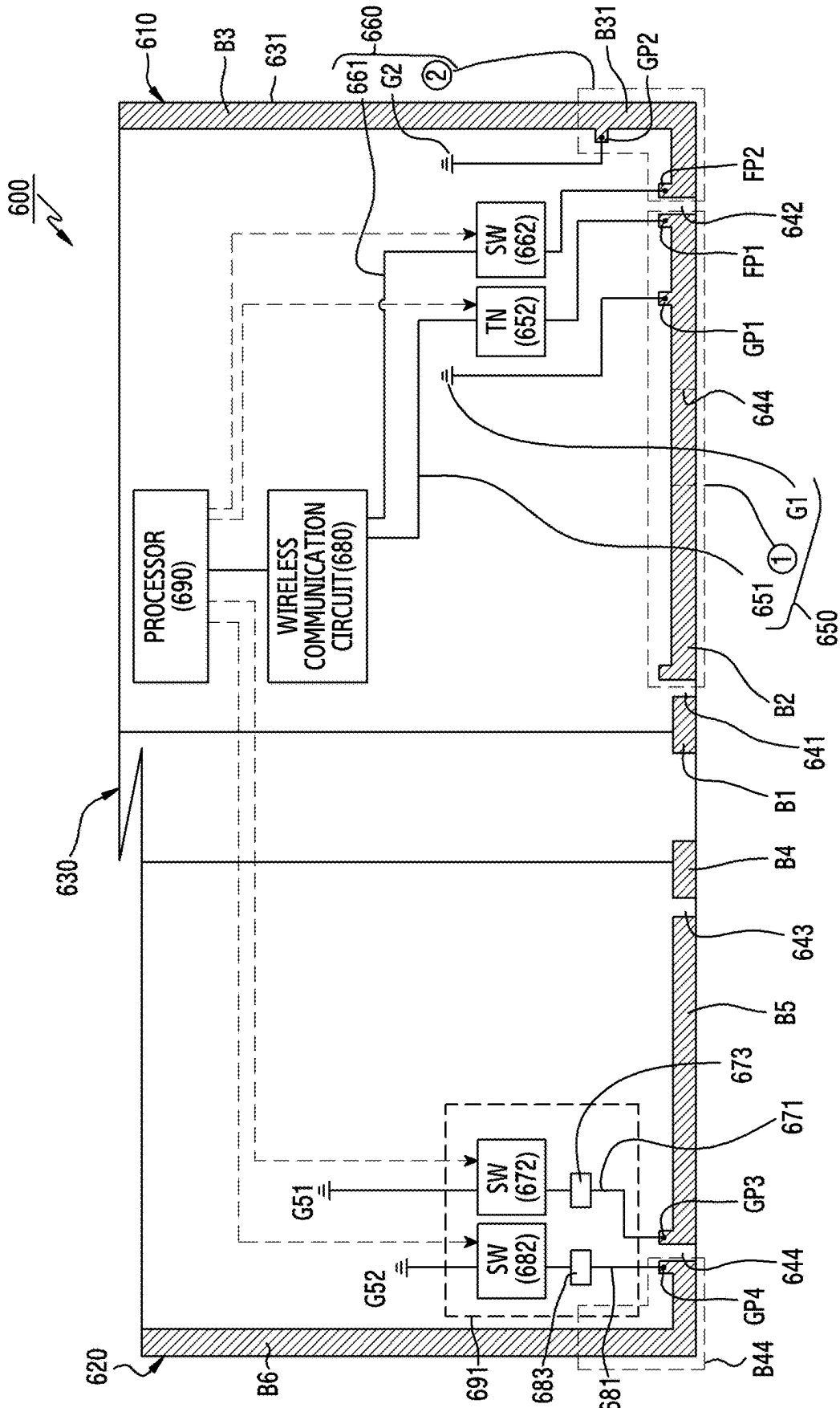


FIG. 6A

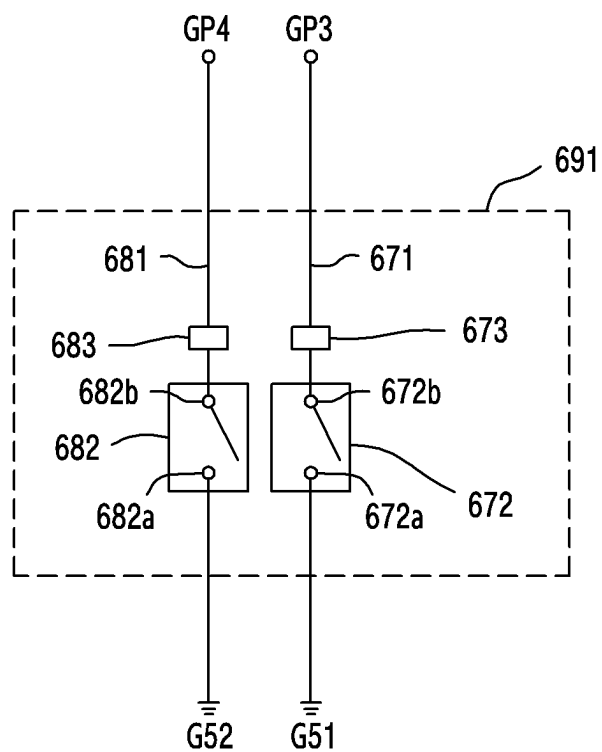


FIG.6B

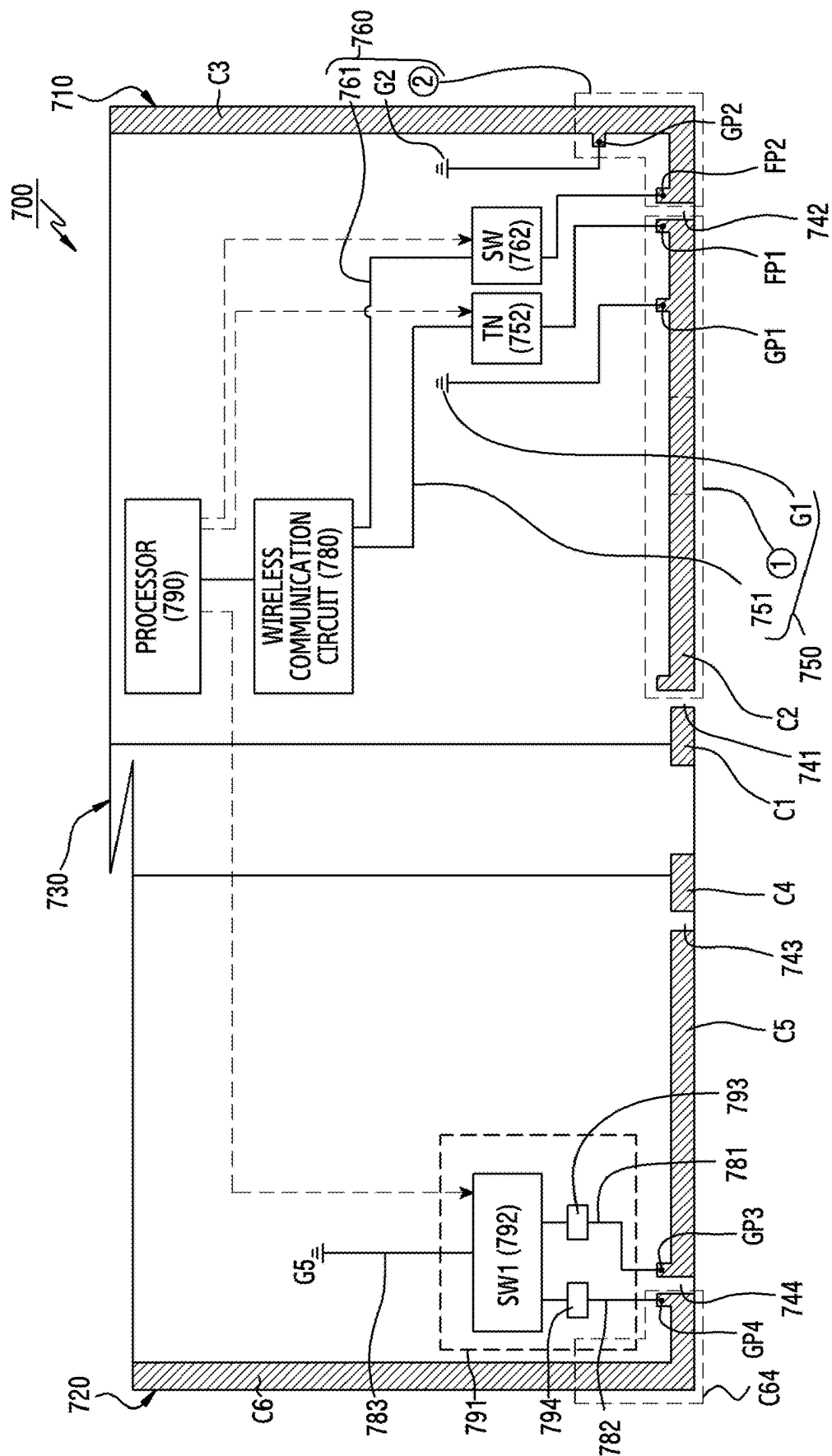


FIG. 7A

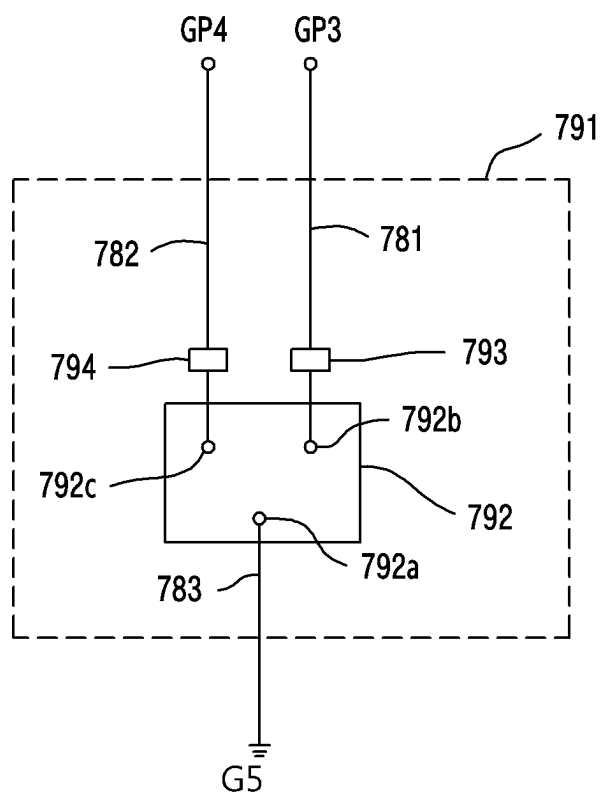


FIG. 7B

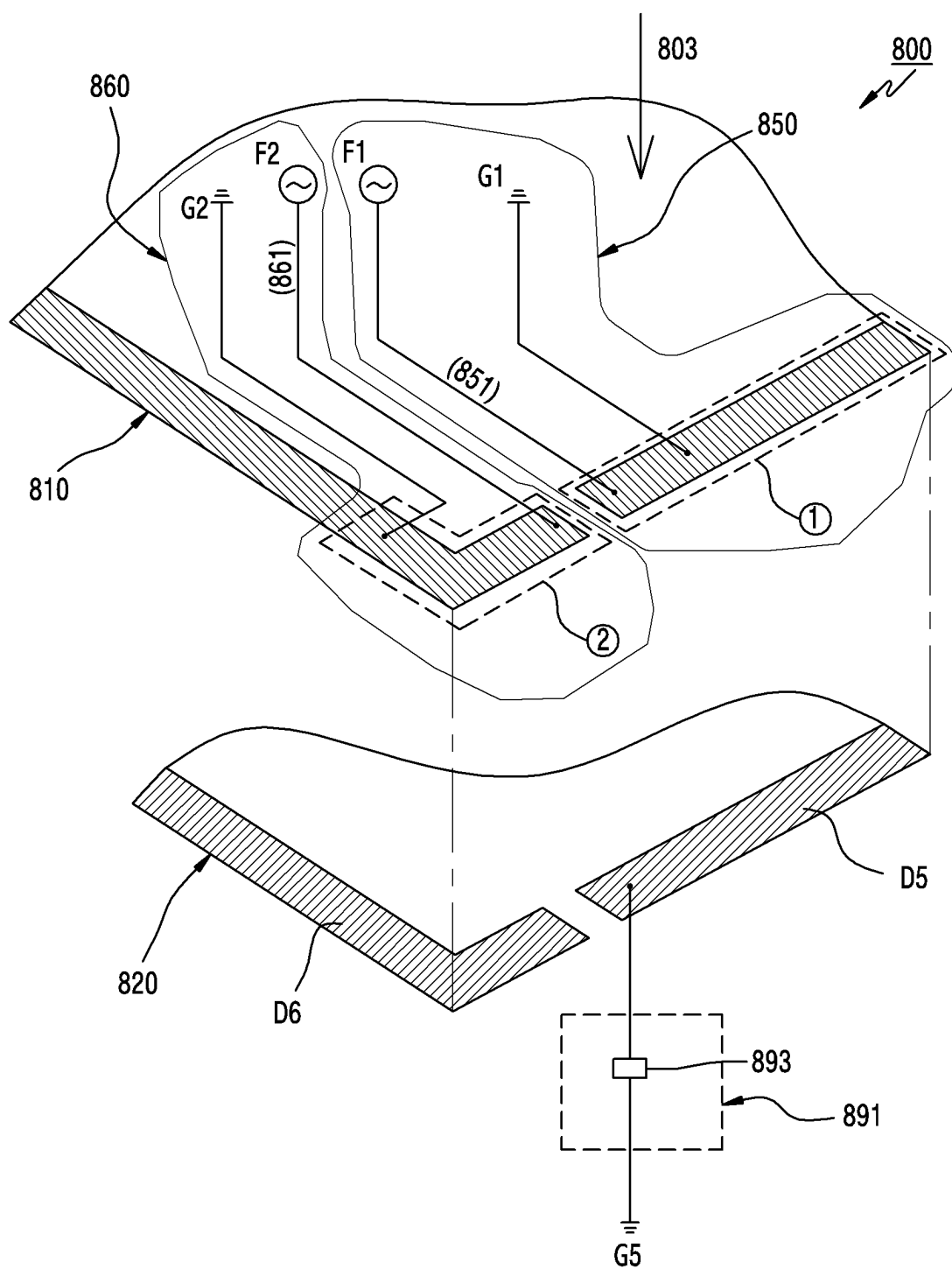


FIG.8

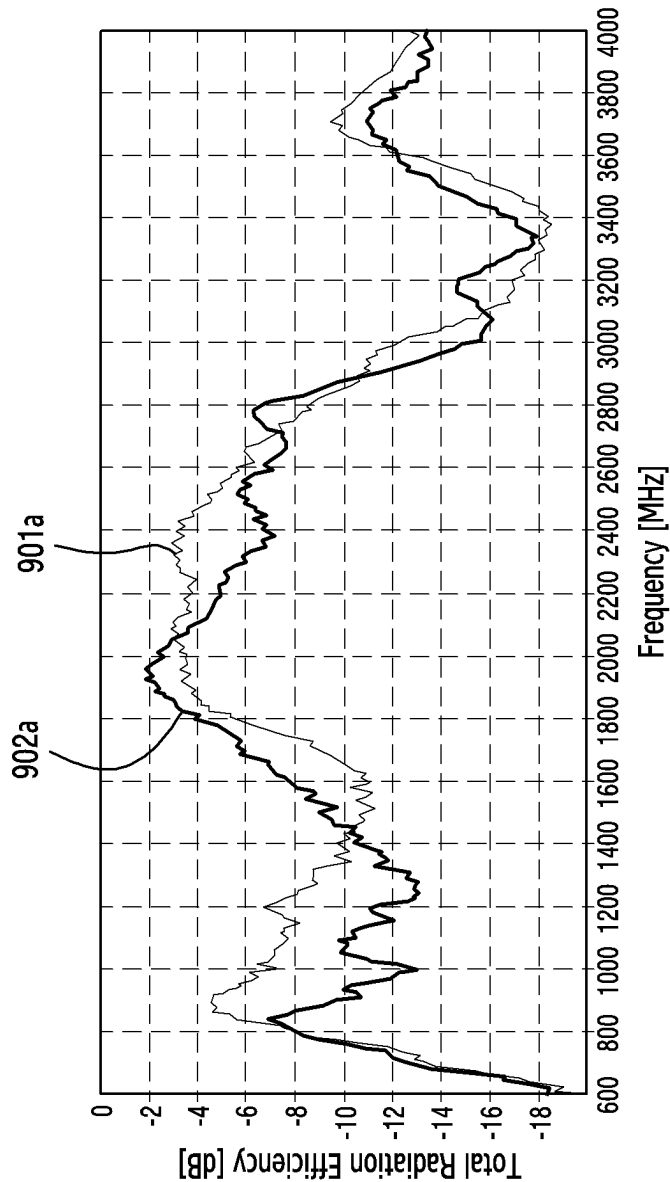


FIG.9A

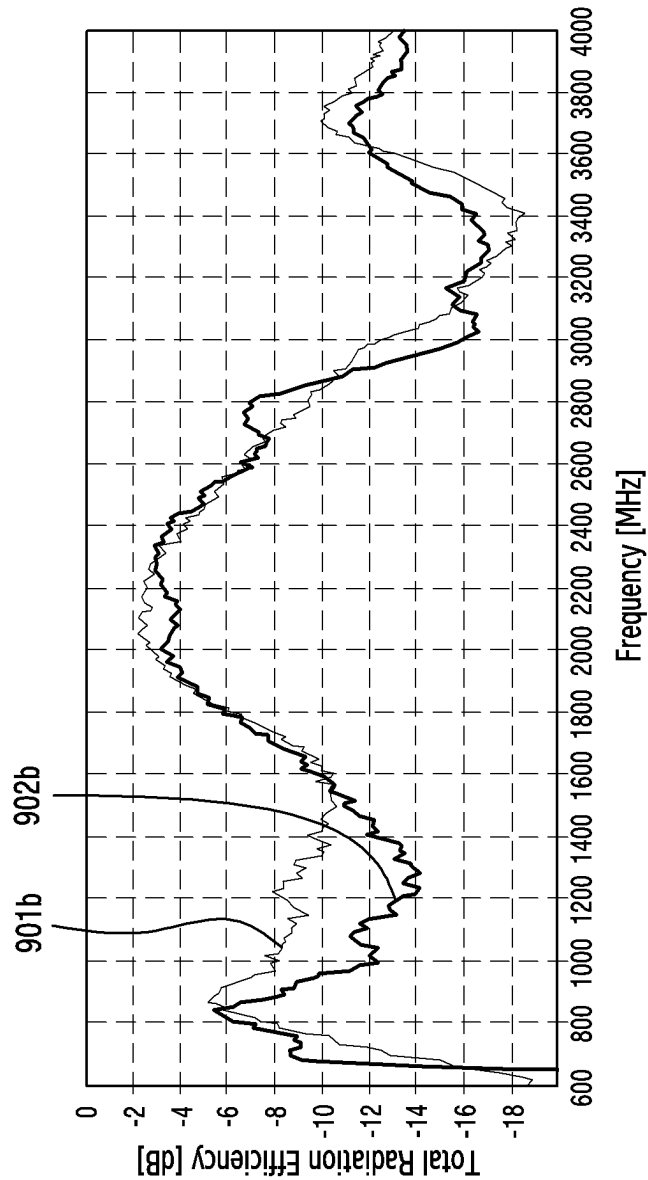


FIG.9B

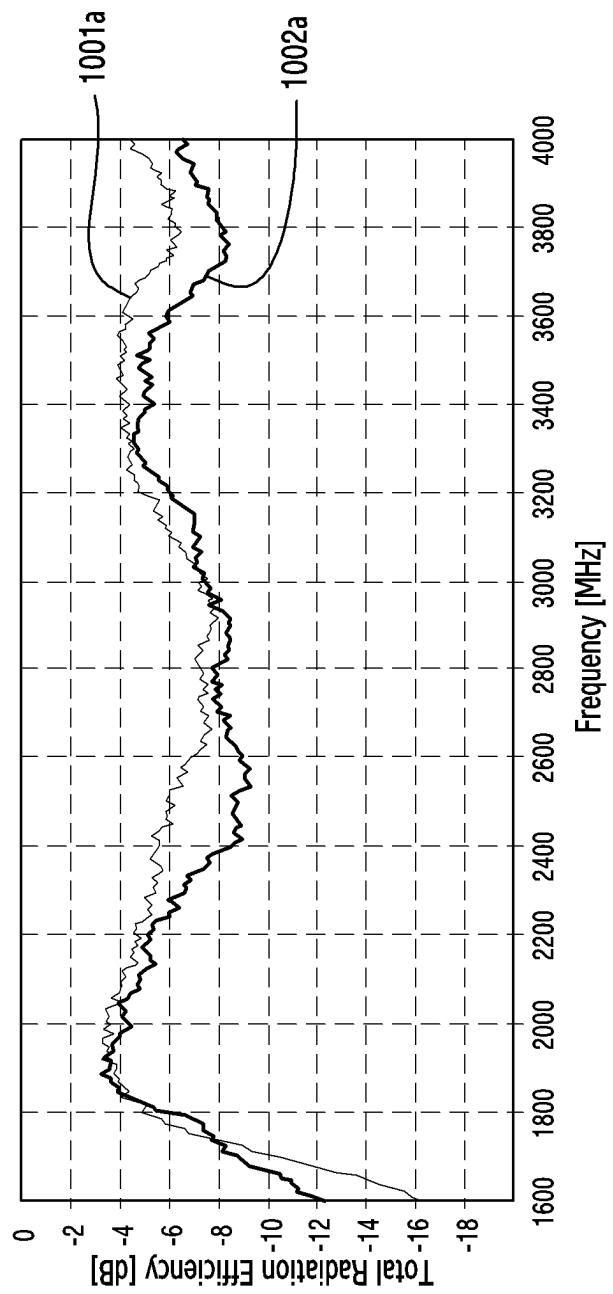


FIG.10A

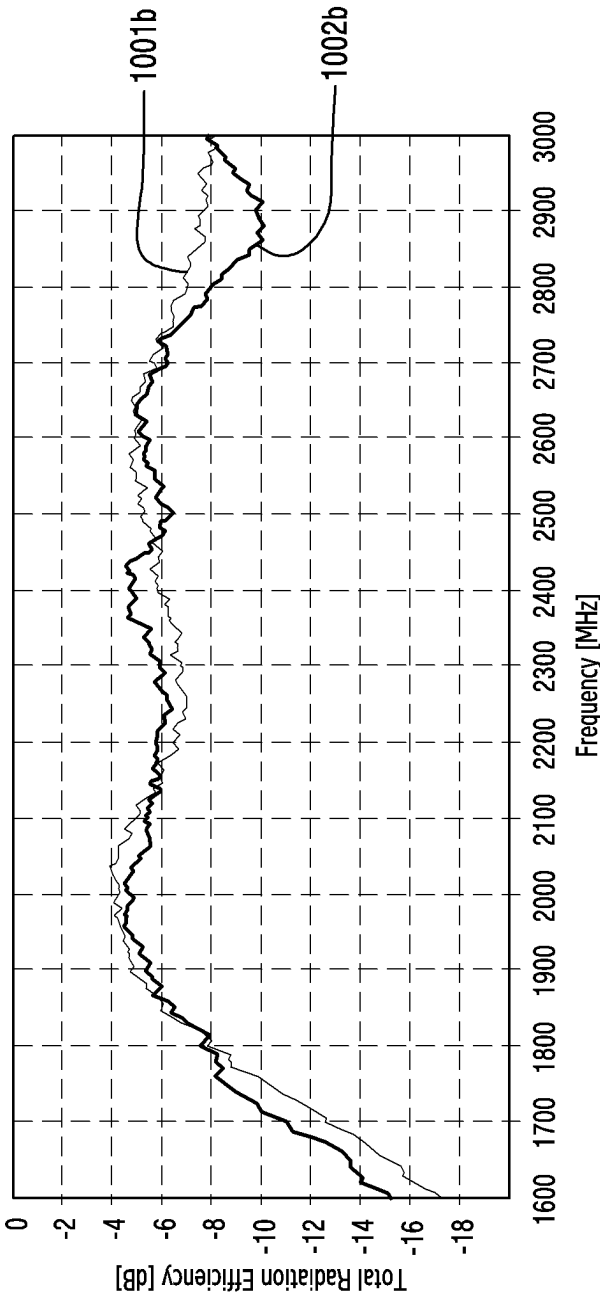


FIG.10B

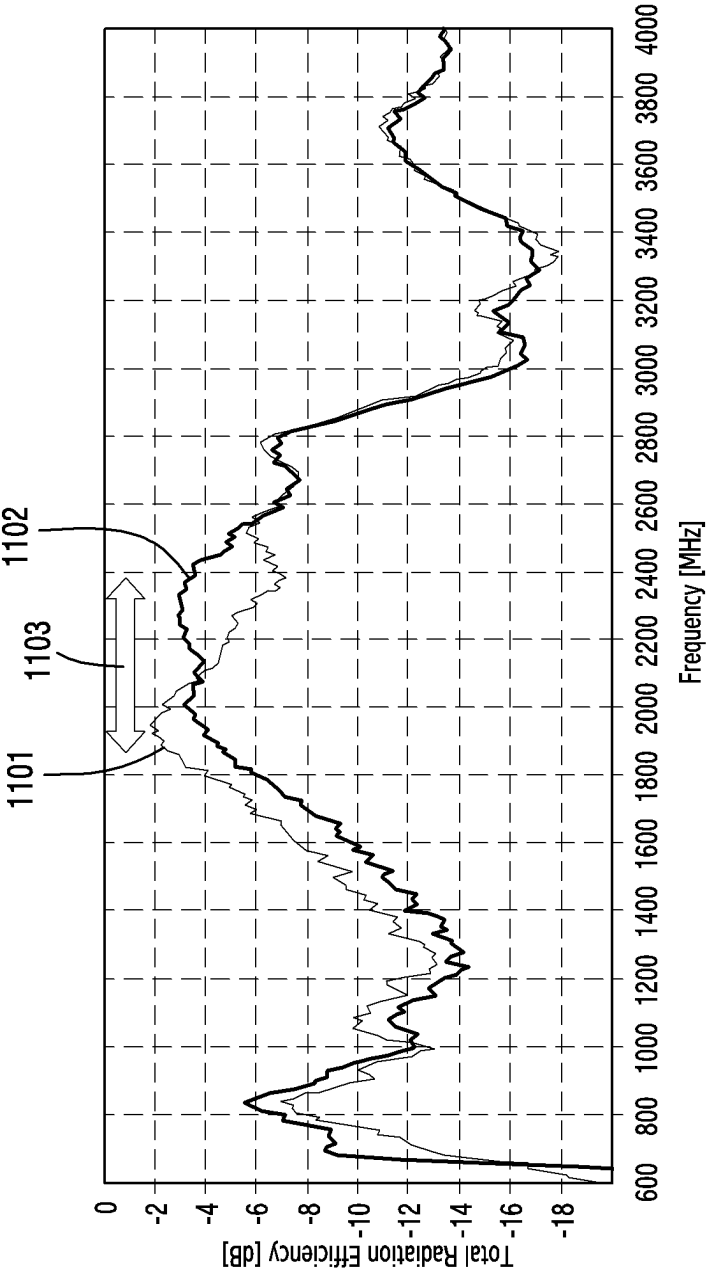


FIG.11

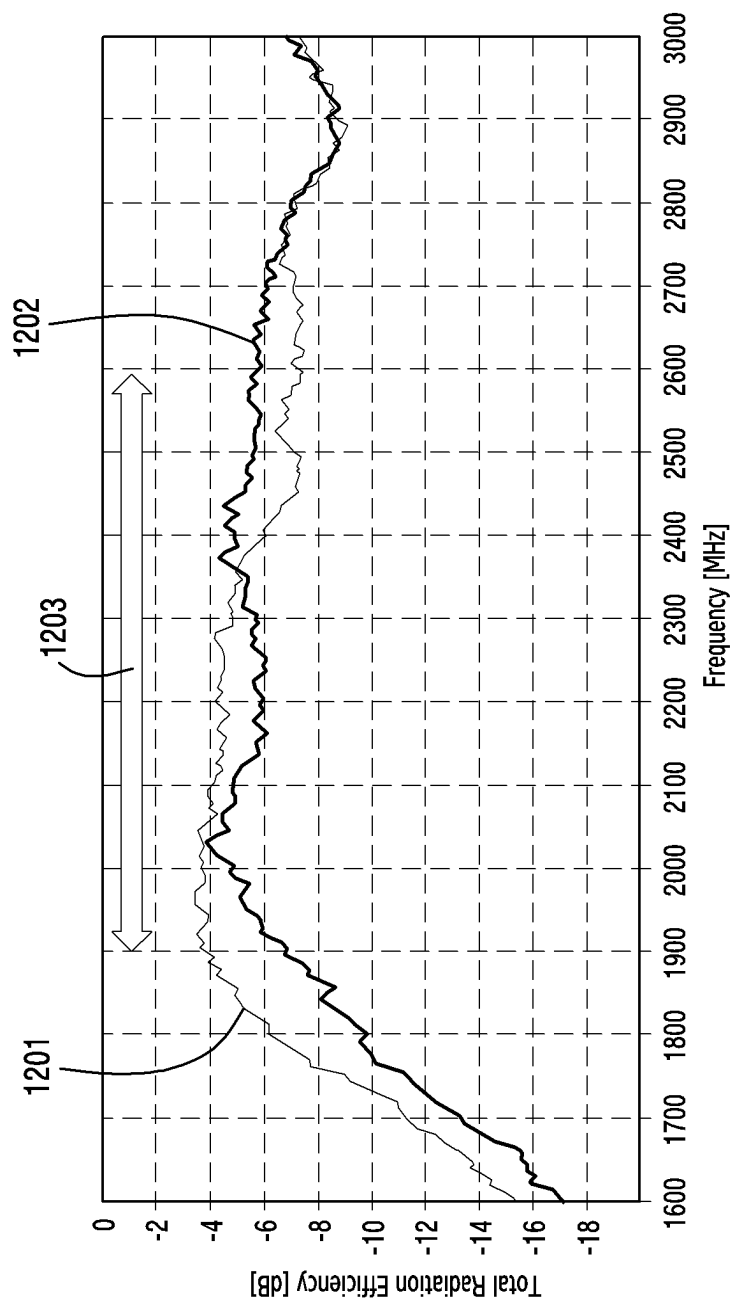


FIG.12

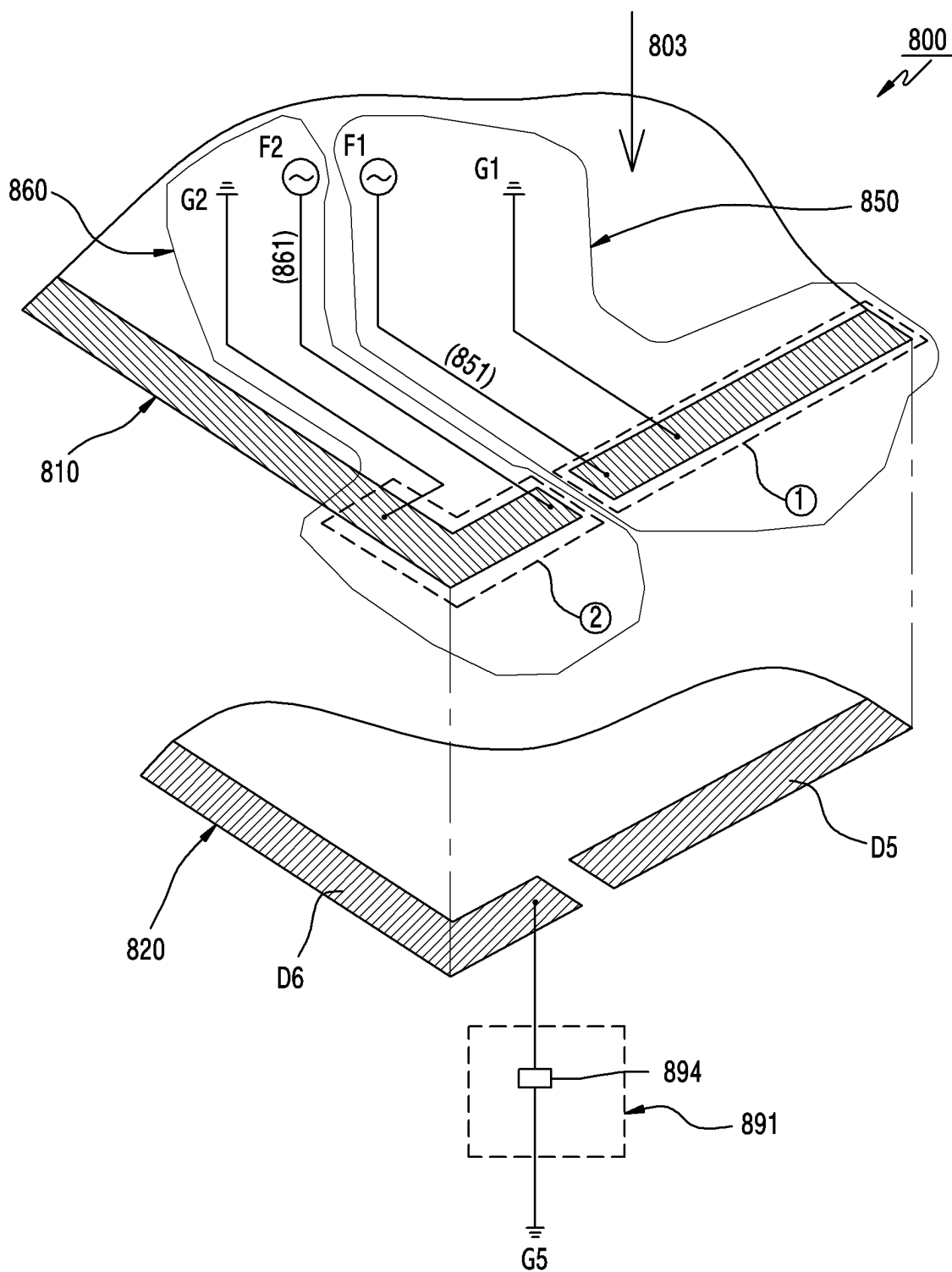


FIG.13

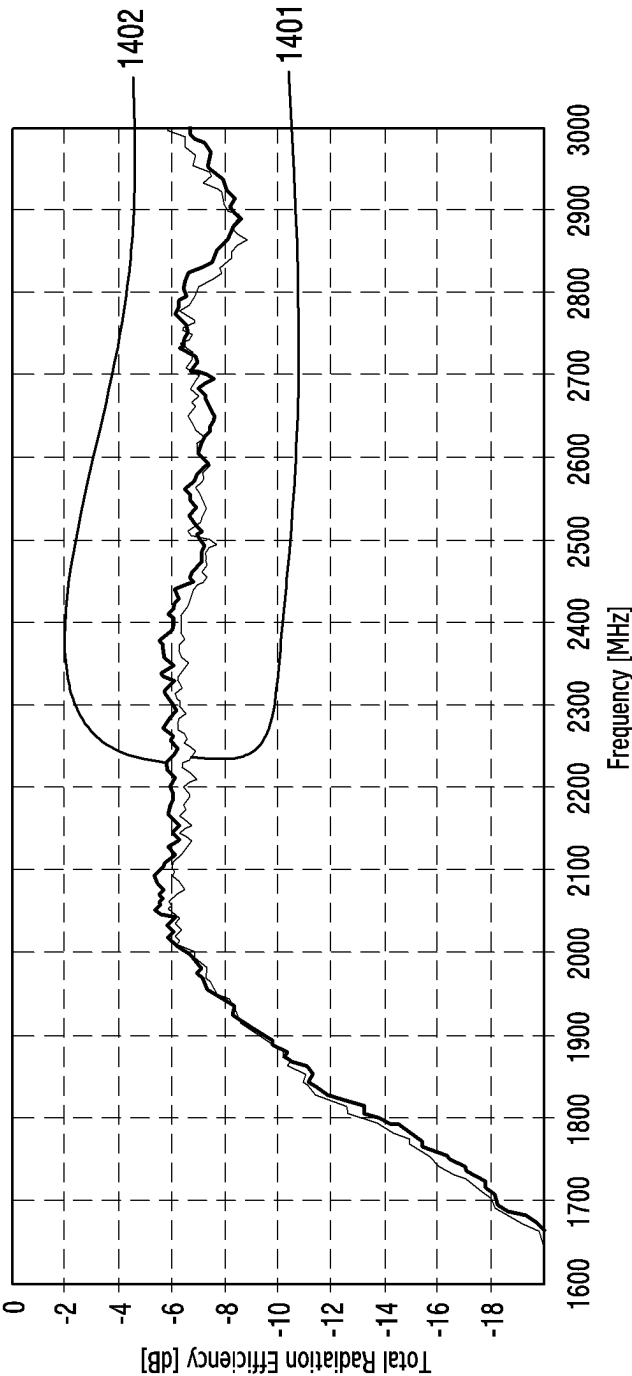


FIG.14

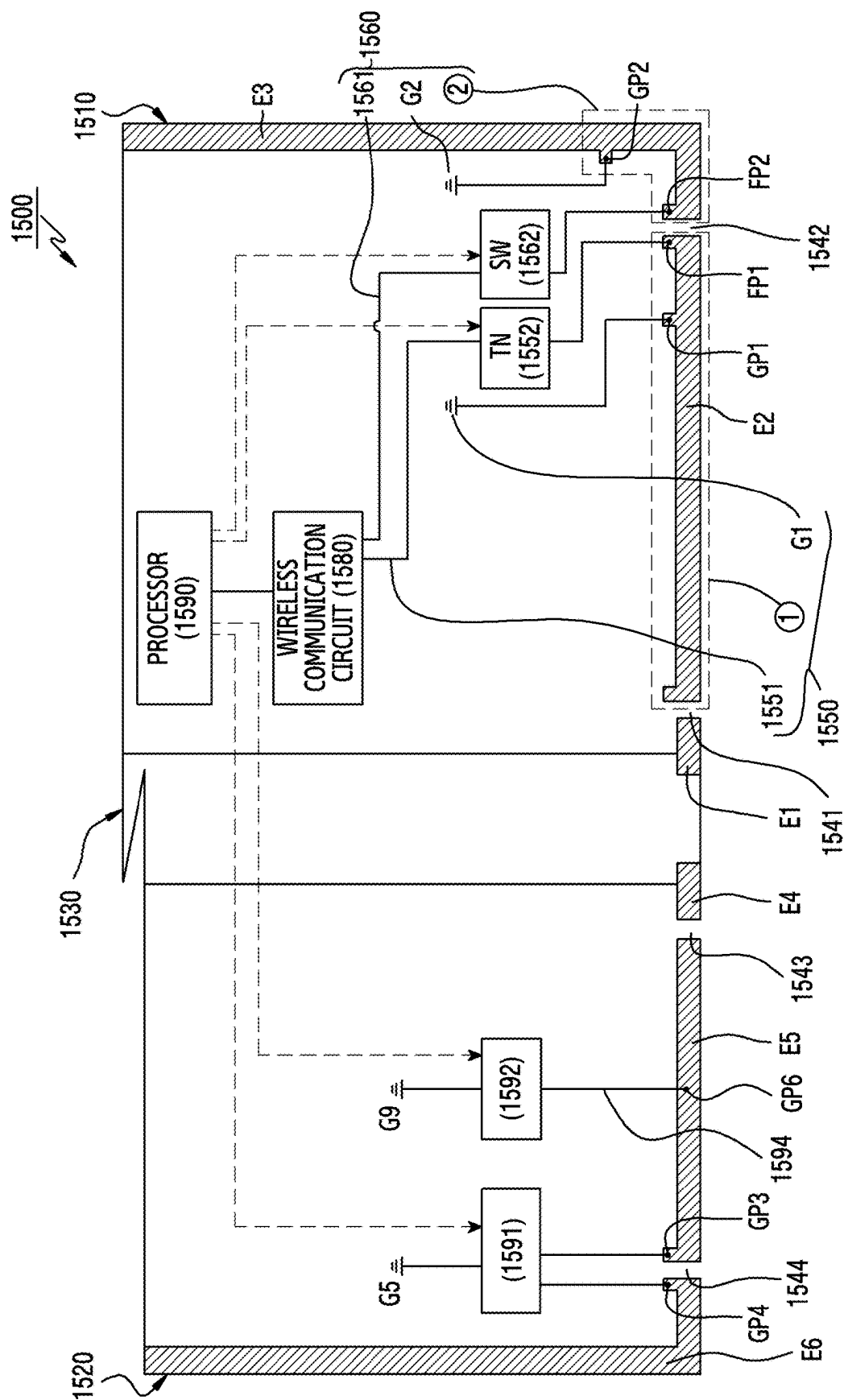


FIG. 15A

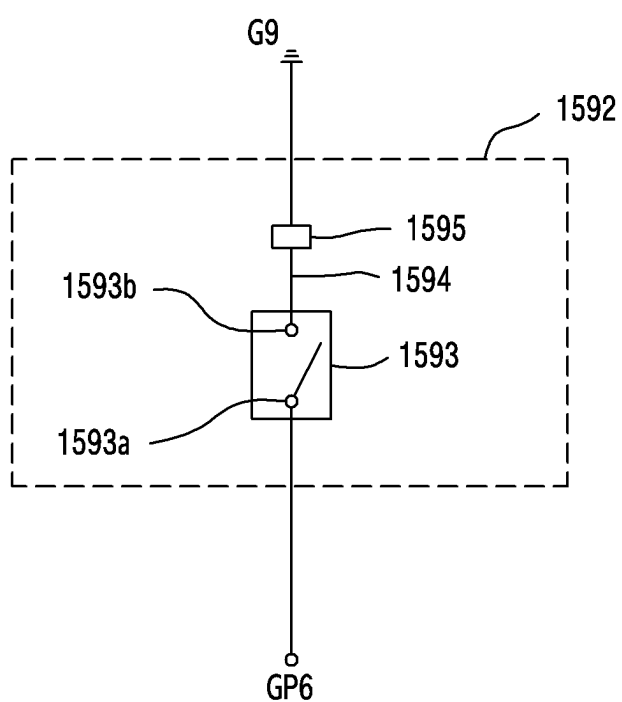


FIG.15B

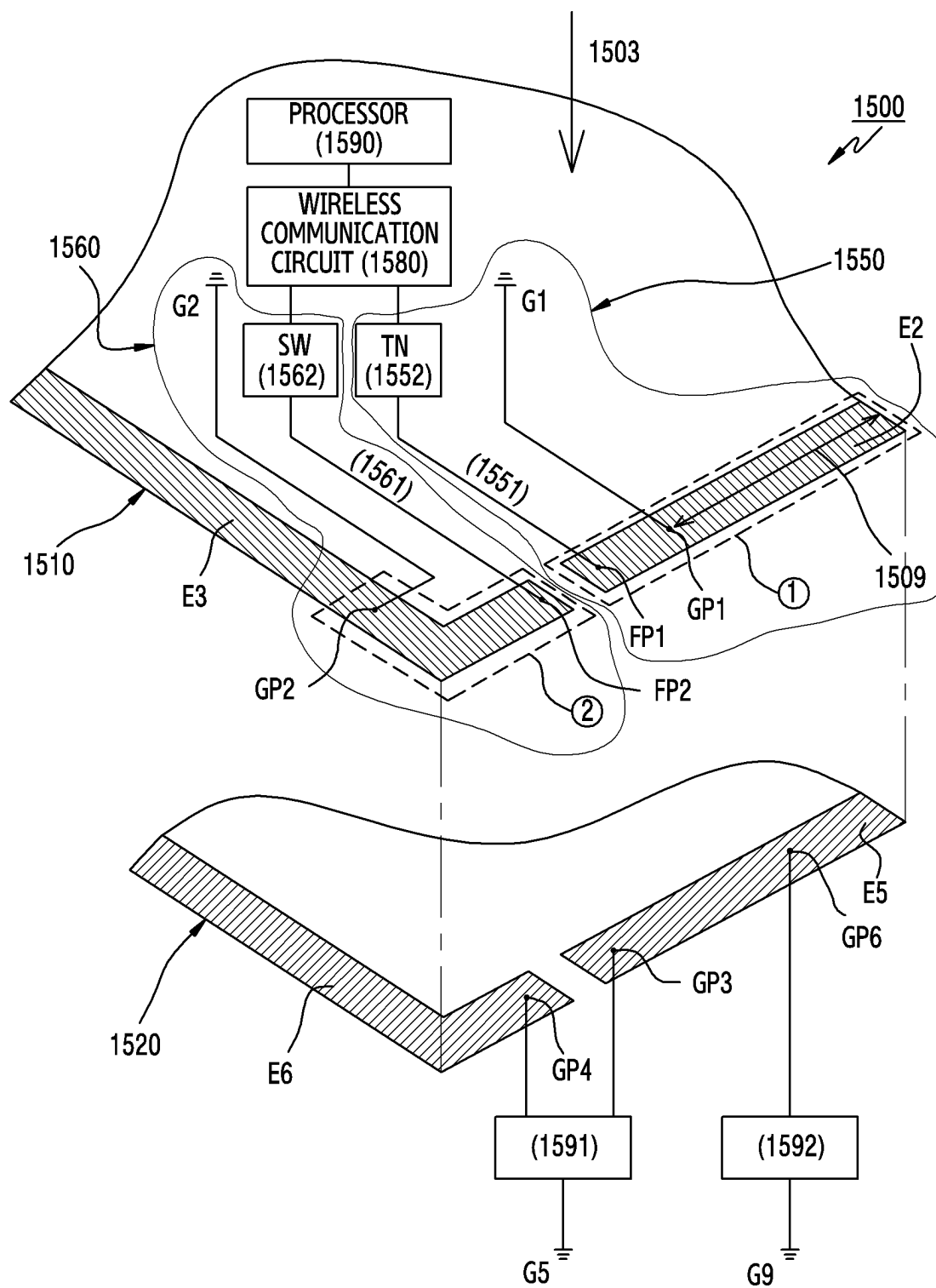


FIG. 15C

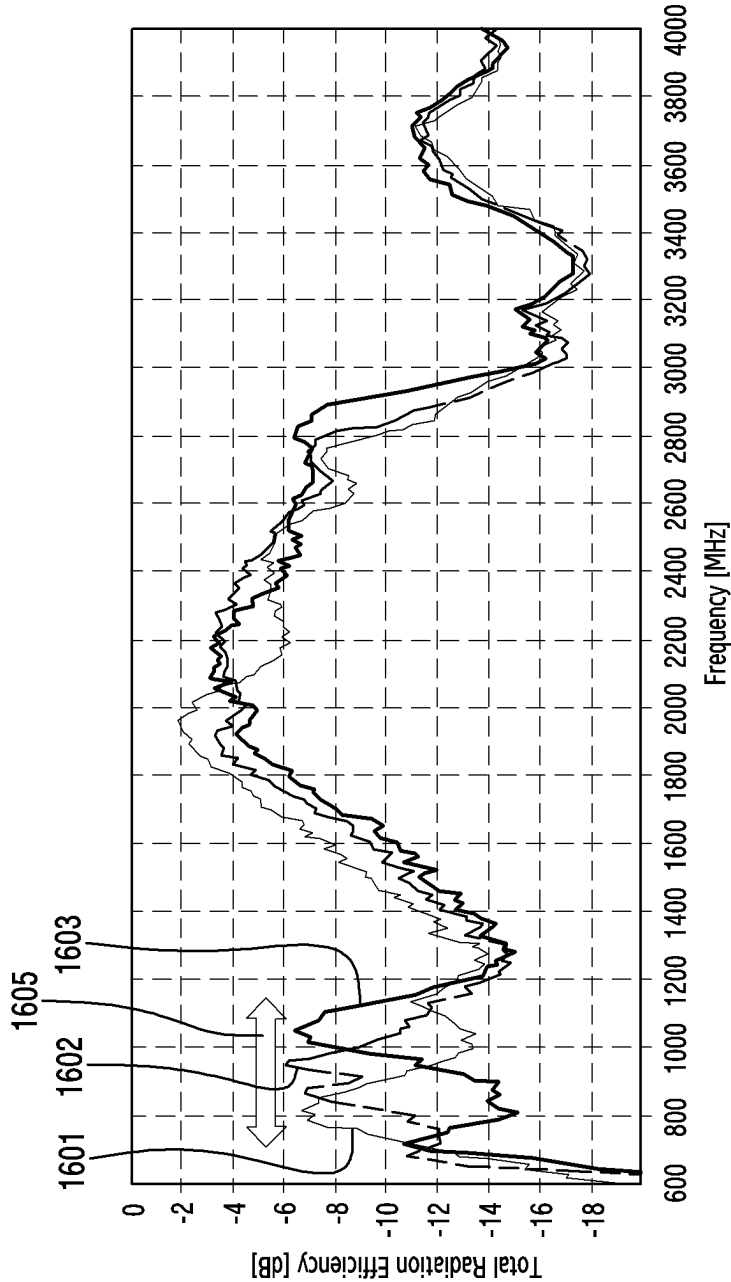


FIG.16

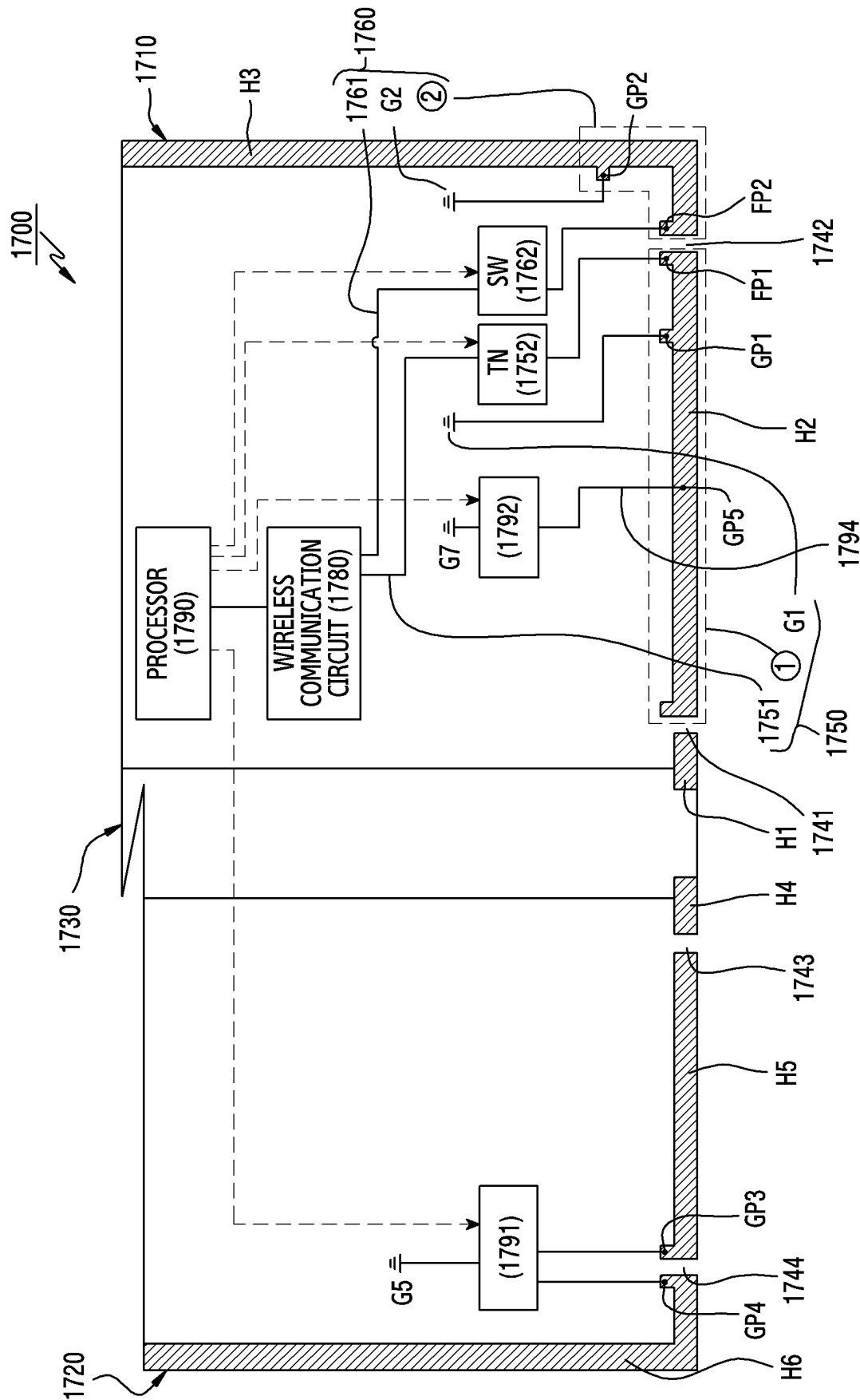


FIG. 17A

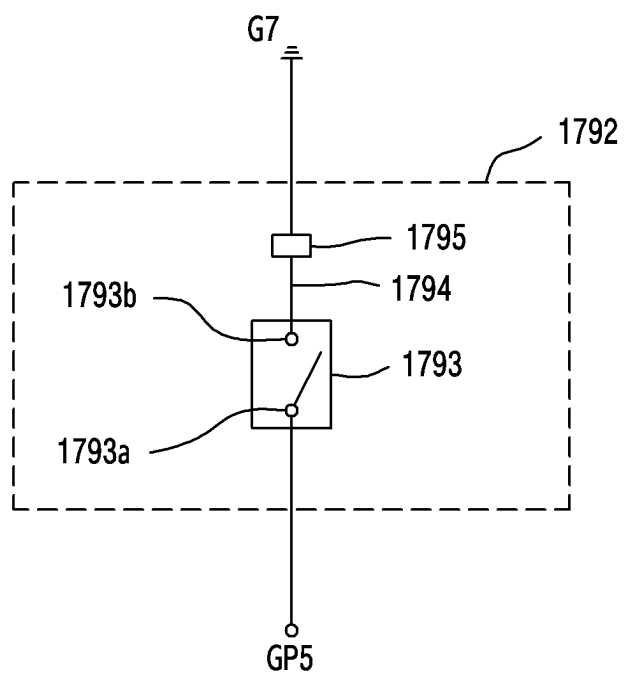


FIG.17B

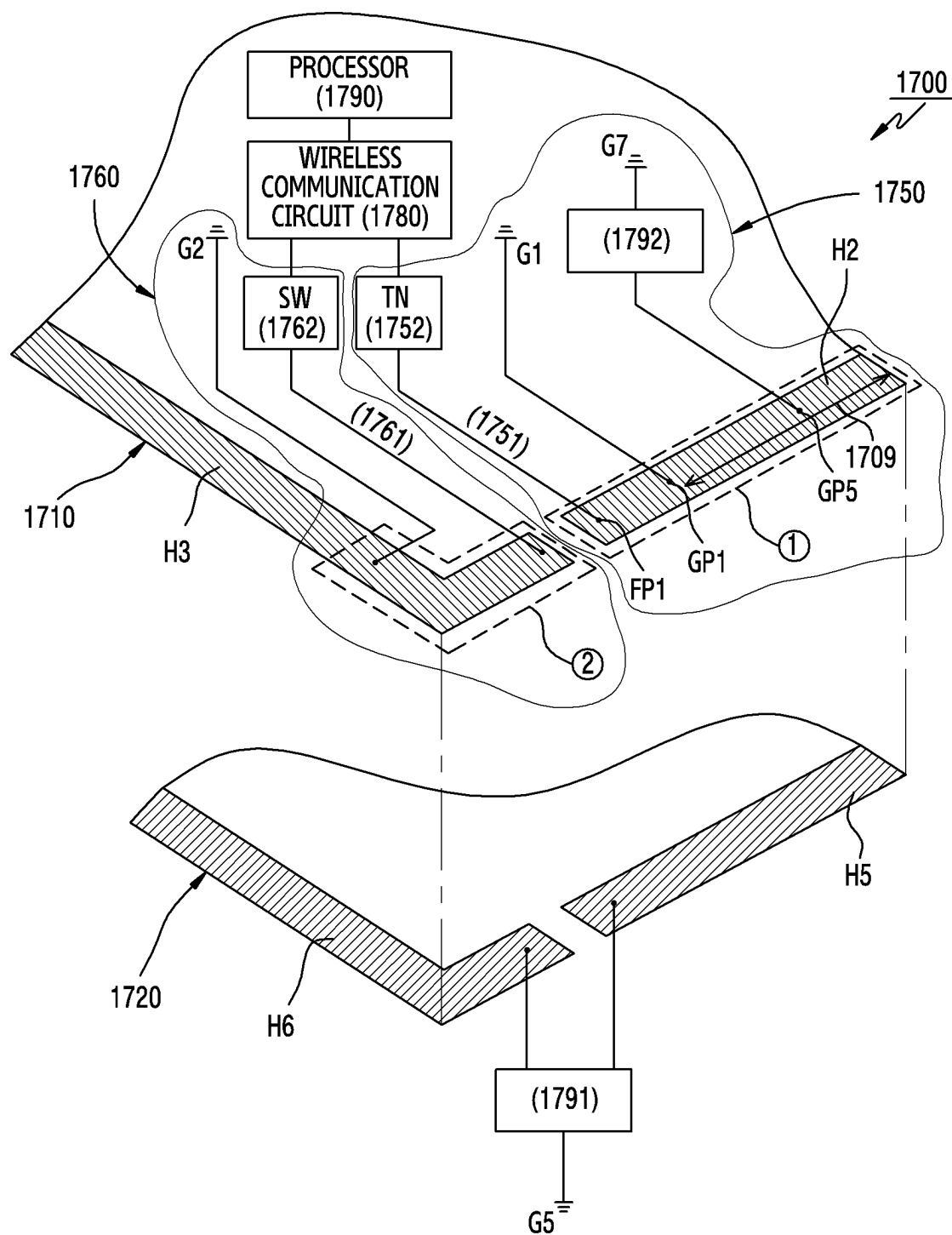


FIG. 17C

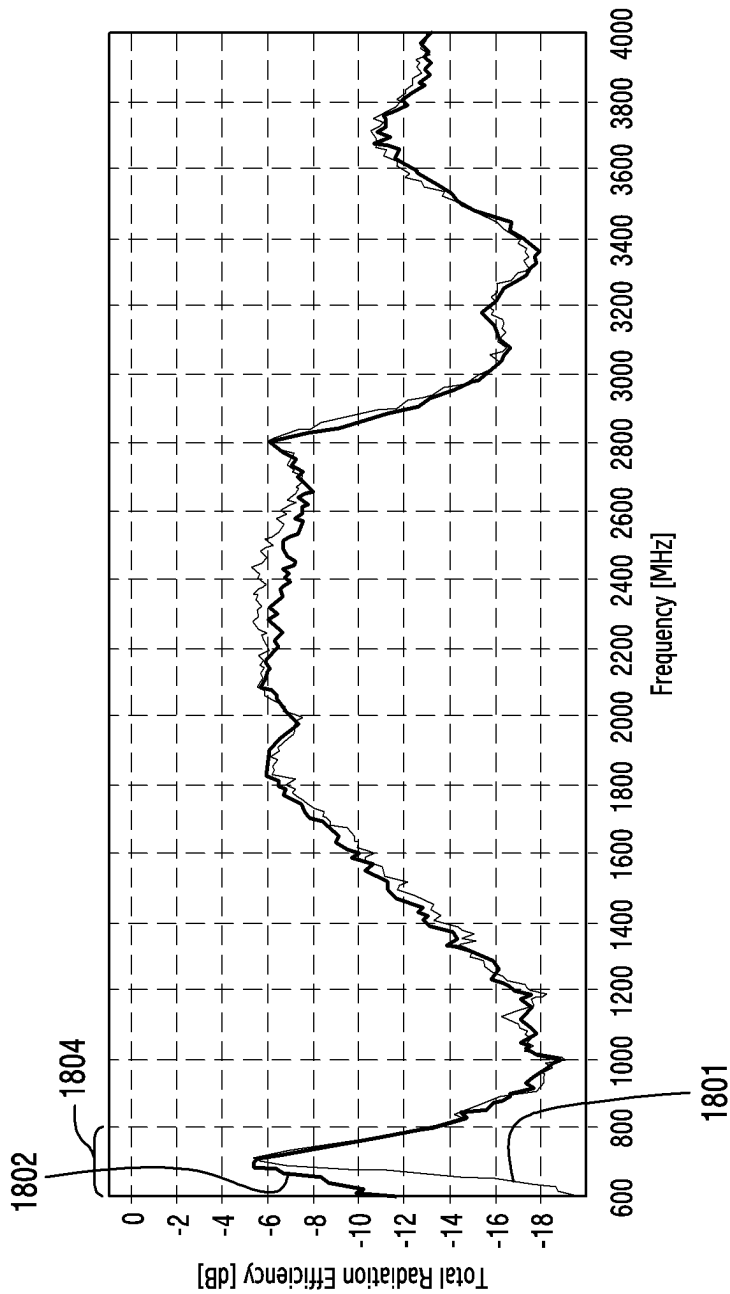


FIG.18

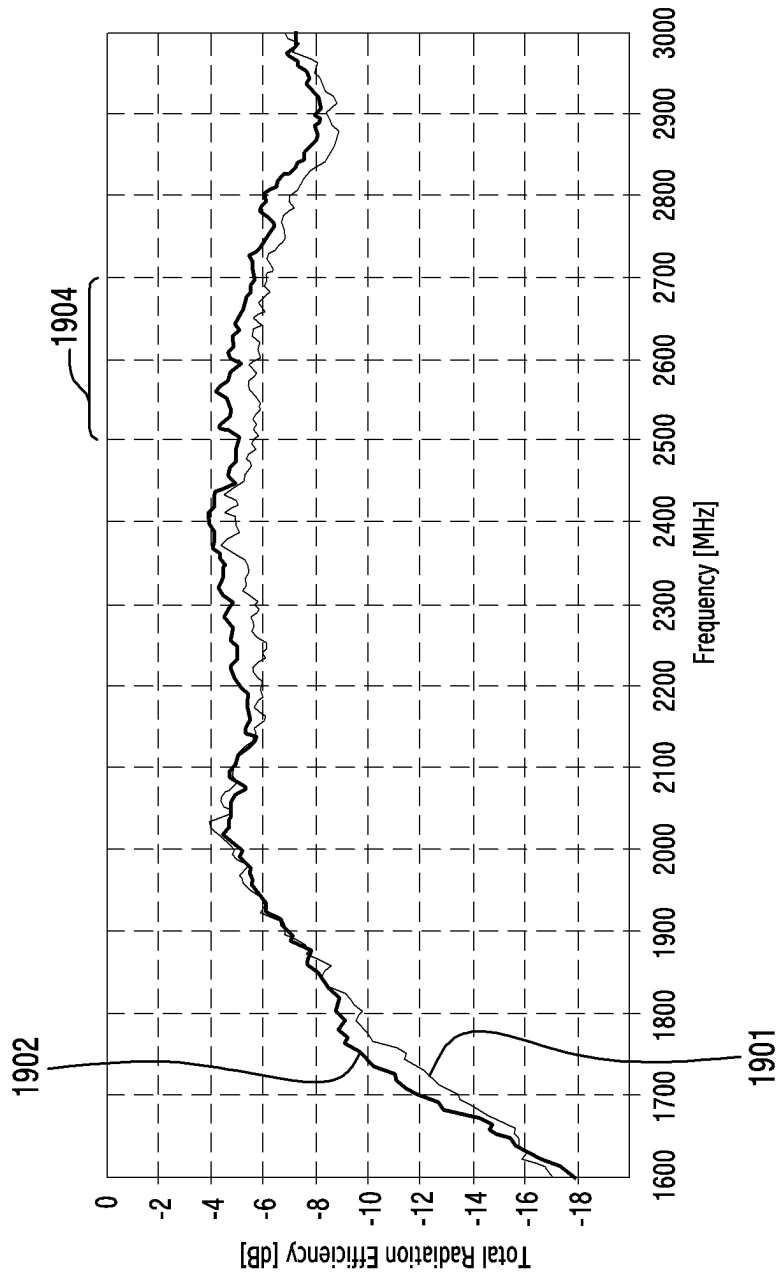


FIG.19

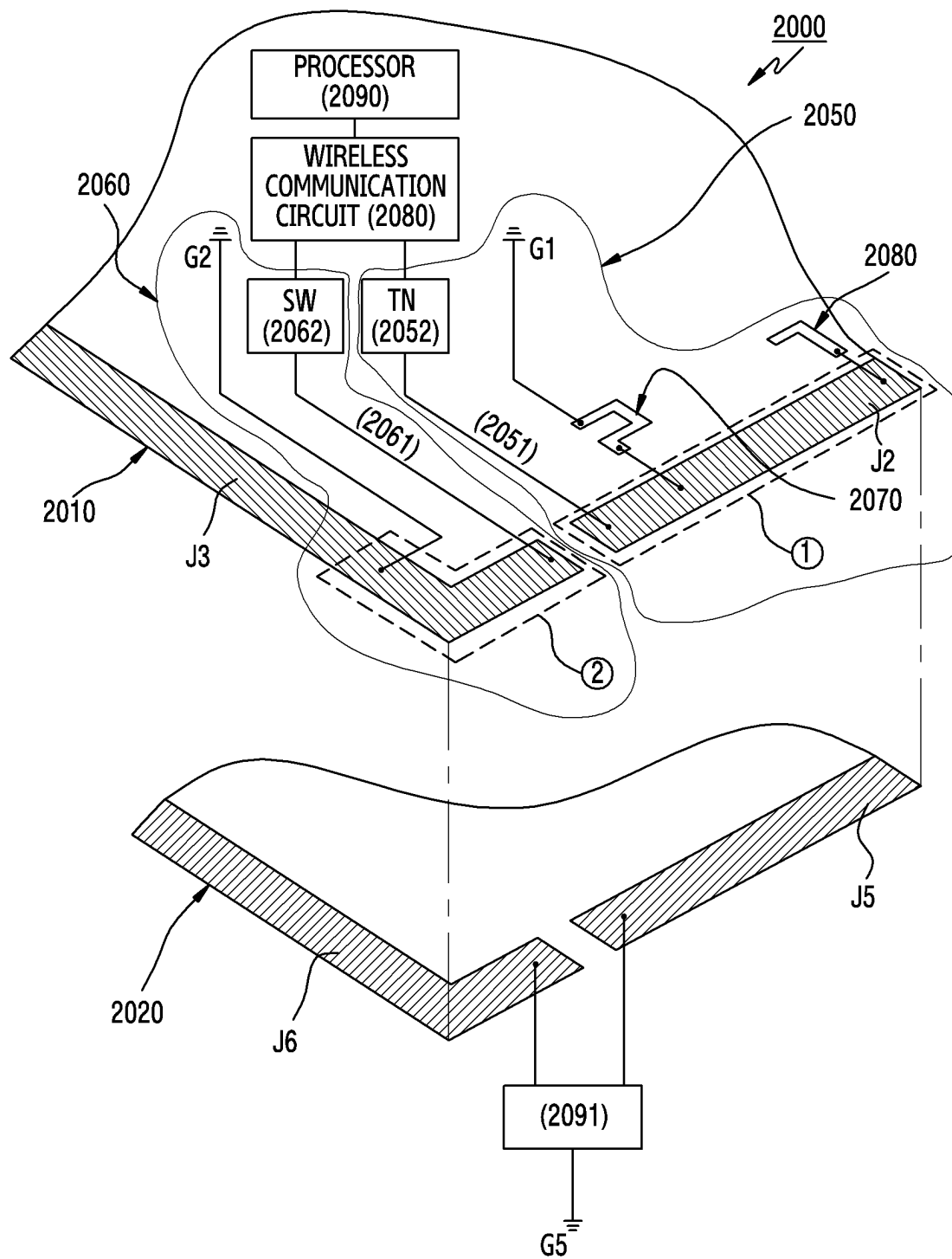


FIG. 20

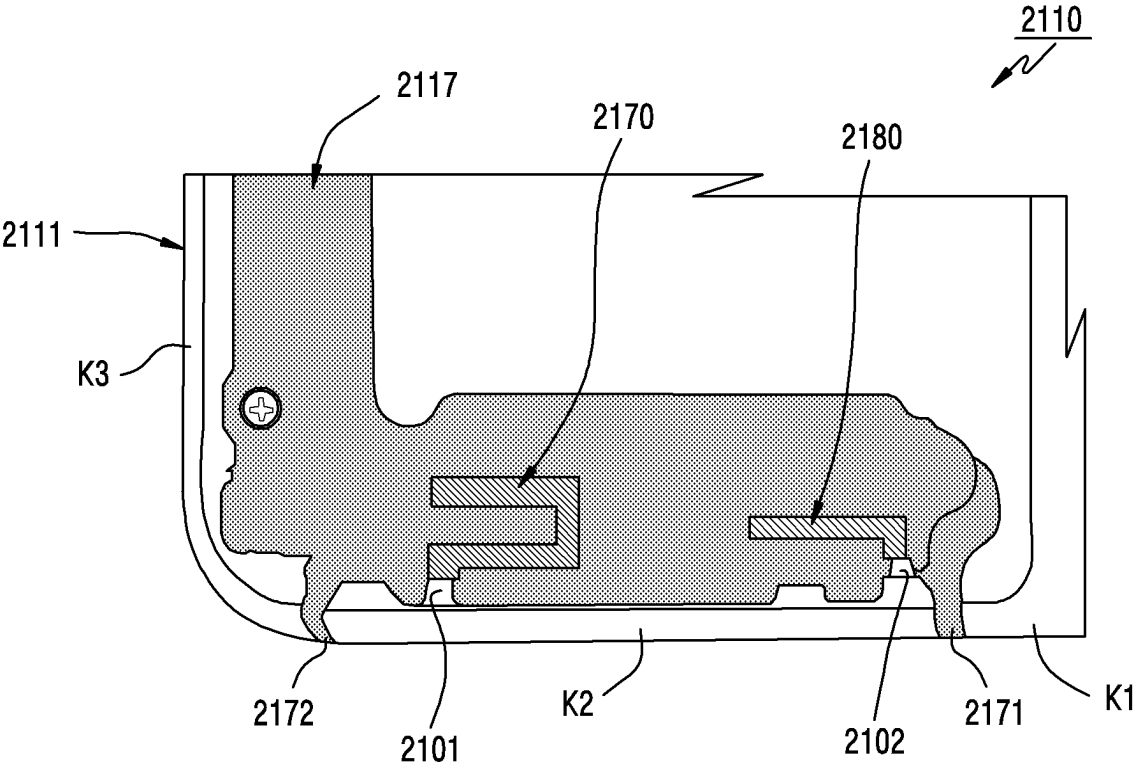


FIG.21

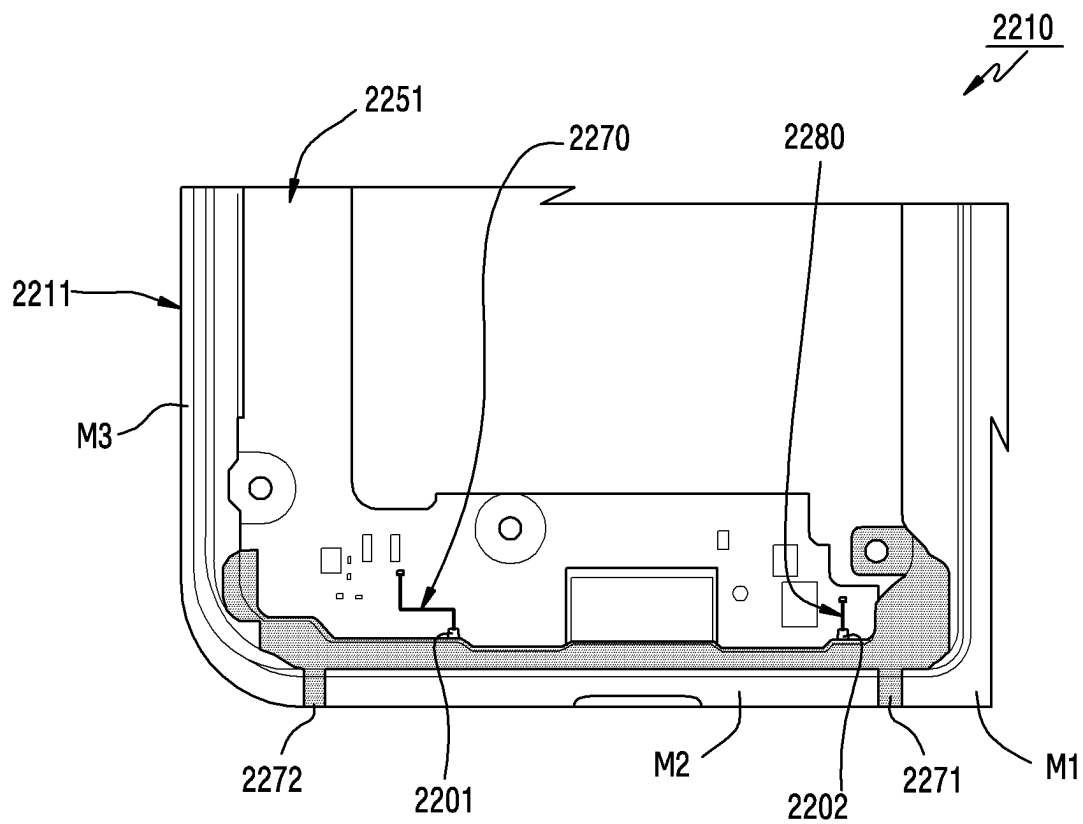


FIG. 22

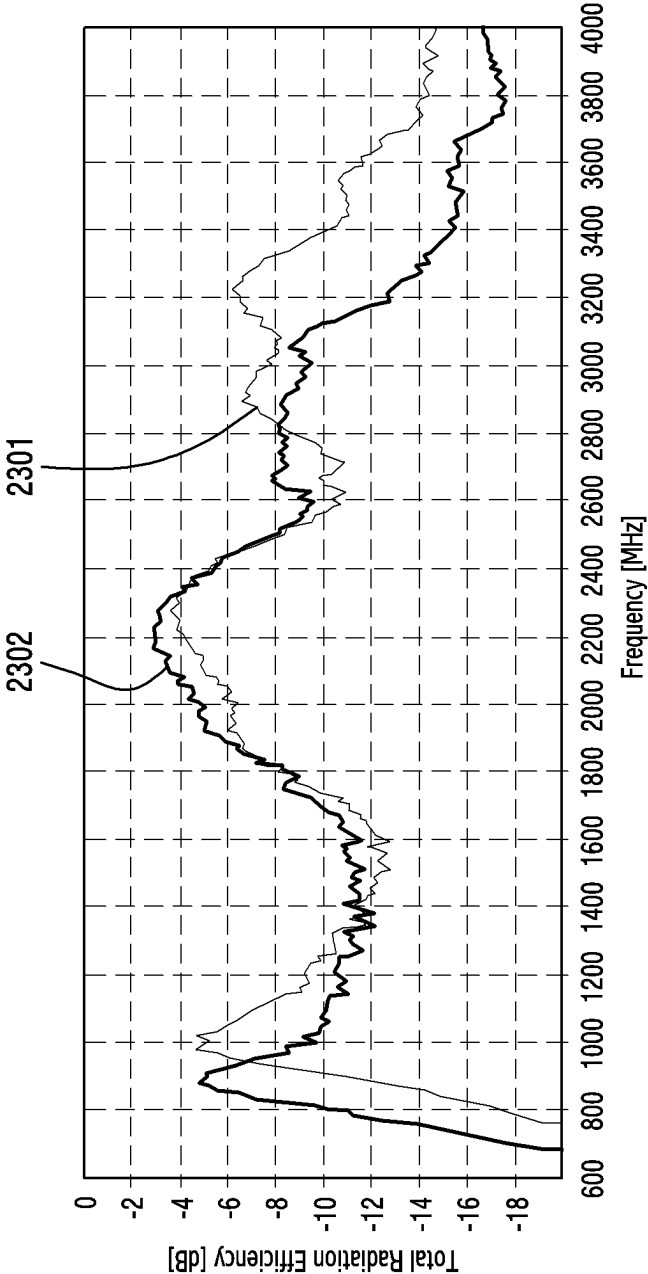


FIG.23

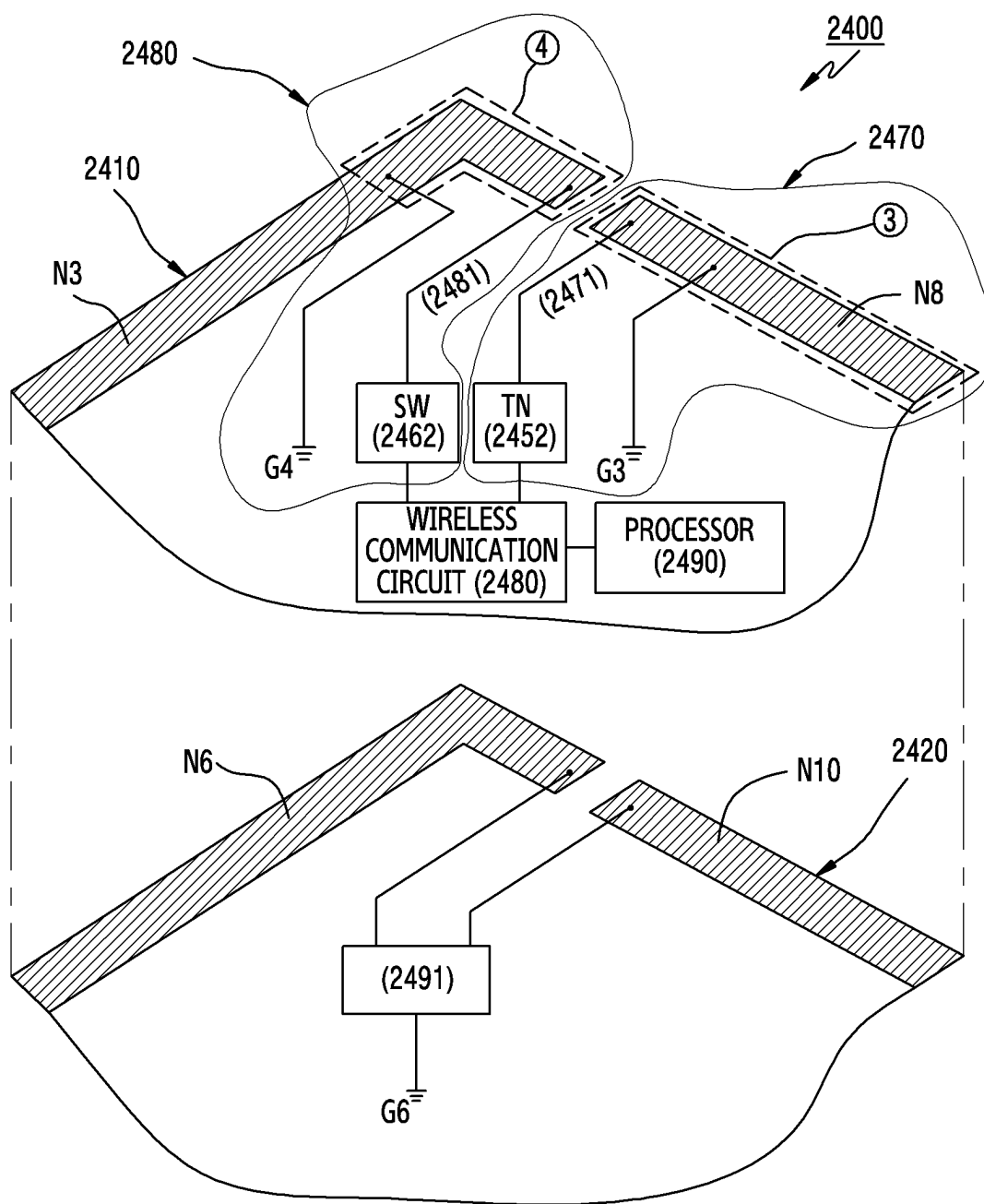


FIG.24

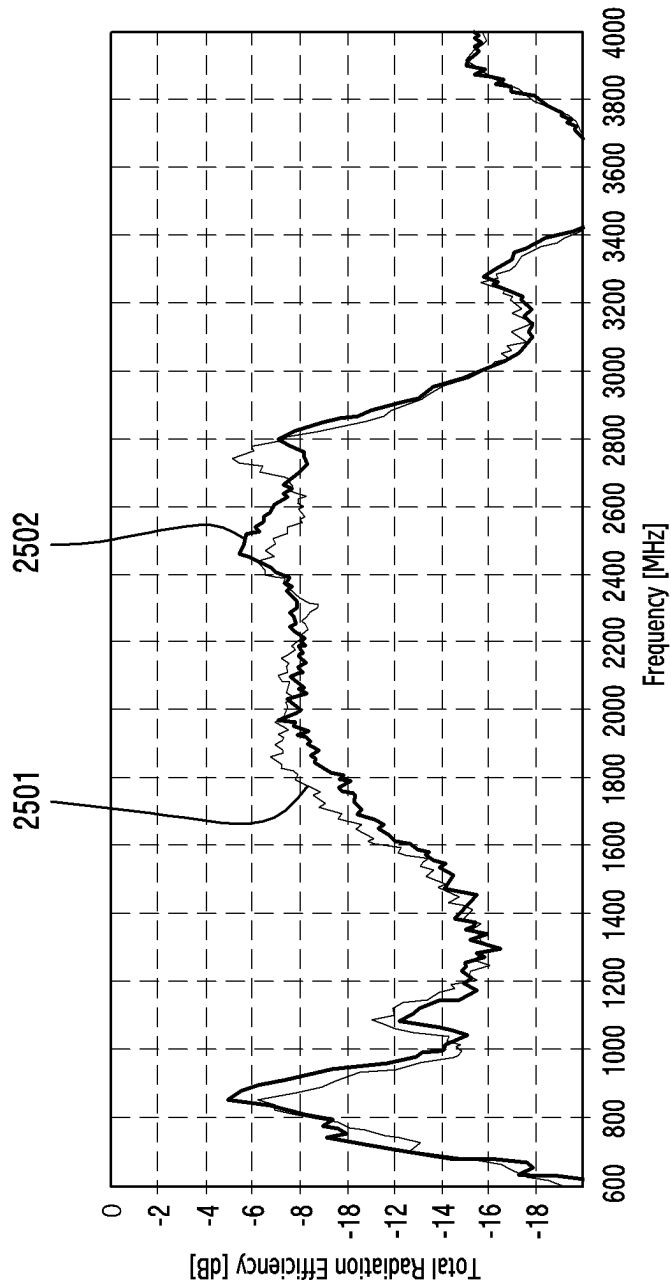


FIG.25

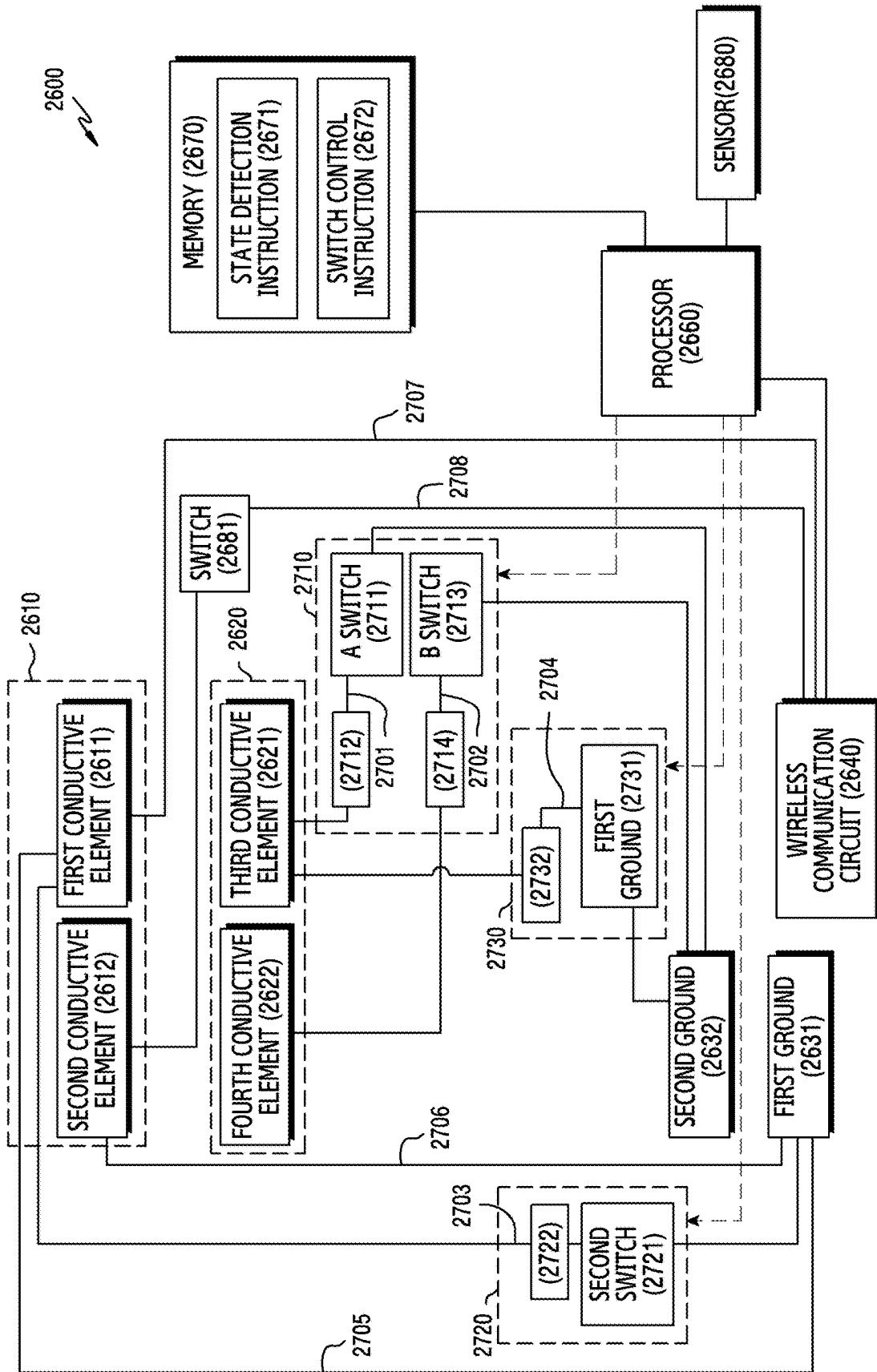


FIG. 26

	LB		MB		HB	
	A SWITCH (2711)	B SWITCH (2713)	A SWITCH (2711)	B SWITCH (2713)	A SWITCH (2711)	B SWITCH (2713)
FIRST CONDUCTIVE ELEMENT (2611)	On	Off	On	Off	-	-
SECOND CONDUCTIVE ELEMENT (2612)	-	-	On	On	On	Off

FIG.27

ANTENNA AND ELECTRONIC DEVICE COMPRISING SAME

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This application is a continuation application of prior application Ser. No. 18/609,428 filed on Mar. 19, 2024; which is a continuation application of prior application Ser. No. 17/431,861 filed on Aug. 18, 2021, which has issued as U.S. Pat. No. 11,962,068 on Apr. 16, 2024; which is a U.S. National Stage application under 35 U.S.C. § 371 of an International application No. PCT/KR2020/002374 filed on Feb. 19, 2020; and which is based on and claims priority of a Korean patent application No. 10-2019-0019038 filed on Feb. 19, 2019 in the Korean Intellectual Property Office, the disclosure of each of which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

[0002] Various embodiments relate to an antenna and an electronic device including the same.

BACKGROUND ART

[0003] As electronic communication technology develops, electronic devices having various functions are emerging. These electronic devices may have a convergence function which performs one or more functions in combination.

[0004] In recent, as a functional gap for each manufacturer remarkably reduces, the electronic device is reinforcing a design aspect and concurrently becoming slimmer to satisfy consumer's purchase needs.

DISCLOSURE OF INVENTION

Technical Problem

[0005] An electronic device of a rotation type (e.g., a foldable type, a slide type, or a swivel type) in which at least two housings are rotatably operated by a connection unit is emerging. For example, the electronic device of the rotation type may be disposed in a manner of overlapping the at least two housings, and due to this operation, an antenna disposed in one housing may be degraded in radiation performance by a metal member disposed at a corresponding position of the other housing which overlaps it.

[0006] Various embodiments of the present disclosure may include an antenna for preventing antenna performance degradation caused by overlapping of housings, and an electronic device including the same.

Solution to Problem

[0007] According to an embodiment of the present disclosure, an electronic device may include a foldable housing which includes, as a foldable housing, a hinge structure, a first housing structure connected to the hinge structure, and including a first surface facing a first direction, a second surface facing a second direction which is opposite to the first direction, and a first side member which surrounds at least in part a space between the first surface and the second surface, and includes a first conductive unit, a first insulating unit, a second conductive unit, a second insulating unit and a third conductive unit sequentially disposed from the hinge structure, and a second housing structure connected to the

hinge structure, including a third surface facing a third direction, a fourth surface facing a fourth direction which is opposite to the third direction, and a second side member which surrounds at least in part a space between the third surface and the fourth surface, and includes a fourth conductive unit, a third insulating unit, a fifth conductive unit, a fourth insulating unit and a sixth conductive unit sequentially disposed from the hinge structure, and folding with the first housing structure based on the hinge structure, the first surface facing the third face in a folded state, and the third direction being the same as the first direction in an unfolded state, and the first insulating unit and the third insulating unit substantially aligning in the folded state, and the second insulating unit and the fourth insulating unit substantially aligning in the folded state, a flexible display extended from the first surface to the third surface, a first printed circuit board disposed between the first surface and the second surface, and including at least one first ground layer, a second printed circuit board disposed between the third surface and the fourth surface, and including at least one second ground layer, at least one wireless communication circuit disposed on the first printed circuit board, and configured to transmit and/or receive a first signal of a first frequency band by electrically connecting to a first position closer to the second insulating unit than the first insulating unit, of the second conductive unit, and a signal of a second frequency band by electrically connecting to a second position close to the second insulating unit of the third conductive unit, and at least one first switch including a first terminal electrically connected to the second ground layer, a second terminal electrically connected to a third position closer to the fourth insulating unit than the third insulating unit, of the fifth conductive unit, and a third terminal electrically connected to a fourth position close to the fourth insulating unit of the sixth conductive unit.

Advantageous Effects of Invention

[0008] According to various embodiments of the present disclosure, a folded state of an electronic device may reduce an influence of at least one conductive unit of a second housing structure on antenna radiation performance of at least one antenna included in a first housing structure.

[0009] Besides, effects attained or predicted with various embodiments of the present disclosure are disclosed directly or implicitly in detailed descriptions of embodiments of the present disclosure. For example, various effects predicted according to various embodiments of the present disclosure shall be disclosed in the detailed descriptions to be explained.

BRIEF DESCRIPTION OF DRAWINGS

[0010] FIG. 1 is a block diagram of an electronic device in a network environment according to embodiments.

[0011] FIG. 2 is a diagram illustrating an unfolded state of an electronic device according to an embodiment.

[0012] FIG. 3 is a diagram illustrating a folded state of an electronic device according to an embodiment.

[0013] FIG. 4 is an exploded perspective view of an electronic device of FIG. 2 or 3.

[0014] FIG. 5 illustrates an electronic device of an unfolded state according to an embodiment.

[0015] FIG. 6A illustrates a circuit of an electronic device of an unfolded state according to an embodiment.

[0016] FIG. 6B illustrates a frequency adjustment circuit of FIG. 6A according to an embodiment.

[0017] FIG. 7A illustrates a circuit of an electronic device of an unfolded state according to various embodiments.

[0018] FIG. 7B illustrates a frequency adjustment circuit of FIG. 7A according to various embodiments.

[0019] FIG. 8 illustrates a circuit related to a folded state of an electronic device according to an embodiment.

[0020] FIG. 9A is a graph showing antenna radiation performance on a frequency distribution for a first antenna system if an electronic device not including a frequency adjustment circuit is unfolded or folded.

[0021] FIG. 9B is a graph showing the antenna radiation performance on the frequency distribution for the first antenna system if the electronic device including the frequency adjustment circuit is unfolded or folded according to an embodiment.

[0022] FIG. 10A is a graph showing antenna radiation performance on a frequency distribution for a second antenna system if an electronic device not including a frequency adjustment circuit is unfolded or folded.

[0023] FIG. 10B is a graph showing the antenna radiation performance on the frequency distribution for the second antenna system if the electronic device including the frequency adjustment circuit is unfolded or folded according to an embodiment.

[0024] FIG. 11 is a graph showing antenna radiation performance on a frequency distribution of a first antenna system, if an electronic device of FIG. 6A in a folded state turns off or on a first switch of a frequency adjustment circuit according to an embodiment.

[0025] FIG. 12 is a graph showing antenna radiation performance on a frequency distribution of a second antenna system, if an electronic device of FIG. 6A in a folded state turns off or on a first switch of a frequency adjustment circuit according to an embodiment.

[0026] FIG. 13 illustrates a circuit related to a folded state of an electronic device according to an embodiment.

[0027] FIG. 14 is a graph illustrating antenna radiation performance on a frequency distribution of a second antenna system, if a sixth conductive unit and a ground are electrically connected and not electrically connected while an electronic device is folded according to an embodiment.

[0028] FIG. 15A illustrates a circuit of an electronic device of an unfolded state according to an embodiment.

[0029] FIG. 15B illustrates a frequency adjustment circuit according to an embodiment.

[0030] FIG. 15C illustrates a circuit of an electronic device of a folded state according to an embodiment.

[0031] FIG. 16 is a graph showing antenna radiation performance on a frequency distribution of a first antenna system in an electronic device of a folded state according to an embodiment.

[0032] FIG. 17A illustrates a circuit of an electronic device of an unfolded state according to an embodiment.

[0033] FIG. 17B illustrates a frequency adjustment circuit according to an embodiment.

[0034] FIG. 17C illustrates a circuit of an electronic device of a folded state according to an embodiment.

[0035] FIG. 18 is a graph showing antenna radiation performance on a frequency distribution of a first antenna system in an electronic device of a folded state according to an embodiment.

[0036] FIG. 19 is a graph showing antenna radiation performance on a frequency distribution of a second antenna system in an electronic device of a folded state according to an embodiment.

[0037] FIG. 20 illustrates a circuit of an electronic device of a folded state according to an embodiment.

[0038] FIG. 21 illustrates an antenna extension structure according to an embodiment.

[0039] FIG. 22 illustrates an antenna extension structure according to various embodiments.

[0040] FIG. 23 is a graph showing antenna radiation performance of a first antenna system according to whether an antenna extension structure is applied or not according to an embodiment.

[0041] FIG. 24 illustrates a circuit of an electronic device of a folded state according to an embodiment.

[0042] FIG. 25 is a graph showing antenna radiation performance on a frequency distribution of an antenna system according to a frequency adjustment circuit while an electronic device of FIG. 24 is folded according to an embodiment.

[0043] FIG. 26 is a block diagram of an electronic device including a frequency adjustment circuit according to an embodiment.

[0044] FIG. 27 is a table for explaining control on a frequency adjustment circuit of FIG. 26 according to an embodiment.

BEST MODE FOR CARRYING OUT THE INVENTION

[0045] Hereinafter, various embodiments of the present document are described with reference to the accompanying drawings.

[0046] FIG. 1 is a block diagram illustrating an electronic device 101 in a network environment 100 according to various embodiments.

[0047] Referring to FIG. 1, the electronic device 101 in the network environment 100 may communicate with an electronic device 102 via a first network 198 (e.g., a short-range wireless communication network), or an electronic device 104 or a server 108 via a second network 199 (e.g., a long-range wireless communication network). According to an embodiment, the electronic device 101 may communicate with the electronic device 104 via the server 108. According to an embodiment, the electronic device 101 may include a processor 120, memory 130, an input device 150, a sound output device 155, a display device 160, an audio module 170, a sensor module 176, an interface 177, a haptic module 179, a camera module 180, a power management module 188, a battery 189, a communication module 190, a subscriber identification module (SIM) 196, or an antenna module 197. In some embodiments, at least one (e.g., the display device 160 or the camera module 180) of the components may be omitted from the electronic device 101, or one or more other components may be added in the electronic device 101. In some embodiments, some of the components may be implemented as single integrated circuitry. For example, the sensor module 176 (e.g., a fingerprint sensor, an iris sensor, or an illuminance sensor) may be implemented as embedded in the display device 160 (e.g., a display).

[0048] The processor 120 may execute, for example, software (e.g., a program 140) to control at least one other component (e.g., a hardware or software component) of the

electronic device **101** coupled with the processor **120**, and may perform various data processing or computation. According to one embodiment, as at least part of the data processing or computation, the processor **120** may load a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to an embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), and an auxiliary processor **123** (e.g., a graphics processing unit (GPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. Additionally, or alternatively, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.

[0049] The auxiliary processor **123** may control at least some of functions or states related to at least one component (e.g., the display device **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). According to an embodiment, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**.

[0050] The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

[0051] The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

[0052] The input device **150** may receive a command or data to be used by other component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input device **150** may include, for example, a microphone, a mouse, a keyboard, or a digital pen (e.g., a stylus pen).

[0053] The sound output device **155** may output sound signals to the outside of the electronic device **101**. The sound output device **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record, and the receiver may be used for an incoming call. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

[0054] The display device **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display device **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display,

hologram device, and projector. According to an embodiment, the display device **160** may include touch circuitry adapted to detect a touch, or sensor circuitry (e.g., a pressure sensor) adapted to measure the intensity of force incurred by the touch.

[0055] The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input device **150**, or output the sound via the sound output device **155** or a headphone of an external electronic device (e.g., an electronic device **102**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.

[0056] The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

[0057] The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly (e.g., wiredly) or wirelessly. According to an embodiment, the interface **177** may include, for example, a high-definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

[0058] A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to an embodiment, the connecting terminal **178** may include, for example, a HDMI connector, a USB connector, a SD card connector, or an audio connector (e.g., a headphone connector).

[0059] The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

[0060] The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

[0061] The power management module **188** may manage power supplied to the electronic device **101**. According to one embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

[0062] The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

[0063] The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the

server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a cellular network, the Internet, or a computer network (e.g., LAN or wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

[0064] The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an embodiment, the antenna module **197** may include an antenna including a radiating element composed of a conductive material or a conductive pattern formed in or on a substrate (e.g., PCB). According to an embodiment, the antenna module **197** may include a plurality of antennas. In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** (e.g., the wireless communication module **192**) from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.

[0065] At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

[0066] According to an embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the electronic devices **102** and **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one

or more of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, or client-server computing technology may be used, for example.

[0067] The electronic device according to various embodiments may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. According to an embodiment of the disclosure, the electronic devices are not limited to those described above.

[0068] It should be appreciated that various embodiments of the present disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” “coupled to,” “connected with,” or “connected to” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

[0069] As used herein, the term “module” may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, “logic,” “logic block,” “part,” or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

[0070] Various embodiments as set forth herein may be implemented as software (e.g., the program **140**) including one or more instructions that are stored in a storage medium (e.g., internal memory **136** or external memory **138**) that is

readable by a machine (e.g., the electronic device **101**). For example, a processor (e.g., the processor **120**) of the machine (e.g., the electronic device **101**) may invoke at least one of the one or more instructions stored in the storage medium, and execute it, with or without using one or more other components under the control of the processor. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a compiler or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.

[0071] According to an embodiment, a method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., PlayStore™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.

[0072] According to various embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities. According to various embodiments, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively, or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to various embodiments, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

[0073] FIG. 2 is a diagram illustrating a flat or unfolded state of an electronic device according to an embodiment. FIG. 3 is a diagram illustrating a folded state of an electronic device according to an embodiment.

[0074] Referring to FIGS. 2 and 3, in an embodiment, an electronic device **20** (e.g., the electronic device **101** of FIG. 1) may include a foldable housing **200**, a hinge cover **230** which covers a foldable unit of the foldable housing **200**, and a flexible or foldable display **300** (hereafter, a “display” **300**, in short) (e.g., the display device **160** of FIG. 1) disposed in a space formed by the foldable housing **200**. According to an embodiment, the foldable housing **200** may

include a front surface **200a** exposing the display **300**, a rear surface **200b** facing away from the front surface **200a**, and side surfaces **200c** and **200d** surrounding a space between the front surface **200a** and the rear surface **200b**.

[0075] According to an embodiment, the foldable housing **200** may include a first housing structure **210** and a second housing structure **220** interconnected by a hinge structure (not shown). For example, in the hinge structure, the first housing structure **210** may be rotatably connected with the second housing structure **220** by the hinge structure.

[0076] According to an embodiment, the first housing structure **210** may include a first surface **2001** facing a first direction **201**, a second surface **2002** facing a second direction **202** which is opposite to the first direction **201**, and a first side surface **200c** which surrounds at least part of a space between the first surface **2001** and the second surface **2002**. The second housing structure **220** may include a third surface **2003** facing a third direction **203**, a fourth surface **2004** facing a fourth direction **204** which is opposite to the third direction **203**, and a second side surface **200d** which surrounds at least part of a space between the third surface **2003** and the fourth surface **2004**. The front surface **200a** of the electronic device **20** may include the first surface **2001** and the third surface **2003**, and the rear surface **200b** of the electronic device **20** may include the second surface **2002** and the fourth surface **2004**. In various embodiments (not shown), the first housing structure **210** may refer to a structure which forms part of the first surface **2001**, the second surface **2002**, and the first side surface **200c**. In various embodiments (not shown), the second housing structure **220** may refer to a structure which forms part of the third surface **2003**, the fourth surface **2004**, and the second side surface **200d**.

[0077] According to an embodiment, the foldable housing **200** may include a transparent plate (not shown) (e.g., a polymer plate including various coating layers) forming the first side **2001** and the third side **2003**. The display **300** may be disposed along the transparent plate, and exposed through the first surface **2001** and the third surface **2003**. The transparent plate may have flexibility allowing a folded state of the electronic device **20**. According to an embodiment, the display **300** may be implemented to include the transparent plate, and the transparent plate may be omitted from the foldable housing **200**.

[0078] According to an embodiment, the first housing structure **210** may include a first rear cover **280** disposed on one side of a folding axis A to form at least part of the second surface **2002**. For example, the first rear cover **280** may have a substantially rectangular periphery **281**, and the periphery **281** may be surrounded by a first side member **211**. According to various embodiments, the first side member **211** and the first rear cover **280** may be integrally formed, and may include the same material.

[0079] According to an embodiment, the second housing structure **220** may include a second rear cover **290** disposed on the other side of the folding axis A to form at least part of the fourth surface **2004**. For example, the second rear cover **290** may have a substantially rectangular periphery **291**, and the periphery **291** may be surrounded by a second side member **221**. According to various embodiments, the second side member **221** and the second rear cover **290** may be integrally formed, and may include the same material.

[0080] According to various embodiments, the first rear cover **280** and/or the second rear cover **290** may be formed

with, for example, a coated or colored glass, a ceramic, a polymer, a metal (e.g., aluminum, stainless steel (STS), or magnesium), or a combination of at least two of those materials.

[0081] According to an embodiment, the first rear cover **280** and the second rear cover **290** may have a substantially symmetrical shape based on the folding axis A. The first rear cover **280** and the second rear cover **290** do not necessarily have the mutually symmetrical shape, and the first rear cover **280** and/or the second rear cover **290** of other various shapes may be provided in another embodiment.

[0082] According to an embodiment, the first housing structure **210** may include the first side member (or a first side bezel structure) **211** which forms the first side surface **200c**, and the second housing structure **220** may include the second side member (or a second side bezel structure) **221** which forms the second side surface **200d**. The first side member **211** and/or the second side member **221** may include a metal or a polymer.

[0083] According to various embodiments, the first side member **211** and the second side member **221** may extend to form a periphery region of the front surface **200a**. For example, the front surface **200a** of the electronic device **20** may be formed by the display **300**, a partial area of the first side member **211** adjacent to the display **300** and a partial area of the second side member **221**.

[0084] According to various embodiments, a partial area (not shown) adjacent to the periphery **281** of the first rear cover **280** in the first side member **211**, and/or a partial area (not shown) adjacent to the periphery **291** of the second rear cover **290** in the second side member **221** may form part of the rear surface **200b**. For example, the rear surface **200b** of the electronic device **20** may be formed by the first rear cover **280**, a partial area of the first side member **211** adjacent to the first rear cover **280**, the second rear cover **290** and a partial area of the second side member **221** adjacent to the second rear cover **290**.

[0085] According to an embodiment, the first side member **211** and the second side member **221** may be disposed on both sides based on the folding axis A, and may have a shape which is generally symmetric with respect to the folding axis A.

[0086] According to an embodiment, the first housing structure **210** may further include a component mounting area **214** extending from the first side member **211** or coupled with the first side member **211** to form the first surface **2001** with the display **300**. Other area than the component mounting area **214** in the first side member **221** may have a shape mutually symmetric with the second side member **221**. At least one component utilizing the first surface **2001** may be disposed in the component mounting area **214**. According to an embodiment, the component mounting area **214** may be formed to have a set area adjacent to one corner of the first side member **211**. According to various embodiments, the arrangement, the shape, and the size of the component mounting area **214** are not limited to the illustrated example. For example, in another embodiment, the component mounting area **214** may be included in other corner of the first side member **211** or in an arbitrary area between an upper corner and a lower corner. Components for performing various functions embedded in the electronic device **20** may be exposed to the first surface **2001** through the component mounting area **214**, or through one or more openings (not shown) provided in the component

mounting area **214**. According to an embodiment, a component **246** disposed in the component mounting area **214** may include at least one of various sensors such as a proximity sensor, a front camera, a light emitting device, and a receiver. For example, the light emitting device may provide state information of the electronic device **20** in the form of light. In another embodiment, the light emitting device may provide a light source interworking with, for example, an operation of the front camera. The light emitting device may include, for example, a light emitting diode (LED), an infrared (IR) LED, and a xenon lamp.

[0087] According to an embodiment, the electronic device **20** may include at least one or more of audio modules **241** and **242**, a key input device **243** or a connector hole **244**.

[0088] According to an embodiment, the audio modules **241** and **242** may include a microphone hole **241** or a speaker hole **242**. A microphone for acquiring external sound may be disposed inside the microphone hole **241**, and a plurality of microphones may be disposed to detect a sound direction in some embodiment. The speaker hole **242** may include an external speaker hole or a call receiver hole. In some embodiment, the speaker hole **242** and the microphone hole **241** may be implemented as a single hole, or a speaker may be included without the speaker hole **242** (e.g., a piezo speaker).

[0089] According to an embodiment, the key input device **243** may be disposed on the side surface **200c** and **200d** of the folder housing **200**. In another embodiment, the electronic device **20** may not include part or whole of the key input device **243** mentioned above, and the key input device **243** which is not included may be implemented in a different form such as a soft key on the display **300**. In some embodiment, the key input device may include a sensor module (e.g., one or more components **245** disposed on the first rear area **282**) disposed on the second surface **2002** of the first housing structure **210**.

[0090] According to an embodiment, the connector hole **244** may include a first connector hole for accommodating a connector (e.g., a USB connector) for transmitting and receiving power and/or data to and from an external electronic device, and/or a second connector hole (e.g., an earphone jack) for accommodating a connector for transmitting and receiving an audio signal to and from an external electronic device. The position or number of the connector holes is not limited to the example shown in FIG. 3 and may be formed differently.

[0091] In another embodiment (not shown), a back surface of a screen display area of the display **300** may include at least one or more of an audio module (e.g., a receiver for a call), a sensor module (e.g., a proximity sensor, or a fingerprint sensor), a camera module (e.g., a front camera) or a light emitting device. In another embodiment (not shown), the display **300** may be coupled to or disposed close to a touch sensing circuit, a pressure sensor for measuring a touch intensity (pressure), and/or a digitizer for detecting a stylus pen of a magnetic field type.

[0092] In an embodiment, the first housing structure **210** and the second housing structure **220** may form a recess which is a space for mounting the display **300**. In the embodiment shown, due to the component mounting area **214**, the recess may have two or more different widths in a direction perpendicular to the folding axis A.

[0093] For example, the recess may include a first width **w1** between a first unit **221a** parallel to the folding axis A of

the second side member **221** and a first unit **211a** formed at the periphery of the component mounting area **214** of the first side member **211**. The recess may include a second width **w2** between a second unit **221b** of the second side member **221** and a second unit **211b** not corresponding to the component mounting area **214** of the first side member **211** and parallel to the folding axis **A**. The second width **w2** may be formed longer than the first width **w1**. According to an embodiment, the first unit **211a** of the first housing structure **210** and the first unit **221a** of the second housing structure **220** having the asymmetric shape may form the first width **w1** of the recess, and the second unit **211b** of the first housing structure **210** and the second unit **221b** of the second housing structure **220** having the symmetrical shape may form the second width **w2**. According to an embodiment, the first unit **221a** and the second unit **221b** of the second housing structure **220** may have different distances from the folding axis **A**. The width of the recess is not limited to the illustrated example. According to various embodiments, the recess may have a plurality of widths due to the shape of the component mounting area **214** or the portion having the asymmetric shape of the first housing structure **210** and the second housing structure **220**.

[0094] According to various embodiments, one or more components may be disposed or visually exposed on the rear surface **200b** of the electronic device **20**. For example, at least part of a sub display **293** may be visually exposed through the second rear area **292** of the second rear cover **290**. For example, one or more components **245** may be visually exposed through the first rear area **282** of the first rear cover **280**. In various embodiments, the one or more components **245** may include a sensor (e.g., a proximity sensor, a heart rate sensor) and/or a rear camera.

[0095] Referring to FIG. 3, the hinge cover **230** may be disposed between the first housing structure **210** and the second housing structure **220**, to cover an internal component (e.g., the hinge structure). According to some embodiment, the hinge structure may be referred to as an element including the hinge cover **230**. In an embodiment, the hinge cover **230** may be covered by part of the first housing structure **210** and the second housing structure **220**, or exposed to the outside, according to the state (e.g., the unfolded state or the folded state) of the electronic device **20**.

[0096] For example, if the electronic device **20** is in the unfolded state as shown in FIG. 2, the hinge cover **230** may be covered by the first housing structure **210** and the second housing structure **220** and not exposed. For example, if the electronic device **20** is in the folded state (e.g., in a fully folded state) as shown in FIG. 3, the hinge cover **230** may be exposed to the outside between the first housing structure **210** and the second housing structure **220**. For example, if the electronic device **20** is in an intermediate state in which the first housing structure **210** and the second housing structure **220** are folded with a certain angle (e.g., a state between the unfolded state and the folded state), the hinge cover **230** may be partially exposed to the outside between the first housing structure **210** and the second housing structure **220**. The exposed area of the hinge cover **230** in the intermediate state may be smaller than the exposed area of the hinge cover **230** in the fully folded state. In an embodiment, the hinge cover **230** may include a curved surface, and the curved surface may form one side surface of the electronic device **20** in the folded state.

[0097] According to various embodiments, the display **300** may indicate a display in which at least some area may be transformed into a flat surface or a curved surface. In an embodiment, referring to FIG. 2, the display **300** may include a folding area **303**, a first area **301** disposed on one side (a right side of the folding area **303**) based on the folding area **303** and a second area **302** disposed on the other side (a left side of the folding area **303**).

[0098] According to various embodiments, the area division of the display **300** shown in FIG. 2 is exemplary, and the display **300** may be divided into a plurality of (e.g., 4 or more or 2) areas according to its structure or function. For example, the area of the display **300** may be divided by the folding area **303** extending in parallel to the **y** axis or the folding axis **A** in the embodiment shown in FIG. 2, but the display **300** may be divided based on other folding area (e.g., a folding area parallel to the **x** axis) or another folding axis (e.g., a folding axis parallel to the **x** axis) in another embodiment.

[0099] According to an embodiment, the first area **301** and the second area **302** of the display **300** may have a shape symmetrical on the whole based on the folding area **303**. According to an embodiment, unlike the first area **301**, the second area **302** may include a cut notch according to the presence of the component mounting area **214**, but other areas may have a symmetrical shape based on the first area **301** and the folding area **303**. For example, the first area **301** and the second area **302** may include a portion having the symmetrical shape based on the folding area **303**, and a portion having the asymmetrical shape.

[0100] According to an embodiment, the angle or distance formed by the first housing structure **210** and the second housing structure **220** may vary depending on the unfolded state, the folded state or the intermediate state of the foldable housing **200**. Hereinafter, operations of the first housing structure **210** and the second housing structure **220** and each area of the display **300** according to the state (e.g., the unfolded state and the folded state) of the electronic device **20** are described.

[0101] According to an embodiment, if the electronic device **20** is in the unfolded state (e.g., see FIG. 2), the first direction **201** of the first surface **2001** of the first housing structure **210** and the third direction **203** of the third surface **2003** of the second housing structure **220** may be the same. For example, in the unfolded state, the first surface **2001** of the first housing structure **210** and the third surface **2003** of the second housing structure **220** may be disposed to form the angle of about 180 degrees and to face the same direction (e.g., the direction of the front surface **200a** of the electronic device **20**). If the electronic device **20** is unfolded, a surface of the first area **301** and a surface of the second area **302** of the display **300** may form the angle of about 180 degrees and face the same direction (e.g., the direction of the front surface **200a** of the electronic device **20**). The folding area **303** of the display **300** may form the same plane as the first area **301** and the second area **302**.

[0102] In an embodiment, if the electronic device **20** is folded (e.g., see FIG. 3), the first housing structure **210** and the second housing structure **220** may be disposed to face each other. For example, in the folded state, the first surface **2001** of the first housing structure **210** and the third surface **2003** of the second housing structure **220** may face each other. In the folded state, the surface of the first area **301** and the surface of the second area **302** of the display **300** may form a narrow

angle (e.g., between about 0 degree and 10 degrees), and face each other. In the folded state, at least part of the folding area **303** may be formed as a curved surface having a certain curvature.

[0103] In an embodiment, if the electronic device **20** is in the intermediate state (e.g., the state between the unfolded state and the folded state), the first housing structure **210** and the second housing structure **220** may be disposed at a certain angle. In the intermediate state, the first surface **2001** of the first housing structure **210** and the third surface **2003** of the second housing structure **220**, or the surface of the first area **301** and the surface of the second area **302** of the display **300** may form an angle which is greater than the folded state and smaller than the unfolded state. In the intermediate state, at least part of the folding area **303** may be formed as a curved surface having a certain curvature, wherein the curvature may be smaller than that of the folded state.

[0104] According to an embodiment, the first side member **211** may include a plurality of first conductive units (not shown) physically or electrically separated. Nonconductive members **271** may be disposed between the plurality of the first conductive units. According to an embodiment, the nonconductive members **271** may extend from a first nonconductive internal structure (not shown) disposed inside the first housing structure **210**. The first internal structure may be coupled with the first side member **211**, and thus the plurality of the first conductive units may be physically separated by the first internal structure. For example, the first internal structure may be formed to be coupled with the first side member **211** by insert injection.

[0105] According to an embodiment, the second side member **221** may include a plurality of second conductive units (not shown) physically or electrically separated. Nonconductive members **272** may be disposed between the plurality of the second conductive units. According to an embodiment, the nonconductive members **272** may extend from a second nonconductive internal structure (not shown) disposed inside the second housing structure **220**. The second internal structure may be coupled with the second side member **221**, and thus the plurality of the second conductive units may be physically separated by the second internal structure. For example, the second internal structure may be formed to be coupled with the second side member **221** by the insert injection.

[0106] According to an embodiment, at least one of the plurality of the first conductive units of the first side member **211** may be electrically connected to a wireless communication circuit (e.g., the wireless communication module **192** of FIG. **1**) and utilized as an antenna radiator (or a radiator, or an antenna) for transmitting or receiving a signal having a selected (or designated) frequency. For example, at least one first conductive unit may be electrically connected to the wireless communication circuit and the ground to operate as an inverted F antenna (IFA). According to some embodiment, at least one of the plurality of the second conductive units of the second side member **221** may be electrically connected to the wireless communication circuit and utilized as an antenna radiator.

[0107] Referring to FIG. **3**, if the electronic device **20** is folded, the second side member **221** and the first side member **211** are adjacent to each other, and wireless communication using at least one of the first conductive units as the antenna radiator may be electrically affected by the

second side member **221**. For example, in the folded state, the second side member **221** may deteriorate wireless communication performance (e.g., antenna radiation performance) which utilizes at least one of the first conductive units as the antenna radiator. For example, electromagnetic wave energy (or electromagnetic field) emitted from at least one of the first conductive units utilized as the antenna radiator may be disrupted by the second side member **221**. According to an embodiment, to reduce this electrical influence, the nonconductive members **271** of the first side member **211** and the nonconductive members **272** of the second side member **221** may be designed to align substantially with each other in the folded state.

[0108] Referring to FIG. **3**, the nonconductive members **271** of the first side member **211** and the nonconductive members **272** of the second side member **221** may be substantially aligned with each other in the folded state and the antenna radiation performance may be improved, but it may be difficult to secure the antenna radiation performance over a set value due to electromagnetic coupling between the first conductive unit and the second conductive unit which are adjacent each other. For example, parasitic resonance may be generated in the second conductive unit by the electric field emitted from the first conductive unit utilized as the antenna radiator in the folded state, and this parasitic resonance (or parasitic component) may degrade the antenna radiation performance in the wireless communication utilizing at least one first conductive unit as the antenna radiator. For example, capacitance may be generated between the first conductive unit and the second conductive unit in the folded state by electromagnetic coupling, and the electromagnetic wave energy (or the electromagnetic field) emitted from the first conductive unit utilized as the antenna radiator may be induced to the second conductive unit. Unlike a forward current flowing in the first conductive unit, a reverse current may be generated in the second conductive unit by the induced electromagnetic wave energy, and thus the antenna radiation performance may be degraded in the wireless communication utilizing at least one of the first conductive units as the antenna radiator. According to an embodiment, the electronic device **20** may include a frequency adjustment circuit for reducing the deterioration of the antenna radiation performance for at least one first conductive unit by the second side member **221** in the folded state. The frequency adjustment circuit may include an electrical element such as inductance, capacitance or conductance in a transmission line for the antenna radiator (e.g., at least one first conductive unit).

[0109] According to various embodiments, the hinge structure (e.g., the hinge cover **230**) may include at least one third conductive unit. At least one third conductive unit of the hinge structure may be electrically connected to the wireless communication circuit (e.g., the wireless communication module **192** of FIG. **1**) and utilized as the antenna radiator.

[0110] FIG. **4** is an exploded perspective view of the electronic device **20** of FIG. **2** or

[0111] **3**.

[0112] Referring to FIG. **4**, in an embodiment, the electronic device **20** may include a display unit **30**, a support member assembly **40**, a substrate unit **450**, a first housing structure **210**, a second housing structure **220**, at least one of a first rear cover **280** or a second rear cover **290**. In this

document, the display unit **30** may be referred to as a display module or a display assembly.

[0113] The display unit **30** may include, for example, a display **300**, and one or more plates or layers **340** on which the display **300** is mounted. In an embodiment, the plate **340** may be disposed between the display **300** and the support member assembly **40**. The display **300** may be disposed on at least part of one surface (e.g., an upper surface based on FIG. 4) of the plate **340**. The plate **340** may be formed in a shape corresponding to the display **300**. For example, a partial area of the plate **340** may be formed in a shape corresponding to a notch **304** of the display **300**.

[0114] According to an embodiment, the support member assembly **40** may include a first support member **410**, a second support member **420**, a hinge structure **401** disposed between the first support member **410** and the second support member **420**, a hinge cover **230** for covering the hinge structure **401** when viewed from outside, and a wiring member **430** (e.g., flexible printed circuit (FPC)) crossing the first support member **410** and the second support member **420**.

[0115] In an embodiment, the support member assembly **40** may be disposed, between the plate **340** and the substrate unit **450**. For example, the first support member **410** may be disposed between a first area **301** of the display **300** and a first substrate (e.g., a first printed circuit board (PCB)) **451**. The second support member **420** may be disposed between a second area **302** of the display **300** and a second substrate (e.g., a second PCB) **452**.

[0116] According to an embodiment, at least part of the wiring member **430** and the hinge structure **401** may be disposed inside the support member assembly **40**. The wiring member **430** may be disposed in a direction (e.g., in the x-axis direction) crossing the first support member **410** and the second support member **420**. The wiring member **430** may be disposed in a direction (e.g., the x-axis direction) perpendicular to a folding axis (e.g., the y axis or the folding axis A of FIG. 2) of the folding area **303** of the display **300**.

[0117] According to an embodiment, the substrate unit **450** may include a first substrate **451** disposed on the side of the first support member **410** and a second substrate **452** disposed on the side of the second support member **420**. The first substrate **451** and the second substrate **452** may be disposed inside a space formed by the support member assembly **40**, the first housing structure **210**, the second housing structure **220**, the first rear cover **280** and the second rear cover **290**. Components for implementing various functions of the electronic device **20** may be mounted on the first substrate **451** and the second substrate **452**.

[0118] According to an embodiment, the first housing structure **210** and the second housing structure **220** may be assembled to couple with both sides of the support member assembly **40** while the display unit **30** is coupled to the support member assembly **40**. According to various embodiments, the first housing structure **210** and the second housing structure **220** may slide on both sides of the support member assembly **40** and be coupled to the support member assembly **40**.

[0119] In an embodiment, the first housing structure **210** may include a first rotation support surface **212**, and the second housing structure **220** may include a second rotation support surface **222** corresponding to the first rotation support surface **212**. The first rotation support surface **212** and

the second rotation support surface **222** may include a curved surface corresponding to the curved surface included in the hinge cover **230**.

[0120] In an embodiment, if the electronic device **20** is in the unfolded state (e.g., see FIG. 2), the first rotation support surface **212** and the second rotation support surface **222** may cover the hinge cover **230**, and the hinge cover **230** may not be exposed to the rear surface of the electronic device **20** or may be exposed least. If the electronic device **20** is in the folded state (e.g., see FIG. 3), the hinge cover **230** may be exposed to a maximum between the first rotation support surface **212** and the second rotation support surface **222**.

[0121] FIG. 5 shows an electronic device of an unfolded state according to an embodiment.

[0122] Referring to FIG. 5, in an embodiment, an electronic device **500** (e.g., the electronic device **102** of FIG. 1 or the electronic device **20** of FIG. 2) may include at least one of a first side member **510** (e.g., the first side member **211** of FIG. 2), a second side member **520** (e.g., the second side member **221** of FIG. 2), a hinge structure **530**, a wireless communication circuit **580** (e.g., the wireless communication module **192** of FIG. 1) or a processor **590** (e.g., the processor **120** of FIG. 1).

[0123] According to an embodiment, the electronic device **500** may include first side surfaces **511** and **512** facing a first direction, second side surfaces **521** and **522** facing a second direction which is opposite to the first direction, and third side surfaces **531** and **532** facing a third direction which is perpendicular to the first direction.

[0124] The first side surfaces **511** and **512** may include, for example, the first side surface **511** formed by the first side member **510** and the first side surface **512** formed by the second side member **520**. If the electronic device **500** is in the unfolded state or the folded state (e.g., see FIG. 3), the first side surfaces **511** and **512** together may form one side surface of the electronic device **500**. The second side surfaces **521** and **522** may include, for example, the second side surface **521** formed by the first side member **510** and the second side surface **522** formed by the second side member **520**. If the electronic device **500** is in the unfolded or the folded state (e.g., see FIG. 3), the second side surfaces **521** and **522** together may form the other side surface of the electronic device **500**.

[0125] The third side surfaces **531** and **532** may include, for example, the third side surface **531** formed by the first side member **510** and the third side surface **532** formed by the second side member **520**. As shown, if the electronic device **500** is in the unfolded state, the third side surfaces **531** and **532** may be disposed to face opposite directions. If the electronic device **500** is in the folded state (e.g., see FIG. 3), the third side surfaces **531** and **532** together may form one side surface of the electronic device **500**.

[0126] According to various embodiments, the first side member **510** may include a plurality of conductive units. For example, the first side member **510** may include at least one of a first conductive unit A1, a first insulating unit **541**, a second conductive unit A2, a second insulating unit **542** and a third conductive unit A3 which are sequentially disposed from one side of the hinge structure **530**. The first side member **510** may include at least one of a seventh conductive unit A7, a fifth insulating unit **545**, an eighth conductive unit A8, and a sixth insulating unit **546** which are sequentially disposed from the other side of the hinge structure **530**. According to an embodiment, the seventh and eighth con-

ductive units A7 and A8 may form part of the first side surface 511. The first and second conductive units A1 and A2 may form part of the second side surface 521. The third conductive unit A3 may include a portion A31 forming the third side surface 531, a portion A32 forming part of the first side surface 511, and a portion A33 forming part of the second side surface 521. In an embodiment, the first insulating unit 541 may be disposed between the first and second conductive units A1 and A2 and form part of the second side surface 521. The first and second conductive units A1 and A2 may be electrically or physically separated by the first insulating unit 541. The second insulating unit 542 may be disposed between the second and third conductive units A2 and A3 and form part of the second side surface 521. The second and third conductive units A2 and A3 may be electrically or physically separated by the second insulating unit 542. The fifth insulating unit 545 may be disposed between the seventh and eighth conductive units A7 and

[0127] A8 and form part of the first side surface 511. The seventh and eighth conductive units A7 and A8 may be electrically or physically separated by the fifth insulating unit 545. The sixth insulating unit 546 may be disposed between the third and eighth conductive units A3 and A8 and form part of the first side surface 511. The third and eighth conductive units A3 and A8 may be electrically or physically separated by the sixth insulating unit 546. According to various embodiments, the first insulating unit 541, the second insulating unit 542, the fifth insulating unit 545 or the sixth insulating unit 546 may be the nonconductive member 271 of FIG. 3. In the folded state of the electronic device 500 (see FIG. 3), the first insulating unit 541 and the third insulating unit 543 may substantially align with each other, and the second insulating unit 542 and the fourth insulating unit 544 may substantially align with each other. In the folded state of the electronic device 500, the fifth insulating unit 545 and the seventh insulating unit 547 may substantially align with each other, and the sixth insulating unit 546 and the eighth insulating unit 548 may substantially align with each other.

[0128] According to various embodiments, the second side member 520 may include a plurality of conductive units. For example, the second side member 520 may include at least one of a fourth conductive unit A4, a third insulating unit 543, a fifth conductive unit A5, a fourth insulating unit 544 and a sixth conductive unit A6 which are disposed in sequence from one side of the hinge structure 530. The second side member 520 may include at least one of a ninth conductive unit A9, a seventh insulating unit 547, a tenth conductive unit A10 and an eighth insulating unit 548 which are disposed in sequence from the other side of the hinge structure 530. According to an embodiment, the ninth and tenth conductive units A9 and A10 may form part of the first side surface 512. The fourth and fifth conductive units A4 and A5 may form part of the second side surface 522. The sixth conductive unit A6 may include a portion A61 forming the third side surface 532, a portion A62 forming part of the first side surface 512, and a portion A63 forming part of the second side surface 522. According to an embodiment, the third insulating unit 543 may be disposed between the fourth and fifth conductive units A4 and A5 and form part of the second side surface 522. The fourth and fifth conductive units A4 and A5 may be electrically or physically separated by the third insulating unit 543. The fourth insulating unit 544 may be disposed between the fifth and sixth

conductive units A5 and A6 and form part of the second side surface 522. The fifth and sixth conductive units A5 and A6 may be electrically or physically separated by the fourth insulating unit 544. The seventh insulating unit 547 may be disposed between the ninth and tenth conductive units A9 and A10 and form part of the first side surface 512. The ninth and tenth conductive units A9 and A10 may be electrically or physically separated by the seventh insulating unit 547. The eighth insulating unit 548 may be disposed between the sixth and tenth conductive units A6 and A10 and form part of the first side surface 512. The sixth and tenth conductive units A6 and A10 may be electrically or physically separated by the eighth insulating unit 548. According to various embodiments, the third insulating unit 543, the fourth insulating unit 544, the seventh insulating unit 547 or the eighth insulating unit 548 may be the nonconductive member 271 of FIG. 3.

[0129] According to various embodiments, at least one of the first, second, third, seventh and eighth conductive units A1, A2, A3, A7, and A8 of the first side member 510 may be electrically connected to the wireless communication circuit 580 (e.g., a radio frequency integrated circuit (RFIC) or an intermediate frequency integrated circuit (IFIC)) and utilized as the antenna radiator.

[0130] In an embodiment, the second conductive unit A2 forming ① part of the second side surface 521 may operate as a first antenna radiator ①. The first antenna radiator ① may be electrically connected to the wireless communication circuit 580 at at least one feeding point (not shown), and may be electrically connected to a ground G1 at at least one grounding point (not shown). According to an embodiment, a first antenna system 550 may include the first antenna radiator ①, the ground G1 electrically connected with the first antenna radiator ① or a first transmission line 551 between the first antenna radiator ① and the wireless communication circuit 580. The first transmission line 551 is a structure for transmitting a signal (voltage, current) of a radio frequency (RF) through the first antenna radiator ①, and may be defined as a conductor system using wave transmission action by an electrical element (e.g., an element having resistance, inductance, conductance, capacitance per unit length). The first transmission line 551 may include wires of various types which interconnect the wireless communication circuit 580 and the first antenna radiator ①. The wireless communication circuit 580 supplies the current to the first antenna radiator ① through the first transmission line 551, and the current may be transmitted along the first antenna radiator ① and flow into the ground G1. Hence, the wireless communication circuit 580 may transmit or receive a wireless electromagnetic wave through the first antenna radiator ①.

[0131] In an embodiment, some portion A34 of the third conductive unit A3 disposed between the second side surface 521 and the third side surface 531 to form part of the third side surface 531 and part of the second side surface 521 may operate as a second antenna radiator ②. The second antenna radiator ② may be electrically connected to the wireless communication circuit 580 at at least one feeding point (not shown), and may be electrically connected to a ground G2 at at least one grounding point (not shown). According to an embodiment, the second antenna system 560 may include the second antenna radiator ②, the ground G2 electrically connected to the second antenna radiator ②, or a second transmission line 561 between the second

antenna radiator ② and the wireless communication circuit 580. The second transmission line 561 is a structure for transmitting an RF signal (voltage, current) through the second antenna radiator ②, and may be defined as a conductor system using the wave transmission action by the electrical element (e.g., an element having resistance, inductance, conductance, capacitance per unit length). The second transmission line 561 may include wires of various types which interconnect the wireless communication circuit 580 and the second antenna radiator ②. The wireless communication circuit 580 supplies current to the second antenna radiator ② through the second transmission line 561, and the current may be transmitted along the second antenna radiator ② and fed into the ground G2. Hence, the wireless communication circuit 580 may transmit or receive a wireless electromagnetic wave through the second antenna radiator ②. The second transmission line 561 may include wires of various types for interconnecting the wireless communication circuit 580 and the second antenna radiator 2.

[0132] In an embodiment, the eighth conductive unit A8 forming part of the first side surface 511 may operate as a third antenna radiator ③. The third antenna radiator ③ may be electrically connected to the wireless communication circuit 580 at at least one feeding point (not shown), and may be electrically connected to a ground G3 at at least one grounding point (not shown). According to an embodiment, the third antenna system 570 may include the third antenna radiator ③, the ground G3 electrically connected to the third antenna radiator ③, or a third transmission line 571 between the third antenna radiator and the wireless communication circuit 580. The third transmission line 571 is a structure for transmitting an RF signal (voltage, current) through the third antenna radiator ③, and may be defined as a conductor system using the wave transmission action by the electrical element (e.g., an element having resistance, inductance, conductance, capacitance per unit length). The third transmission line 571 may include wires of various types for interconnecting the wireless communication circuit 580 and the third antenna radiator ③. The wireless communication circuit 580 supplies current to the third antenna radiator ③ through the third transmission line 571, and the current may be transmitted along the third antenna radiator ③ and fed into the ground G3. Thus, the wireless communication circuit 580 may transmit or receive a wireless electromagnetic wave through the third antenna radiator ③.

[0133] In an embodiment, some portion A35 of the third conductive unit A3 disposed between the first side surface 511 and the third side surface 531 to form part of the third side surface 531 and part of the first side surface 511 may operate as a fourth antenna radiator ④. The fourth antenna radiator ④ may be electrically connected to the wireless communication circuit 580 at at least one feeding point (not shown), and may be electrically connected to a ground G4 at at least one grounding point (not shown). According to an embodiment, the fourth antenna system 580 may include the fourth antenna radiator ④, the ground G4 electrically connected to the fourth antenna radiator ④, or a fourth transmission line 581 between the fourth antenna radiator ④ and the wireless communication circuit 580. The fourth transmission line 581 is a structure for transmitting an RF signal (voltage, current) through the fourth antenna radiator ④, and may be defined as a conductor system using the wave transmission action by the electrical element (e.g., an element having resistance, inductance, conductance, capaci-

tance per unit length). The fourth transmission line 581 may include wires of various types for interconnecting the wireless communication circuit 580 and the fourth antenna radiator ④. The wireless communication circuit 580 supplies the current to the fourth antenna radiator ④ through the fourth transmission line 581, and the current may be transmitted along the fourth antenna radiator ④ and fed into the ground G4. Hence, the wireless communication circuit 580 may transmit or receive the wireless electromagnetic wave through the fourth antenna radiator ④.

[0134] According to an embodiment, at least part of the ground G1, G2, G3, or G4 may be a first ground or a first ground layer disposed on a first PCB 501 (e.g., the first substrate 451 of FIG. 4) on which the wireless communication circuit 580 or the processor 590 is disposed.

[0135] According to an embodiment, reflection characteristics and impedance of the antenna radiator (e.g., the first antenna radiator ①, the second antenna radiator ②, the third antenna radiator ③, or the fourth antenna radiator ④) are related to the antenna radiation performance, and may vary depending on its shape, size, and material. The radiation characteristics of the antenna radiator may include an antenna radiation pattern (or an antenna pattern) which is a directional function representing a relative distribution of the power radiated from the antenna radiator, and a polarization state (or antenna polarization) of the wave radiated from the antenna radiator. The impedance of the antenna radiator may be related to the power transfer from the transmitter (e.g., the wireless communication circuit 580) to the antenna radiator or the power transfer from the antenna radiator to the receiver (e.g., the wireless communication circuit 580). The impedance of the antenna radiator may be designed to match impedance of the transmission line to minimize reflection at a connection between the transmission line (e.g., the first transmission line 551, the second transmission line 561, the third transmission line 571 or the fourth transmission line 581) and the antenna radiator (e.g., the first antenna radiator ①, the second antenna radiator ②, the third antenna radiator ③ or the fourth antenna radiator ④). Thus, maximum power transmission (or power loss minimization) or efficient signal transmission through the antenna radiator may be possible. Such impedance matching may lead to efficient signal flow at a specific frequency.

[0136] According to an embodiment, if the electronic device 500 is folded (e.g., see FIG. 3), the antenna radiator (e.g., the first antenna radiator ①, the second antenna radiator ②, the third antenna radiator ③ or the fourth antenna radiator ④) formed as at least part of the first side member 510 may be close to at least part of the second side member 520. For example, in the folded state of the electronic device 500, when viewed from above the first housing structure 510, the first antenna radiator ① may at least partially overlap the fifth conductive unit A5, and the second antenna radiator ② may at least partially overlap the sixth conductive unit A6. For example, in the folded state of the electronic device 500, when viewed from above the first housing structure 510, the third antenna radiator ③ may at least partially overlap the tenth conductive unit A10, and the fourth antenna radiator ④ may at least partially overlap the sixth conductive unit A6. In the folded state of the electronic device 500, the second side member 520 may affect the antenna radiation performance of the antenna radiator (e.g., the first antenna radiator ①, the second antenna radiator ②, the third antenna radiator ③ or the fourth antenna radiator ④).

④) formed as at least part of the first side member 510. For example, the antenna radiation performance of the first antenna radiator ① may be degraded by the fifth conductive unit A5 which is close to the first antenna radiator ① if the electronic device 500 is folded. In the folded state of the electronic device 500, parasitic resonance may be generated at the fifth conductive unit A5 by an electric field emitted from the first antenna radiator ①, and this parasitic resonance (or parasitic component) may deteriorate the antenna radiation performance of the first antenna radiator ①. Capacitance may be generated between the first antenna radiator ① and the fifth conductive unit A5 by the electromagnetic coupling in the folded state of the electronic device 500, and the electromagnetic wave energy (or electromagnetic field) emitted from the first antenna radiator ① may be induced into the fifth conductive unit A5. Unlike a forward current flowing in the first antenna radiator ①, a reverse current may be generated in the fifth conductive unit A5 by the induced electromagnetic wave energy, and thus the antenna radiation performance related to the first antenna radiator ① may be deteriorated. The antenna radiation performance of the second antenna radiator ② or the fourth antenna radiator ④ may be degraded by the sixth conductive unit A6 close to the second antenna radiator ② or fourth antenna radiator ④ in the folded state of the electronic device 500, similarly to the degradation of the antenna radiation performance of the first antenna radiator ①. The antenna radiation performance of the third antenna radiator ③ may be deteriorated by the tenth conductive unit A10 which is close to the third antenna radiator ③ in the folded state of the electronic device 500, similarly to the deterioration of the antenna radiation performance of the first antenna radiator ①. The deterioration of the antenna radiation performance may indicate a state that it is difficult to transmit or receive a signal having a selected or designated frequency (or a signal of a frequency designated in the corresponding communication mode) through the antenna radiator (e.g., the first antenna radiator ①, the second antenna radiator ②, the third antenna radiator ③ or the fourth antenna radiator ④). According to various embodiments of the present disclosure, the electronic device 500 may include at least one frequency adjustment circuit for reducing the influence of the second side member 520 on the first antenna radiator ①, the second antenna radiator ②, the third antenna radiator ③ or the fourth antenna radiator ④ in the folded state.

[0137] According to an embodiment, the frequency adjustment circuit may include an electrical element having a component such as inductance, capacitance or conductance acting on the antenna system (e.g., the first antenna system 550, the second antenna system 560, the third antenna system 570 or the fourth antenna system 580). For example, the frequency adjustment circuit may apply the electrical element to the antenna system, and this phenomenon may be referred to as loading of the electrical element (or a loading effect of the electrical element). For example, the frequency adjustment circuit may generate inductance loading (or inductance loading effect) which acts the inductance (or inductance component) on the antenna system. According to an embodiment, the frequency adjustment circuit may include various elements such as a lumped element or a passive element.

[0138] According to an embodiment, the electronic device 500 may include a first frequency adjustment circuit 591

which is electrically connected to the second side member 520 and is related to the first antenna radiator ① and/or the second antenna radiator ②. The first frequency adjustment circuit 591 may be electrically connected to the fifth conductive unit A5 disposed close to the first antenna radiator ① in the folded state of the electronic device 500, and a ground G5. The first frequency adjustment circuit 591 may be electrically connected to the sixth conductive unit A6 disposed close to the second antenna radiator ② in the folded state of the electronic device 500, and the ground G5. The ground G5 may be, for example, a second ground or a second ground layer included in a second PCB 502 (e.g., the second substrate 452 of FIG. 4) on which the first frequency adjustment circuit 591 is disposed. According to an embodiment, in the folded state of the electronic device 500, the first frequency adjustment circuit 591 may include an electrical element (e.g., an inductor) having the component such as inductance, capacitance or conductance acting on the first antenna system 550 and/or the second antenna system 560, and thus reduce the influence of the second side member 520 on the antenna radiation performance of the first antenna radiator ① and/or the second antenna radiator ②.

[0139] For example, in the folded state of the electronic device 500 (see FIG. 3), the second side member 520 is close to the first antenna radiator ① and/or the second antenna radiator ② to generate the capacitance (or capacitance component) (e.g., parasitic capacitance), and a parasitic resonance frequency may be generated due to the capacitance. According to an embodiment, the first frequency adjustment circuit 591 may adjust the parasitic resonance frequency not to include the parasitic resonance frequency generated in the folded state of the electronic device 500 in a resonance frequency band of the first antenna radiator ① and/or the second antenna radiator ②. The parasitic resonance frequency generated in the folded state of the electronic device 500 may be moved out of the resonance frequency band of the first antenna radiator ① and/or the second antenna radiator ② by the first frequency adjustment circuit 591, thus reducing the degradation of the antenna radiation performance due to the parasitic resonance frequency.

[0140] According to an embodiment, part of the second side member 520 electrically connected to the first frequency adjustment circuit 591 may operate as an additional antenna radiator for transmitting or receiving a signal having the selected or designated frequency, together with the first antenna radiator ① and/or the second antenna radiator ② in the folded state of the electronic device 500 (see FIG. 3). In an embodiment, part of the side member 520 operating as the additional antenna radiator may include the fifth conductive unit A5 at least partially overlapping the first antenna radiator ①, and/or a part A64 of the sixth conductive unit A6 at least partially overlapping with the second antenna radiator ②, when viewed from above the first housing structure 510 in the folded state of the electronic device 500. For example, the fifth conductive unit A5 may operate as the antenna radiator electromagnetically coupled with the first antenna radiator ① and/or the second antenna radiator ② in the folded state of the electronic device 500. For example, the part A64 of the sixth conductive unit A6 may operate as the antenna radiator electromagnetically coupled with the second antenna radiator ② and/or the first antenna radiator ① in the folded state of the electronic device 500. According to an embodiment, the first frequency

adjustment circuit 591 may have inductance (e.g., inductance component) for reducing capacitance (e.g., parasitic capacitance) which may occur if the fifth conductive unit A5 is close to the first antenna radiator ①. According to an embodiment, the first frequency adjustment circuit 591 may have inductance (or inductance component) for reducing capacitance (e.g., parasitic capacitance) which may occur if the part A64 of the sixth conductive unit A6 is close to the second antenna radiator ②. For example, the first frequency adjustment circuit 591 electrically connected to the ground G5 may be electrically connected to part (e.g., the fifth conductive unit A5 and/or the part A64 of the sixth conductive unit A6) of the second side member 520 and operate as an inductor. According to an embodiment, the first frequency adjustment circuit 591 may adjust the impedance of part (e.g., the fifth conductive unit A5 and/or the part A64 of the sixth conductive unit A6) of the second side member 520 to match the impedance of the transmission line with respect to the first antenna radiator ① and/or the second antenna radiator ②. For example, the inductance of the first frequency adjustment circuit 591 may adjust the impedance of the part (e.g., the fifth conductive unit A5 and/or the part A64 of the sixth conductive unit A6) of the second side member 520.

[0141] According to an embodiment, the first frequency adjustment circuit 591 in the folded state of the electronic device 500 (see FIG. 3) may reduce the degradation of the antenna radiating performance due to the parasitic resonance frequency by moving the parasitic resonance frequency generated by the second side member 520 which is close to first the antenna radiator ① and/or the second antenna radiator 2. In addition, the first frequency adjustment circuit 591 may operate part (e.g., the fifth conductive unit A5 and/or the part A64 of the sixth conductive unit A6) of the second side member 520 as an additional antenna radiator coupled with the first antenna radiator ① and/or the second antenna radiator ② by adjusting the impedance.

[0142] According to an embodiment, the first frequency adjustment circuit 591 may include a lumped element (or a passive element) such as an inductor, and/or a switch. For example, if the switch is turned on by the control of the processor 590, the fifth conductive unit A5 and/or the sixth conductive unit A6 may be electrically connected to the ground G5 through the lumped element. According to an embodiment, if the lumped element electrically interconnects the fifth conductive unit A5 and the ground G5 in the folded state of the electronic device 500, the lumped element may have a function of reducing the degradation of the antenna radiation performance due to the parasitic resonance frequency by moving the parasitic resonance frequency generated due to the fifth conductive unit A5 which is close to the first antenna radiator ①, and a function of operating the fifth conductive unit A5 as the additional antenna radiator coupled with the first antenna radiator ① and/or the second antenna radiator ② by adjusting the impedance.

[0143] According to an embodiment, if the lumped element electrically interconnects the sixth conductive unit A6 and the ground G5 in the folded state of the electronic device 500, the lumped element may have a function of reducing the degradation of the antenna radiation performance due to the parasitic resonance frequency by moving the parasitic resonance frequency generated due to the sixth conductive unit A6 which is close to the second antenna radiator ②, and a function of operating the part A64 of the sixth conductive

unit A6 as the additional antenna radiator coupled with the first antenna radiator ① and/or the second antenna radiator ② by adjusting the impedance.

[0144] According to various embodiments, the first frequency adjustment circuit 591 may move the resonance frequency of the first antenna radiator ① and/or the second antenna radiator ② to a designated frequency, or move it as designated in the folded state of the electronic device 500 (see FIG. 3).

[0145] For example, in the folded state of the electronic device 500 (see FIG. 3), if the switch included in the first frequency adjustment circuit 591 is turned on under the control of the processor 590, the fifth conductive unit A5 and/or the sixth conductive unit A6 may be electrically connected to the ground G5 through the lumped element. The lumped element may reduce the resonance frequency of the parasitic component which may generate in the frequency band of the first antenna radiator ① and/or the second antenna radiator ② if the second side member 520 is close to the first antenna radiator ① and/or the second antenna radiator ②. For example, in the folded state of the electronic device 500, if the switch included in the first frequency adjustment circuit 591 is turned off by the control of the processor 590, the fifth conductive unit A5 and/or the sixth conductive unit A6 may be electrically separated from the ground G5. If the fifth conductive unit A5 and/or the sixth conductive unit A6 is electrically separated from the ground G5, the impedance (or, the condition of the parasitic resonance) may be changed to thus reduce the antenna radiation performance degradation of the first antenna system 550 and/or the second antenna system 560 due to the parasitic resonance. Hence, according to on or off of the switch included in the first frequency adjustment circuit 591, the resonance frequency of the first antenna radiator ① and/or the resonance frequency of the second antenna radiator ② may be moved in the folded state of the electronic device 500.

[0146] According to an embodiment, the electronic device 500 may include a second frequency adjustment circuit 592 electrically connected to the second side member 520 and related to the third antenna radiator ③ and/or the fourth antenna radiator ④. The second frequency adjustment circuit 592 may be electrically connected to the tenth conductive unit A10 disposed close to the third antenna radiator ③ in the folded state of the electronic device 500, and a ground G6. The ground G6 may be, for example, a second ground or a second ground layer included in the second PCB 502 (e.g., the second substrate 452 of FIG. 4) on which the second frequency adjustment circuit 592 is disposed. According to an embodiment, the second frequency adjustment circuit 592 may be electrically connected with the sixth conductive unit A6 disposed close to the fourth antenna radiator ④ in the folded state of the electronic device 500, and the ground G6. According to an embodiment, the second frequency adjustment circuit 592 may include an electrical element (e.g., an inductor) having the component such as inductance, capacitance or conductance acting on the third antenna system 570 and/or the fourth antenna system 580, and thus reduce influence of the second side member 520 on the antenna radiation performance of the third antenna radiator ③ and/or the fourth antenna radiator 4.

[0147] For example, in the folded state of the electronic device 500 (see FIG. 3), the second side member 520 is close

to the third antenna radiator ③ and/or the fourth antenna radiator ④ and the capacitance (or capacitance component) (e.g., parasitic capacitance) may occur, and a parasitic resonance frequency may be generated due to the capacitance. According to an embodiment, the second frequency adjustment circuit 592 may adjust the parasitic resonance frequency not to include the parasitic resonance frequency generated in the folded state of the electronic device 500 in a parasitic resonance frequency of the third antenna radiator ③ and/or the fourth antenna radiator ④. The parasitic resonance frequency generated in the folded state of the electronic device 500 may be moved out of the resonance frequency band of the third antenna radiator ③ and/or the fourth antenna radiator ④ by the second frequency adjustment circuit 592, thus reducing the degradation of the antenna radiation performance due to the parasitic resonance frequency.

[0148] According to an embodiment, part of the second side member 520 electrically connected to the second frequency adjustment circuit 592 may operate as an additional antenna radiator for transmitting or receiving a signal having a selected or designated frequency, together with the third antenna radiator ③ and/or the fourth antenna radiator ④ in the folded state of the electronic device 500 (see FIG. 3). In an embodiment, the part of the side member 520 operating as the additional antenna radiator may include the tenth conductive unit A10 at least partially overlapping the third antenna radiator ③, and/or part A65 of the sixth conductive unit A6 at least partially overlapping the fourth antenna radiator ④. For example, the tenth conductive unit A10 may operate as the antenna radiator electromagnetically coupled with the third antenna radiator ③ and/or the fourth antenna radiator ④ in the folded state of the electronic device 500. For example, the sixth conductive unit A6 may operate as the antenna radiator electromagnetically coupled with the fourth antenna radiator ④ and/or the third antenna radiator ③ in the folded state of the electronic device 500. According to an embodiment, the second frequency adjustment circuit 592 may have inductance (e.g., inductance component) for reducing capacitance (e.g., parasitic capacitance) which may occur if the tenth conductive unit A10 is close to the third antenna radiator ③. According to an embodiment, the second frequency adjustment circuit 592 may have inductance (e.g., inductance component) for reducing capacitance (e.g., parasitic capacitance) which may occur if the part A65 of the sixth conductive unit A6 is close to the fourth antenna radiator ④. For example, the second frequency adjustment circuit 592 electrically connected to the ground G6 may be electrically connected to part (e.g., the tenth conductive unit A10 and/or the part A65 of the sixth conductive unit A6) of the second side member 520 and operate as an inductor. According to an embodiment, the second frequency adjustment circuit 592 may adjust impedance of part (e.g., the tenth conductive unit A10 and/or the part A65 of the sixth conductive unit A6) of the second side member 520 to match impedance of a transmission line related to the third antenna radiator ③ and/or the fourth antenna radiator ④. For example, the inductance of the second frequency adjustment circuit 592 may adjust the impedance of the part (e.g., the tenth conductive unit A10 and/or the part A65 of the sixth conductive unit A6) of the second side member 520.

[0149] According to an embodiment, the second frequency adjustment circuit 592 in the folded state of the

electronic device 500 (see FIG. 3) may function to reduce the degradation of the antenna radiating performance due to the parasitic resonance frequency by moving the parasitic resonance frequency generated by the second side member 520 which is close to the third antenna radiator ③ and/or the fourth antenna radiator ④. In addition, the second frequency adjustment circuit 592 may function to operate the part (e.g., the tenth conductive unit A10 and/or the part A65 of the sixth conductive unit A6) of the second side member 520 as the additional antenna radiator coupled with the third antenna radiator ③ and/or the fourth antenna radiator ④ by adjusting the impedance.

[0150] According to an embodiment, the second frequency adjustment circuit 592 may include a lumped element (or a passive element) such as an inductor, and/or a switch. For example, if the switch is turned on by control of the processor 590, the tenth conductive unit A10 and/or the sixth conductive unit A6 may be electrically connected to the ground G6 through the lumped element. According to an embodiment, if the lumped element electrically interconnects the tenth conductive unit A10 and the ground G6 in the folded state of the electronic device 500, the lumped element may have a function of reducing the degradation of the antenna radiation performance due to the parasitic resonance frequency by moving the parasitic resonance frequency generated by the tenth conductive unit A10 which is close to the third antenna radiator ③, and a function of operating the tenth conductive unit A10 as the additional antenna radiator coupled with the third antenna radiator ③ and/or the fourth antenna radiator ④ by adjusting the impedance. According to an embodiment, if the lumped element electrically interconnects the sixth conductive unit A6 and the ground G6 in the folded state of the electronic device 500, the lumped element may have a function of reducing the degradation of the antenna radiation performance due to the parasitic resonance frequency by moving the parasitic resonance frequency generated by the sixth conductive unit A6 which is close to the fourth antenna radiator ④, and a function of operating the part A65 of the sixth conductive unit A6 as the additional antenna radiator coupled to the fourth antenna radiator ④ and/or the third antenna radiator ③ by adjusting the impedance.

[0151] According to various embodiments, the second frequency adjustment circuit 592 may move the resonance frequency of the third antenna radiator ③ and/or the fourth antenna radiator ④ to a designated frequency, or move it as designated in the folded state of the electronic device 500. For example, if the switch included in the second frequency adjustment circuit 592 is turned on by the control of the processor 590, the tenth conductive unit A10 and/or the sixth conductive unit A6 may be electrically connected to the ground G6 through the lumped element. The lumped element may reduce the resonance frequency of the parasitic component which may generate in the frequency band for the third antenna radiator ③ and/or the fourth antenna radiator ④ if the second side member 520 is close to the third antenna radiator ③ and/or the fourth antenna radiator ④. For example, in the folded state of the electronic device 500, if the switch included in the second frequency adjustment circuit 592 is turned off by the control of the processor 590, the tenth conductive unit A10 and/or the sixth conductive unit A6 may be electrically separated from the ground G6. If the tenth conductive unit A10 and/or the sixth conductive unit A6 is electrically separated from the ground G6, the

impedance (or the parasitic resonance condition) may be changed to thus reduce the antenna radiation performance degradation of the third antenna system 570 and/or the fourth antenna system 580 due to the parasitic resonance. Hence, according to on or off of the switch included in the second frequency adjustment circuit 592, the resonance frequency of the third antenna radiator ③ and/or the resonance frequency of the fourth antenna radiator ④ may be moved in the folded state of the electronic device 500.

[0152] According to various embodiments, the electronic device 500 may further include at least one of a third frequency adjustment circuit 593 electrically connected to the first antenna radiator ① and a ground G7, a fourth frequency adjusting circuit 594 electrically connected to the third antenna radiator ③ and a ground G8, a fifth frequency adjusting circuit 595 electrically connected to the fifth conductive unit A5 and a ground G9, or a sixth frequency adjustment circuit 596 electrically connected to the tenth conductive unit A10 and a ground G10. The ground G7 or G8 may be, for example, the first ground or the first ground layer included in the first PCB 501 (e.g., the first substrate 451 of FIG. 4) on which the third frequency adjustment circuit 593, the fourth frequency adjustment circuit 594, the wireless communication circuit 580 or the processor 590 is disposed. The ground G9 or G10 may be, for example, the second ground or the second ground layer included in the second PCB 502 (e.g., the second substrate 452 of FIG. 4) on which the first frequency adjustment circuit 591, the second frequency adjustment circuit 592, the fifth frequency adjustment circuit 595 or the sixth frequency adjustment circuit 596 is disposed.

[0153] According to an embodiment, the third frequency adjustment circuit 593 or the fifth frequency adjustment circuit 595 may reduce influence of at least part (e.g., the fifth conductive unit A5) of the second side member 520 on the antenna radiation performance of the first antenna radiator ① and/or the second antenna radiator ②, or move the resonance frequency of the first antenna radiator ① and/or the second antenna radiator ②, in the folded state of the electronic device 500. According to an embodiment, the fourth frequency adjustment circuit 594 or the sixth frequency adjustment circuit 596 may reduce influence of at least part (e.g., the tenth conductive unit A10) of the second side member 520 on the antenna radiation performance of the third antenna radiator ③ and/or the fourth antenna radiator ④, or move the resonance frequency of the third antenna radiator ③ and/or the fourth antenna radiator ④, in the folded state of the electronic device 500. According to an embodiment, the third or fifth frequency adjustment circuit 593 or 595 may include an electrical element having a component such as inductance, capacitance or conductance acting on the first antenna system 550 and/or the second antenna system 560. According to an embodiment, the fourth or sixth frequency adjustment circuit 594 or 596 may include an electrical element having a component such as inductance, capacitance or conductance acting on the third antenna system 570 and/or the fourth antenna system 580.

[0154] According to various embodiments, the electronic device 500 may further include a seventh frequency adjustment circuit (not shown) electrically connected to the second antenna (2) and the ground G2. The seventh frequency adjustment circuit may, in the folded state of the electronic device 500, reduce influence of at least part (e.g., the sixth

conductive unit A6) of the second side member 520 on the antenna radiation performance of the first antenna radiator ① and/or the second antenna radiator ②, or move the resonance frequency of the first antenna radiator ① and/or the second antenna radiator ②.

[0155] According to various embodiments, the electronic device 500 may further include an eighth frequency adjustment circuit (not shown) electrically connected to the fourth antenna radiator ④ and the ground G4. The eighth frequency adjustment circuit may, in the folded state of the electronic device 500, reduce influence of at least part (e.g., the sixth conductive unit A6) of the second side member 520 on the antenna radiation performance of the third antenna radiator ③ and/or the fourth antenna radiator ④, or move the resonance frequency of the third antenna radiator ③ and/or the fourth antenna radiator ④.

[0156] According to various embodiments, at least part of the first conductive unit A1 adjacent to the hinge structure 530 and a conductive unit (not shown) included in the hinge structure 530 electrically connected to the first conductive unit A1 may operate as a fifth antenna radiator. According to various embodiments, at least part of the seventh conductive unit A7 adjacent to the hinge structure 530 and a conductive unit (not shown) included in the hinge structure 530 electrically connected to the seventh conductive unit A7 may operate as a sixth antenna radiator. According to various embodiments (not shown), at least part of a conductive unit (not shown) of the hinge structure 530 adjacent to the first conductive unit A1 forming part of the second side surface 521 may operate as a seventh antenna radiator. According to various embodiments (not shown), a conductive unit (not shown) of the hinge structure 530 adjacent to the seventh conductive unit A7 forming part of the first side surface 511 may operate as an eighth antenna radiator. According to various embodiments, the electronic device 500 may further include a frequency adjustment circuit including an electrical element having inductance, capacitance or conductance acting on a fifth antenna system for the fifth antenna radiator, a sixth antenna system for the sixth antenna, a seventh antenna system for the seventh antenna or an eighth antenna system for the eighth antenna.

[0157] FIG. 6A illustrates a circuit of an electronic device of an unfolded state according to an embodiment. FIG. 6B illustrates a frequency adjustment circuit according to an embodiment.

[0158] Referring to FIG. 6A, in an embodiment, an electronic device 600 (e.g., the electronic device 500 of FIG. 5) may include at least one of a first side member 610 (e.g., the first side member 510 of FIG. 5), a second side member 620 (e.g., the second side member 520 of FIG. 5), a hinge structure 630 (e.g., the hinge structure 530 of FIG. 5), at least one wireless communication circuit 680 (e.g., the wireless communication circuit 580 of FIG. 5), a processor 690 (e.g., the processor 590 of FIG. 5) or a frequency adjustment circuit 691 (e.g., the first frequency adjustment circuit 591 of FIG. 5).

[0159] According to an embodiment, the first side member 610 may include at least one of a first conductive unit B1 (e.g., the first conductive unit A1 of FIG. 5), a first insulating unit 641 (e.g., the first insulating unit 541 of FIG. 5), a second conductive unit B2 (e.g., the second conductive unit A2 of FIG. 5), a second insulating unit 642 (e.g., the second insulating unit 542 of FIG. 5), and a third conductive unit B3

(e.g., the third conductive unit A3 of FIG. 5) which are sequentially disposed from the hinge structure 630.

[0160] According to an embodiment, the second side member 620 may include at least one of a fourth conductive unit B4 (e.g., the fourth conductive unit A4 of FIG. 5), a third insulating unit 543 (e.g., the third insulating unit 543 of FIG. 5), a fifth conductive unit B5 (e.g., the fifth conductive unit A5 of FIG. 5), a fourth insulating unit 644 (e.g., the fourth insulating unit 544 of FIG. 5), and a sixth conductive unit B6 (e.g., the sixth conductive unit A6 of FIG. 5) which are sequentially disposed from the hinge structure 630.

[0161] According to an embodiment, a first antenna radiator ① (e.g., the first antenna radiator ① of FIG. 5) may be formed as at least part of the second conductive unit B2 included in the first side member 610. A second antenna radiator ② (e.g., the second antenna radiator ② of FIG. 5) may be formed as at least part of the third conductive unit B3 included in the first side member 610. The first antenna radiator ① may be electrically connected to the at least one wireless communication circuit 680 at at least one feeding point, and may be electrically connected to a ground G1 (e.g., the ground G1 of FIG. 5) at at least one grounding point. The second antenna radiator ② may be electrically connected to the wireless communication circuit 680 at at least one feeding point, and may be connected to a ground G2 (e.g., the ground G2 of FIG. 5) at at least one grounding point.

[0162] According to an embodiment, the at least one wireless communication circuit 680 may be electrically connected to a first position (e.g., a first feeding point FP1) of the second conductive unit B2 closer to the second insulating unit 642 than the first insulating unit 641. The at least one wireless communication circuit 680 may transmit and/or receive a first signal of a first frequency band via the first antenna radiator ① formed as at least part of the second conductive unit B2.

[0163] According to an embodiment, the at least one wireless communication circuit 680 may be electrically connected to a second position (e.g., a second feeding point FP2) of the third conductive unit B3 adjacent to the second insulating unit 642. The at least one wireless communication circuit 680 may transmit and/or receive a second signal of a second frequency band via the second antenna radiator ② formed as at least part of the third conductive unit B3.

[0164] For example, the first feeding point FP1 of the first antenna radiator ① or the second feeding point FP2 of the second antenna radiator ② may be positioned near (e.g., within 10 mm) the second insulating unit 642 between the first antenna radiator ① and the second antenna radiator ②. The first grounding point GP1 (or a seventh position) of the first antenna radiator ① may be positioned away from the first feeding point FP1 toward the hinge structure 630. For example, the first grounding point GP1 of the first antenna radiator ① may be positioned between the first feeding point FP1 and a connector hole 644 (e.g., the connector hole 244 of FIG. 2). For example, the second grounding point GP2 (or an eighth position) of the second antenna radiator ② may be disposed in a portion B31 (e.g., the portion A31 of FIG. 5) forming a third side surface 631 (e.g., the third side surface 531 of FIG. 5).

[0165] According to an embodiment, the electronic device 600 may include a first antenna system (or a first antenna device) 650 electrically connected to the at least one wireless communication circuit 680. The first antenna system

650 may include the first antenna radiator ①, the ground G1 electrically connected to the first antenna radiator ①, and a first transmission line 651 (e.g., the first transmission line 551 of FIG. 5) between the first antenna radiator ① and the at least one wireless communication circuit 680.

[0166] According to an embodiment, the first antenna system 650 may include a tuner 652 connected to the first transmission line 651. The tuner 652 is a frequency adjustment circuit for adjusting a resonance frequency of the first antenna radiator ①, and may include, for example, one or more switching elements, or elements such as series and/or parallel tunable capacitors for implementing various capacitances. According to an embodiment, the tuner 652 may move the resonance frequency of the first antenna radiator ① to a designated frequency under control of the processor 690 (or the wireless communication circuit 680) to support a corresponding communication mode, or move the resonance frequency of the first antenna radiator ① as designated.

[0167] According to an embodiment, the electronic device 600 may include a second antenna system (or a second antenna device) 660 electrically connected to the at least one wireless communication circuit 680. The second antenna system 660 may include the second antenna radiator ②, the ground G2 electrically connected to the second antenna radiator and a second transmission line 661 (e.g., the second transmission line 561 of FIG. 5) between the second antenna radiator ② and the at least one wireless communication circuit 680.

[0168] According to an embodiment, the second antenna system 660 may include a switch 662 connected to the second transmission line 661. The second antenna radiator ② may be electrically connected to the at least one wireless communication circuit 680 if the switch 662 is turned on, and may not be electrically connected to (to e.g., electrically separated from) the at least one wireless communication circuit 680 if the switch 662 is turned off. According to an embodiment, the processor 690 may control the switch 662 based on the communication mode. For example, the processor 690 may turn on the switch 662 in a first communication mode, and the at least one wireless communication circuit 680 may transmit and/or receive a signal of a designated frequency related to the first communication mode via the second antenna radiator ②. For example, the processor 690 may turn off the switch 662 in a second communication mode which utilizes a different frequency from the first communication mode.

[0169] According to various embodiments, the processor 690 may control the switch 662 according to the unfolded state or the folded state of the electronic device 600 (e.g., see FIG. 3). For example, the processor may turn on the switch 662 in the unfolded state of the electronic device 600, and turn off the switch 662 in the folded state of the electronic device 600.

[0170] According to an embodiment, a frequency band usable by the electronic device 600 may include a low band (LB) (about 600~1 GHz), a middle band (MB) (about 1~2.2 GHz), a high band (HB) (about 2.2~2.7 GHz), or an ultra high band (UHB) (about 2.7~3.6 GHz). According to various embodiments, the electronic device 600 may further utilize other frequency band. According to an embodiment, the first antenna system 650 or the second antenna system 660 may correspond to a transmission line supporting the communication mode using the LB, the MB, the HB, or the

UHB. For example, the first antenna system **650** or the second antenna system **660** may support various communication services such as global system for mobile communication (GSM), long term evolution (LTE) or 5th generation (5G) network utilizing a corresponding frequency band.

[0171] According to an embodiment, the second side member **620** may include a fifth conductive unit **B5** (e.g., the fifth conductive unit **A5** of FIG. 5) disposed close to the first antenna radiator **①**, and a sixth conductive unit **B6** (e.g., the sixth conductive unit **A6** of FIG. 5) disposed close to the second antenna radiator **②**, in the folded state of the electronic device **600** (e.g., see FIG. 3).

[0172] Referring to FIGS. 6A and 6B, in an embodiment, the frequency adjustment circuit **691** (e.g., the first frequency adjustment circuit **591** of FIG. 5) may include at least part of a first electrical path (or a first conductive path) **671** electrically connected between the fifth conductive unit **B5** disposed close to the first antenna radiator **①** and a ground **G51** (e.g., the ground **G5** of FIG. 5), in the folded state of the electronic device **600** (e.g., FIG. 3). The frequency adjustment circuit **691** may include a first switch **672** disposed on the first electrical path **671**. According to an embodiment, the first switch **672** may include a first terminal **672a** electrically connected to the ground **G51**, and a second terminal **672b** electrically connected to a third position (e.g., a third grounding point **GP3**) closer to the fourth insulating unit **644** than the third insulating unit **643**. For example, the fifth conductive unit **B5** may be electrically connected to the first conductive path **671** at the third grounding point **GP3** positioned near (e.g., within 10 mm) the fourth insulating unit **644** between the fifth and sixth conductive units **B5** and **B6**.

[0173] According to an embodiment, the frequency adjustment circuit **691** may further include at least one first lumped element (or a first passive element) **673** such as an inductor disposed on the first electrical path **671**. For example, the first lumped element **673** may be connected to a wire between the first switch **672** and the third grounding point **GP3**. According to various embodiments (not shown), the first lumped element **673** may be connected to a wire between the first switch **672** and the ground **G51**.

[0174] Referring to FIGS. 6A and 6B, in an embodiment, the frequency adjustment circuit **691** may include at least part of a second electrical path (or a second conductive path) **681** electrically connected between the second conductive unit **B6** disposed close to the second antenna radiator in the folded state of the electronic device **600** (e.g., see FIG. 3) and a ground **G52** (e.g., the ground **G5** of FIG. 5). The frequency adjustment circuit **691** may include a second switch **682** disposed on the second electrical path **681**. According to an embodiment, the second switch **682** may include a third terminal **682a** electrically connected to the ground **G52**, and a fourth terminal **682b** electrically connected to a fourth position (e.g., a fourth grounding point **GP4**) close to the fourth insulating unit **644**. For example, the sixth conductive unit **B6** may be electrically connected to the second electrical path **681** at the fourth grounding point **GP4** positioned near (e.g., within 10 mm) the fourth insulating unit **644** between the fifth and sixth conductive units **B5** and **B6**.

[0175] According to an embodiment, the frequency adjustment circuit **691** may further include at least one second lumped element (or a second passive element) **683** such as an inductor disposed on the second electrical path **681**. For

example, the second lumped element **683** may be connected to a wire between the second switch **682** and the fourth grounding point **GP4**. According to various embodiments (not shown), the second lumped element **683** may be connected to a wire between the second switch **682** and the ground **G52**.

[0176] According to an embodiment, in the folded state of the electronic device **600** (see FIG. 3), the frequency adjustment circuit **691** may include an electrical element (e.g., an inductor) having a component such as inductance, capacitance or conductance acting on the first antenna system **650** and/or the second antenna system **660**, and thus reduce influence of the second side member **620** on the antenna radiation performance of the first antenna radiator **①** and/or the second antenna radiator **2**.

[0177] For example, the processor **690** may turn on the first switch **672** in the folded state of the electronic device **600** (see FIG. 3), and the fifth conductive unit **B5** may be electrically connected to the ground **G51** through the first lumped element **673**. In the folded state of the electronic device **600**, the first lumped element **673** may reduce the influence of the fifth conductive unit **B5** on the antenna radiation performance of the first antenna radiator **①** and/or the antenna radiation performance of the second antenna radiator **②**. In the folded state of the electronic device **600**, the second side member **620** is close to the first antenna radiator **①** and/or the second antenna radiator **②** and capacitance (or capacitance component) (e.g., parasitic capacitance) may occur, and a parasitic resonance frequency may be generated due to the capacitance. According to an embodiment, the first lumped element **673** may adjust the parasitic resonance frequency not to include the parasitic resonance frequency generated in the folded state of the electronic device **600** in the resonance frequency band of the first antenna radiator **①** and/or the second antenna radiator **②**. The parasitic resonance frequency generated in the folded state of the electronic device **600** may be moved out of the resonance frequency band of the first antenna radiator **①** and/or the second antenna radiator **②** by the first lumped element **673**, thus reducing degradation of the antenna radiation performance due to the parasitic resonance frequency.

[0178] According to an embodiment, the fifth conductive unit **B5** electrically connected to the first lumped element **673** may operate as an additional antenna radiator for transmitting or receiving a signal having a selected or designated frequency, together with the first antenna radiator **①** and/or the second antenna radiator **②**, in the folded state of the electronic device **600** (see FIG. 3). For example, the fifth conductive unit **B5** may operate as an antenna radiator electromagnetically coupled with the first antenna radiator **①** and/or the second antenna radiator **②** in the folded state of the electronic device **600**. According to an embodiment, the first lumped element **673** may have inductance (e.g., inductance component) for reducing capacitance (e.g., parasitic capacitance) which may occur if the fifth conductive unit **B5** is close to the first antenna radiator **①**. For example, the first lumped element **673** electrically connected to the ground **G51** may be electrically connected to the fifth conductive unit **B5** to operate as an inductor. According to an embodiment, the first lumped element **673** may adjust impedance of the fifth conductive unit **B5** to match impedance of the transmission line related to the first antenna radiator **①** and/or the second antenna radiator **2**. For

example, the inductance of the first lumped element **673** may adjust the impedance of the fifth conductive unit **B5**.

[0179] For example, the processor **690** may turn on the second switch **682** in the folded state of the electronic device **600** (see FIG. 3), and the sixth conductive unit **B6** may be electrically connected to the ground **G52** through the second lumped element **683**. The second lumped element **683** may reduce influence of the sixth conductive unit **B6** on the antenna radiation performance of the first antenna radiator **①** and/or the antenna radiation performance of the second antenna radiator **②**, in the folded state of the electronic device **600**. In the folded state of the electronic device **600**, the second side member **620** is close to the first antenna radiator **①** and/or the second antenna radiator **②** and capacitance (or capacitance component) (e.g., parasitic capacitance) may occur, and a parasitic resonance frequency may be generated due to the capacitance. According to an embodiment, the second lumped element **683** may adjust the parasitic resonance frequency not to include the parasitic resonance frequency generated in the folded state of the electronic device **600** in the resonance frequency band of the first antenna radiator **①** and/or the second antenna radiator **②**. The parasitic resonance frequency generated in the folded state of the electronic device **600** may be moved out of the resonance frequency band of the first antenna radiator **①** and/or the second antenna radiator **②** by the second lumped element **683**, thus reducing the degradation of the antenna radiation performance due to the parasitic resonance frequency.

[0180] According to an embodiment, part **B64** (e.g., the part **A64** of the sixth conductive unit **A6** of FIG. 5) of the sixth conductive unit **B6** electrically connected to the second lumped element **683** may operate as an additional antenna radiator for transmitting or receiving a signal having a selected or designated frequency, together with the first antenna radiator **①** and/or the second antenna radiator **②** in the folded state of the electronic device **600** (see FIG. 3). For example, the part **B64** of the sixth conductive unit **B6** may operate as the antenna radiator electromagnetically coupled with the first antenna radiator **①** and/or the second antenna radiator **②** in the folded state of the electronic device **600**. According to an embodiment, the second lumped element **683** may have inductance (or inductance component) for reducing capacitance (e.g., parasitic capacitance) which may occur if the part **B64** of the sixth conductive unit **B6** is close to the second antenna radiator **②**. For example, the second lumped element **683** electrically connected to the ground **G52** may be electrically connected to the sixth conductive unit **B6** to operate as an inductor. According to an embodiment, the second lumped element **683** may adjust impedance of the part **B64** of the sixth conductive unit **B6** to match the impedance of the transmission line related to the first antenna radiator **①** and/or the second antenna radiator **②**. For example, the inductance of the second lumped element **683** may adjust the impedance of the part **B64** of the sixth conductive unit **B6**.

[0181] According to an embodiment, the frequency adjustment circuit **691** in the folded state of the electronic device **600** (see FIG. 3) may function to reduce degradation of the antenna radiating performance due to the parasitic resonance frequency by moving the parasitic resonance frequency generated by the second side member **620** which is close to the first antenna radiator **①** and/or the second antenna radiator **②**, and function to operate part (e.g., the fifth

conductive unit **B5** and/or the part **B64** of the sixth conductive unit **B6**) of the second side member **620** as the additional antenna radiator coupled with the first antenna radiator **①** and/or the second antenna radiator **②** by adjusting the impedance.

[0182] According to various embodiments, the frequency adjustment circuit **691** may move the resonance frequency of the first antenna radiator **①** and/or the resonance frequency of the second antenna radiator **④** to a designated frequency in the folded state of the electronic device **600** (see FIG. 3), or move it as designated, according to on or off of the first switch **672** or the second switch **673**.

[0183] For example, in the folded state of the electronic device **600** (see FIG. 3), if the first switch **672** is turned on under control of the processor **690**, the first lumped element **673** may be electrically connected to the fifth conductive unit **B5**, to reduce a resonance frequency of the parasitic component which may occur in the frequency band related to the first antenna radiator **①** if the fifth conductive unit **B5** is close to the first antenna radiator **①**, and/or a resonance frequency of the parasitic component which may occur in the frequency band related to the second antenna radiator **②** if the sixth conductive unit **B6** is close to the second antenna radiator **②**. In the folded state of the electronic device **600**, if the first switch **672** is turned off by the control of the processor **690**, the fifth conductive unit **B5** may be electrically separated from the ground **G51**. If the fifth conductive unit **B5** is electrically separated from the ground **G51**, the impedance (or the condition of the parasitic resonance) may be changed to reduce the antenna radiation performance degradation of the first antenna system **650** and/or the second antenna system **660** due to the parasitic resonance.

[0184] For example, in the folded state of the electronic device **600** (see FIG. 3), if the second switch **682** is turned on under the control of the processor **690**, the second lumped element **683** may be electrically connected to the sixth conductive unit **B6**, to reduce the resonance frequency of the parasitic component which may occur in the frequency band related to the second antenna radiator **②** if the sixth conductive unit **B6** is close to the second antenna radiator **②**, and/or the resonance frequency of the parasitic component which may occur in the frequency band related to the first antenna radiator **①** if the fifth conductive unit **B5** is close to the second antenna radiator **①**. In the folded state of the electronic device **600**, if the second switch **682** is turned off by the control of the processor **690**, the sixth conductive unit **B6** may be electrically separated from the ground **G52**. If the sixth conductive unit **B6** is electrically separated from the ground **G52**, the impedance (or, the condition of the parasitic resonance) may be changed to reduce the antenna radiation performance degradation of the first antenna system **650** and/or the second antenna system **660** due to the parasitic resonance.

[0185] FIG. 7A illustrates a circuit of an electronic device in an unfolded state according to various embodiments. FIG. 7B illustrates a frequency adjustment circuit of FIG. 7A according to various embodiments.

[0186] Referring to FIG. 7A, in an embodiment, an electronic device **700** (e.g., the electronic device **500** of FIG. 5) may include at least one of a first side member **710** (e.g., the first side member **510** of FIG. 5), a second side member **720** (e.g., the second side member **620** of FIG. 6A), a hinge structure **730** (e.g., the hinge structure **530** of FIG. 5), at least one wireless communication circuit **780** (e.g., the wireless

communication circuit **580** of FIG. 5), a processor **790** (e.g., the processor **590** of FIG. 5) or a frequency adjustment circuit **791** (e.g., the first frequency adjustment circuit **591** of FIG. 5).

[0187] According to an embodiment, the first side member **710** may include at least one of a first conductive unit **C1** (e.g., the first conductive unit **A1** of FIG. 5), a first insulating unit **741** (e.g., the first insulating unit **541** of FIG. 5), a second conductive unit **C2** (e.g., the second conductive unit **A2** of FIG. 5), a second insulating unit **742** (e.g., the second insulating unit **542** of FIG. 5), and a third conductive unit **C3** (e.g., the third conductive unit **A3** of FIG. 5) which are sequentially disposed from the hinge structure **730**.

[0188] According to an embodiment, the second side member **720** may include at least one of a fourth conductive unit **C4** (e.g., the fourth conductive unit **A4** of FIG. 5), a third insulating unit **743** (e.g., the third insulating unit **543** of FIG. 5), a fifth conductive unit **C5** (e.g., the fifth conductive unit **A5** of FIG. 5), a fourth insulating unit **744** (e.g., the fourth insulating unit **544** of FIG. 5), and a sixth conductive unit **C6** (e.g., the sixth conductive unit **A6** of FIG. 5) which are sequentially disposed from the hinge structure **730**.

[0189] According to an embodiment, a first antenna radiator **①** (e.g., the first antenna radiator **①** of FIG. 5) may be formed as at least part of the second conductive unit **C2** included in the first side member **710**. A second antenna radiator **②** (e.g., the second antenna radiator **②** of FIG. 5) may be formed as at least part of the third conductive unit **C3** included in the first side member **710**. A first antenna system **750** (e.g., the first antenna system **650** of FIG. 6A) may include the first antenna radiator **①** (e.g., the first antenna radiator **①** of FIG. 6A), a ground **G1** (e.g., the ground **G1** of FIG. 6A) electrically connected with the first antenna radiator **①**, a first transmission line **751** (e.g., the first transmission line **651** of FIG. 6A) between the first antenna radiator **①** and at least one wireless communication circuit **780** or a tuner **752** (e.g., the tuner **652** of FIG. 6A). A second antenna system **760** (e.g., the second antenna system **660** of FIG. 6A) may include the second antenna radiator **②** (e.g., the second antenna radiator **②** of FIG. 6A), a ground **G2** (e.g., the ground **G2** of FIG. 6A) electrically connected to the second antenna radiator **②**, a second transmission line **761** (e.g., the second transmission line **661** of FIG. 6A) between the second antenna radiator **②** and the at least one wireless communication circuit **780** or a switch **762** (e.g., the switch **662** of FIG. 6A). The first antenna radiator **①** or the second antenna radiator **②** may be formed as at least part of the first side member **710**. Since the first antenna system **750** or the second antenna system **760** is substantially the same as FIGS. 6A and 6B, its detailed descriptions are omitted.

[0190] According to an embodiment, the at least one wireless communication circuit **780** may be electrically connected to a first position (e.g., a first feeding point **FP1**) of the second conductive unit **C2** closer to the second insulating unit **742** than the first insulating unit **741**. The at least one wireless communication circuit **780** may transmit and/or receive a first signal of a first frequency band through the first antenna radiator **①** formed as at least part of the second conductive unit **C2**. For example, the first feeding point **FP1** may be positioned near (e.g., within about 10 mm) the second insulating unit **742**.

[0191] According to an embodiment, the at least one wireless communication circuit **780** may be electrically connected to a second position (e.g., a second feeding point **FP2**) of the third conductive unit **C3** close to the second insulating unit **742**. The at least one wireless communication circuit **780** may transmit and/or receive a second signal of a second frequency band through the second antenna radiator **②** formed as at least part of the third conductive unit **C3**. For example, the second feeding point **FP2** may be positioned near (e.g., within about 10 mm) the second insulating unit **742**.

[0192] According to an embodiment, the second side member **720** may include a fifth conductive unit **C5** (e.g., the fifth conductive unit **A5** of FIG. 5) disposed close to the first antenna radiator **①**, and a sixth conductive unit **C6** (e.g., the sixth conductive unit **A6** of FIG. 5) disposed close to the second antenna radiator **②**, in the folded state of the electronic device **700** (e.g., see FIG. 3).

[0193] Referring to FIGS. 7A and 7B, in an embodiment, the frequency adjustment circuit **791** (e.g., the first frequency adjustment circuit **591** of FIG. 5) may include at least one first switch **792** including a first terminal **792a** electrically connected to a ground **G5** (e.g., the ground **G5** of FIG. 5), a second terminal **792b** electrically connected to a third position (e.g., a third grounding point **GP3**) of the fifth conductive unit **C5** closer to the fourth insulating unit **744** than the third insulating unit **743**, and a third terminal **792c** electrically connected to a fourth position (e.g., a fourth grounding point **GP4**) of the sixth conductive unit **C6** close to the fourth insulating unit **744**. For example, the third grounding point **GP3** or the fourth grounding point **GP4** may be positioned near (e.g., within about 10 mm) the fourth insulating unit **744**.

[0194] According to an embodiment, the frequency adjustment circuit **791** may further include at least one first lumped element (or a first passive element) **793** disposed on a first electrical path **781** electrically connected between the third grounding point **GP3** and the ground **G5**. For example, the at least one first lumped element **793** may be connected between the second terminal **792b** of the at least one first switch **792** and the third grounding point **GP3**.

[0195] According to an embodiment, the frequency adjustment circuit **791** may further include at least one second lumped element (or a second passive element) **794** disposed on a second electrical path **782** electrically connected between the fourth grounding point **GP4** and the ground **G5**. For example, the at least one second lumped element **794** may be connected between the third terminal **792c** of the at least one first switch **792** and the fourth grounding point **GP4**.

[0196] According to an embodiment, the first switch **792** may electrically connect the first terminal **792a** with at least one of the second terminal **792b** and the third terminal **792c**, under control of the processor **790**. For example, if the first switch **792** electrically connects the first terminal **792a** and the second terminal **792b**, the fifth conductive unit **C5** may be electrically connected to the ground **G5** through the first lumped element **793**. For example, if the first switch **792** electrically connects the first terminal **792a** and the third terminal **792c**, the sixth conductive unit **C6** may be electrically connected to the ground **G5** through the second lumped element **794**. According to an embodiment (not shown), in place of the first lumped element **793** and the second lumped element **794**, a lumped element connected to an electrical

path **783** electrically connected between the first switch **792** and the ground **G5** may be included.

[0197] According to an embodiment, in the folded state of the electronic device **700** (see FIG. 3), the frequency adjustment circuit **791** may include an electrical element (e.g., an inductor) having the component such as inductance, capacitance or conductance acting on the first antenna system **750** and/or the second antenna system **760**, thus reducing influence of the second side member **720** on antenna radiation performance of the first antenna radiator **①** and/or the second antenna radiator **②**.

[0198] For example, the fifth conductive unit **C5** may be electrically connected to the ground **G5** through the first lumped element **793** in the folded state of the electronic device **700** (see FIG. 3). The first lumped element **793** may reduce the influence of the fifth conductive unit **C5** on the antenna radiation performance for the first antenna radiator **①** and/or the antenna radiation performance for the second antenna **②**, in the folded state of the electronic device **700**. In the folded state of the electronic device **700**, the side member **720** may be close to the first antenna radiator **①** and/or the second antenna radiator **②** to generate parasitic capacitance, and the parasitic resonance may generate a parasitic resonance frequency.

[0199] According to an embodiment, the first lumped element **793** may adjust the parasitic resonance frequency not to include the parasitic resonance frequency generated in the folded state of the electronic device **700** in a resonance frequency band of the first antenna radiator **①** and/or the second antenna radiator **②**. The parasitic resonance frequency generated in the folded state of the electronic device **700** may be moved out of the resonance frequency band of the first antenna radiator **①** and/or the second antenna radiator **②** by the first lumped element **793**, thus reducing degradation of the antenna radiation performance due to the parasitic resonance frequency.

[0200] According to an embodiment, the fifth conductive unit **C5** electrically connected to the first lumped element **793** may operate as an additional antenna radiator for transmitting or receiving a signal having a selected or designated frequency, together with the first antenna radiator **①** and/or the second antenna radiator **②** in the folded state of the electronic device **700** (see FIG. 3). For example, the fifth conductive unit **C5** may operate as the antenna radiator electromagnetically coupled with the first antenna radiator **①** and/or the second antenna radiator **②** in the folded state of the electronic device **700**. According to an embodiment, the first lumped element **793** may have inductance for reducing the parasitic capacitance which may occur if the fifth conductive unit **C5** is close to the first antenna radiator **①**. For example, the first lumped element **793** electrically connected to the ground **G5** may be electrically connected to the fifth conductive unit **C5** to operate as an inductor. According to an embodiment, the first lumped element **793** may adjust impedance of the fifth conductive unit **C5** to match impedance of a transmission line related to the first antenna radiator **①** and/or the second antenna radiator **②**. For example, the inductance of the first lumped element **793** may adjust the impedance of the fifth conductive unit **C5**.

[0201] For example, in the folded state of the electronic device **700** (see FIG. 3), the sixth conductive unit **C6** may be electrically connected to the ground **G5** through the second lumped element **794**. In the folded state of the electronic device **700**, the second lumped element **794** may reduce

influence of the sixth conductive unit **C6** on the antenna radiation performance for the first antenna radiator **①**) and/or the antenna radiation performance for the second antenna radiator **②**. In the folded state of the electronic device **700**, the second side member **720** may be close to the first antenna radiator **①** and/or the second antenna radiator **②**) to generate parasitic capacitance, and the parasitic capacitance may generate a parasitic resonance frequency. According to an embodiment, the second lumped element **794** may adjust the parasitic resonance frequency not to include the parasitic resonance frequency generated in the folded state of the electronic device **700** in the resonance frequency band of the first antenna radiator **①** and/or the second antenna radiator **②**. The parasitic resonance frequency generated in the folded state of the electronic device **700** may be moved out of the resonance frequency band of the first antenna radiator **①**) and/or the second antenna radiator **2** by the second lumped element **794**, thus reducing degradation of the antenna radiation performance due to the parasitic resonance frequency.

[0202] According to an embodiment, part **C64** (e.g., the part **A64** of the sixth conductive unit **A6** of FIG. 5) of the sixth conductive unit **C6** electrically connected to the second lumped element **794** may operate as an additional antenna radiator for transmitting or receiving a signal having a selected or designated frequency, together with the first antenna radiator **①** and/or the second antenna radiator **②** in the folded state of the electronic device **700** (see FIG. 3). For example, the part **C64** of the sixth conductive unit **C6** may operate as the antenna radiator electromagnetically coupled to the first antenna radiator **①** and/or the second antenna radiator **②** in the folded state of the electronic device **700**. According to an embodiment, the second lumped element **794** may have inductance (or inductance component) for reducing capacitance (e.g., parasitic capacitance) which may occur if the part **C64** of the sixth conductive unit **C6** is close to the second antenna radiator **②**. For example, the second lumped element **794** electrically connected to the ground **G5** may be electrically connected to the sixth conductive unit **C6** to operate as an inductor. According to an embodiment, the second lumped element **794** may adjust impedance of the part **C64** of the sixth conductive unit **C6** to match impedance of a transmission line related to the first antenna radiator **①**) and/or the second antenna radiator **②**. For example, the inductance of the second lumped element **794** may adjust the impedance of the part **C64** of the sixth conductive unit **C6**.

[0203] According to an embodiment, the frequency adjustment circuit **791** in the folded state of the electronic device **700** (see FIG. 3) may function to reduce degradation of the antenna radiating performance due to the parasitic resonance frequency by moving the parasitic resonance frequency generated by the second side member **720** which is close to the first antenna radiator **①** and/or the second antenna radiator **②**, and function to operate the part (e.g., the fifth conductive unit **C5** and/or the part **C64** of the sixth conductive unit **C6**) of the second side member **720** as the additional antenna radiator coupled with the first antenna radiator **①** and/or the second antenna radiator **②** by adjusting the impedance.

[0204] According to various embodiments, the frequency adjustment circuit **791** may move the resonance frequency of the first antenna radiator **①** and/or the second antenna radiator **④** to a designated frequency in the folded state of

the electronic device 700 (see FIG. 3), or move it as designated, according to on or off of the switch 792.

[0205] For example, in the folded state of the electronic device 700 (see FIG. 3), according to the switching state of the first switch 792, the fifth conductive unit C5 may be electrically connected to the G5 through the first lumped element 793, or the sixth conductive unit C6 may be electrically connected to the ground G5 through the second lumped element 794. Impedance characteristics of the first antenna system 750 or the second antenna system 760 may vary according to the switching state of the first switch 792, and thus the resonance frequency of the first antenna radiator ① and/or the resonance frequency of the second antenna radiator ② may be moved.

[0206] FIG. 8 illustrates a circuit related to a folded state of an electronic device according to an embodiment. FIG. 9A is a graph showing antenna radiation performance on a frequency distribution for a first antenna system if an electronic device not including a frequency adjustment circuit is unfolded or folded. FIG. 9B is a graph showing the antenna radiation performance on the frequency distribution for the first antenna system if the electronic device including the frequency adjustment circuit is unfolded or folded. FIG. 10A is a graph showing antenna radiation performance on a frequency distribution for a second antenna system if an electronic device not including a frequency adjustment circuit is unfolded or folded. FIG. 10B is a graph showing the antenna radiation performance on the frequency distribution for the second antenna system if the electronic device including the frequency adjustment circuit is unfolded or folded according to an embodiment.

[0207] Referring to FIG. 8, in various embodiments, an electronic device 800 may be the electronic device 500 of FIG. 5, the electronic device 600 of FIG. 6A or the electronic device 700 of FIG. 7A. The electronic device 800 may include a first antenna system 850 or a second antenna system 860. In an embodiment, the first antenna system 850 may include a first antenna radiator ① formed as at least part of a first side member 810, a ground G1 electrically connected to the first antenna radiator ①, and a first transmission line 851 between a first feed unit F1 and the first antenna radiator ①. In an embodiment, the second antenna system 860 may include a second antenna radiator ② formed as at least part of the first side member 810, a ground G2 electrically connected to the second antenna radiator ②, and a second transmission line 861 between a second feed unit F2 and the second antenna radiator ②. According to various embodiments, the first antenna system 850 may include the first antenna system 550 of FIG. 5, the first antenna system 650 of FIG. 6A, or the first antenna system 750 of FIG. 7A. According to various embodiments, the second antenna system 860 may include the second antenna system 560 of FIG. 5, the second antenna system 660 of FIG. 6A, or the second antenna system 760 of FIG. 7A.

[0208] According to an embodiment, when viewed in a third direction 803 (e.g., the third direction 203 of FIG. 2) in the folded state of the electronic device 600, the first antenna radiator ① may be disposed by at least overlapping a fifth conductive unit D5 (e.g., the fifth conductive unit B5 of FIG. 6A, or the fifth conductive unit C5 of FIG. 7A) of the second side member 820. When viewed in the third direction (e.g., the third direction 203 of FIG. 2) in the folded state of the electronic device 600, the second antenna radiator ② may

be disposed by at least overlapping a sixth conductive unit D6 (e.g., the sixth conductive unit B6 of FIG. 6A or the sixth conductive unit C6 of FIG. 7A) of the second side member 820.

[0209] According to an embodiment, by controlling a frequency adjustment circuit 891 (e.g., the frequency adjustment circuit 691 of FIG. 6A, or the frequency adjustment circuit 791 of FIG. 7A) in the folded state, the electronic device 800 may electrically connect the fifth conductive unit D5 to the ground G5 (e.g., the ground G51 of FIG. 6A, or the ground G5 of FIG. 7A) through a first lumped element 893 (e.g., the first lumped element 673 of FIG. 6A, or the first lumped element 793 of FIG. 7A). In the folded state, the electronic device 800 may control the frequency adjustment circuit 891, to electrically separate the sixth conductive unit D6 and the ground G5.

[0210] Referring to FIG. 9A, 901a indicates the antenna radiation performance on the frequency distribution of the first antenna system 850, if the electronic device not including the frequency adjustment circuit 891 is unfolded. 902a indicates the antenna radiation performance on the frequency distribution for the first antenna system 850, if the electronic device not including the frequency adjustment circuit 891 is folded. Referring to 901a and 902a, the antenna radiation performance of the first antenna system 850 in at least some frequency band deteriorates if the electronic device is changed from the unfolded state to the folded state.

[0211] Referring to FIG. 9B, 901b indicates the antenna radiation performance on the frequency distribution of the first antenna system 850, if the electronic device 800 including the frequency adjustment circuit 891 is unfolded (see FIG. 2). 902b indicates the antenna radiation performance on the frequency distribution for the first antenna system 850, if the electronic device 800 including the frequency adjustment circuit 891 has the circuit state of FIG. 8 in the folded state. Referring to FIG. 8, in the folded state of the electronic device 800, the first lumped element 893 may reduce the influence of the second side member 820 on the antenna radiation performance for the first antenna system 850 in at least some frequency band.

[0212] Referring to 901b and 902b, the antenna radiation performance for the first antenna system 850 in at least some frequency band does not deteriorate even if the electronic device 800 is changed from the unfolded state to the folded state, or its deterioration degree may be smaller than the case not including the frequency adjustment circuit 891.

[0213] Referring to FIG. 10A, 1001a indicates the antenna radiation performance on the frequency distribution for the second antenna system 860, if the electronic device not including the frequency adjustment circuit 891 is unfolded. 1002a indicates the antenna radiation performance on the frequency distribution for the second antenna system 860, if the electronic device not including the frequency adjustment circuit 891 is folded. Referring to 1001a and 1002a, the antenna radiation performance of the second antenna system in at least some frequency band deteriorates, if the electronic device is changed from the unfolded state to the folded state.

[0214] Referring to FIG. 10B, 1001b indicates the antenna radiation performance on the frequency distribution of the second antenna system 860, if the electronic device 800 including the frequency adjustment circuit 891 is unfolded. 902b indicates the antenna radiation performance on the frequency distribution of the second antenna system 860, if

the electronic device **800** including the frequency adjustment circuit **891** has the circuit state of FIG. **8** in the folded state.

[0215] Referring to FIG. **8**, in the folded state of the electronic device **800**, the first lumped element **893** may reduce the influence of the second side member **820** on the antenna radiation performance of the second antenna system **860** in at least some frequency band. Referring to **1001b** and **1002b**, the antenna radiation performance of the second antenna system **860** in at least some frequency band does not deteriorate even if the electronic device **800** is changed from the unfolded state to the folded state, or its deterioration degree may be smaller than the case not including the adjustment circuit **891**.

[0216] In various embodiments (not shown), the electronic device **800** may electrically connect the sixth conductive unit **D6** to the ground (e.g., the ground **G52** of FIG. **6A**, or the ground **G5** of FIG. **7A**) through the second lumped element (e.g., the second lumped element **683** of FIG. **6A**, or the second lumped element **794** of FIG. **7A**), by controlling the frequency adjustment circuit **891** (e.g., the frequency adjustment circuit **691** of FIG. **6A**, or the frequency adjustment circuit **791** of FIG. **7A**) in the folded state. According to various embodiments, the second side member **820** may be electrically connected to the ground (e.g., the ground **G51** or **G52** of FIG. **6A**, or the ground **G5** of FIG. **7A**) through at least one lumped element (e.g., the first lumped element **673** and the second lumped element of FIG. **6A**, or the first lumped element **793** and the second lumped element **794** of FIG. **7A**) by the control of the frequency adjustment circuit **891**. Hence, the loading of the electrical element by the at least one lumped element may reduce the influence of the second side member **820** on the antenna radiation performance of the first antenna system **850** and/or the second antenna system **860** in at least some frequency band in the folded state of the electronic device **800**.

[0217] FIG. **11** is a graph showing antenna radiation performance on a frequency distribution of a first antenna system **650**, if a first switch **672** of a frequency adjustment circuit **691** is turned off and on in a folded state of an electronic device **600** of FIG. **6A** according to an embodiment.

[0218] Referring to FIGS. **6A** and **11**, **1101** indicates the antenna radiation performance on the frequency distribution for the first antenna system **650**, if the first switch **672** of the frequency adjustment circuit **691** is turned off in the folded state of the electronic device **600**. **1102** indicates the antenna radiation performance on the frequency distribution of the first antenna system **650**, if the first switch **672** of the frequency adjustment circuit **691** is turned on in the folded state of the electronic device **600**. According to an embodiment, if the first switch **672** is on, the third conductive element **B4** may be electrically connected to the ground **G51** through the first lumped element **673** (e.g., see FIG. **8**).

[0219] According to an embodiment, if the first switch **672** is turned on, the first lumped element **673** may act loading of an electrical element (e.g., inductance loading) on the first antenna system **650**. Hence, the antenna radiation performance on the frequency distribution for the first antenna system **650** may differ if the first switch **672** is turned off or on (see **1101** and **1102**).

[0220] For example, the antenna radiation performance for the first antenna system **650** in a first frequency band (e.g., the LB or the MB) may be higher if the first switch **672** is

turned off. For example, the antenna radiation performance for the first antenna system **650** in a second frequency band (e.g., the HB) different from the first frequency band may be higher if the first switch **672** is turned on. According to an embodiment, the resonance frequency of the first antenna system **650** (or, the first antenna radiator **①**) in the folded state of the electronic device **600** may be generated in the first frequency band if the first switch **672** is turned off, and the resonance frequency of the first antenna system **650** (or, the first antenna radiator **①**) may be generated in the second frequency band if the first switch **672** is turned on.

[0221] According to an embodiment, the electronic device **600** in the folded state may change the frequency band utilizing the first antenna radiator **①** according to on or off of the first switch **672** (see **1103**). For example, if the first switch **672** is off, the electronic device **600** may be set to transmit or receive the electromagnetic wave via the first antenna radiator **①** in the first frequency band. For example, if the first switch **672** is turned on, the electronic device **600** may be set to transmit or receive the electromagnetic wave via the first antenna radiator **①** in the second frequency band.

[0222] According to various embodiments, referring to FIGS. **7A** and **11**, **1101** may indicate the antenna radiation performance on the frequency distribution of the first antenna system **750** if the first terminal **792a** and the second terminal **792b** are not electrically connected by controlling the first switch **792** of the frequency adjustment circuit **791** in the folded state of the electronic device **700**. According to various embodiments, referring to FIGS. **7A** and **11**, **1102** may indicate the antenna radiation performance on the frequency distribution of the first antenna system **750** if the first terminal **792a** and the second terminal **792b** are electrically connected by controlling the first switch **792** of the frequency adjustment circuit **791** in the folded state of the electronic device **700**. In an embodiment, if the first terminal **792a** and the second terminal **792b** are electrically connected, the first lumped element **793** may act the loading (e.g., the inductance loading) of the electrical element on the first antenna system **750**. For example, not electrically connecting the first terminal **792a** and the second terminal **792b** in the frequency adjustment circuit **791** may be substantially the same as turning off the first switch **672** in the frequency adjustment circuit **691** of FIG. **6A**. For example, electrically connecting the first terminal **792a** and the second terminal **792b** in the frequency adjustment circuit **791** may be substantially the same as turning on the first switch **672** in the frequency adjustment circuit **691** of FIG. **6A**.

[0223] FIG. **12** is a graph showing antenna radiation performance on a frequency distribution of a second antenna system **660**, if a first switch **672** of a frequency adjustment circuit **691** is turned off and on in a folded state of an electronic device **600** of FIG. **6A** according to an embodiment.

[0224] Referring to FIGS. **6A** and **12**, **1201** indicates the antenna radiation performance on the frequency distribution of the second antenna system **660** if the first switch **672** of the frequency adjustment circuit **691** is turned off in the folded state of the electronic device **600**. **1202** indicates the antenna radiation performance on the frequency distribution of the second antenna system **660** if the first switch **672** of the frequency adjustment circuit **691** is turned on in the folded state of the electronic device **600**. According to an embodiment, if the first switch **672** is on, the third conduc-

tive element B3 may be electrically connected to the ground G51 through the first lumped element 673 (e.g., see FIG. 8).

[0225] According to an embodiment, if the first switch 672 is turned on, the first lumped element 673 may act the loading of the electrical element (e.g., the inductance loading) on the second antenna system 660. Hence, the antenna radiation performance on the frequency distribution for the second antenna system 660 may differ if the first switch 672 is turned off or on (see 1201 and 1202).

[0226] For example, the antenna radiation performance for the first antenna system 650 in the first frequency band (e.g., about 1600~ about 2350 MHz) may be higher if the first switch 672 is turned off. For example, the antenna radiation performance for the second antenna system 660 in the second frequency band different from the first frequency band (e.g., 2350~about 2700 MHz) may be higher if the first switch 672 is turned on. According to an embodiment, the resonance frequency of the second antenna system 660 (or, the second antenna radiator ②) in the folded state of the electronic device 600 may be generated in the first frequency band if the first switch 672 is turned off, and the resonance frequency of the second antenna system 660 (or, the second antenna radiator ②) if the first switch 672 is turned on may be generated in the second frequency band.

[0227] According to an embodiment, the electronic device 600 may change the frequency band utilizing the second antenna radiator ② in the folded state according to on or off of the first switch 672 (see 1203). For example, if the first switch 672 is turned off, the electronic device 600 may be set to transmit or receive the electromagnetic wave through the second antenna radiator ② in the first frequency band. For example, if the first switch 672 is turned on, the electronic device 600 may be set to transmit or receive the electromagnetic wave through the second antenna radiator ② in the second frequency band.

[0228] According to various embodiments, referring to FIGS. 7A and 12, 1201 may indicate the antenna radiation performance on the frequency distribution of the second antenna system 760 if the first terminal 792a and the second terminal 792b are not electrically connected by controlling the first switch 792 of the frequency adjustment circuit 791 in the folded state of the electronic device 700. According to various embodiments, referring to FIGS. 7A and 11, 1202 may indicate the antenna radiation performance on the frequency distribution of the second antenna system 760 if the first terminal 792a and the second terminal 792b are electrically connected by controlling the first switch 792 of the frequency adjustment circuit 791 in the folded state of the electronic device 700. In an embodiment, if the first terminal 792a and the second terminal 792b are electrically connected, the first lumped element 793 may act the loading of the electrical element (e.g., the inductance loading) on the first antenna system 750. For example, not electrically connecting the first terminal 792a and the second terminal 792b in the frequency adjustment circuit 791 may be substantially the same as turning off the first switch 672 in the frequency adjustment circuit 691 of FIG. 6A. For example, electrically connecting the first terminal 792a and the second terminal 792b in the frequency adjustment circuit 791 may be substantially the same as turning on the first switch 672 in the frequency adjustment circuit 691 of FIG. 6A.

[0229] FIG. 13 illustrates a circuit related to a folded state of an electronic device 800 according to an embodiment. FIG. 14 is a graph illustrating antenna radiation performance

on a frequency distribution of a second antenna system 860, if a sixth conductive unit D6 and a ground G5 are electrically connected and are not electrically connected in a folded state of an electronic device 800 according to an embodiment.

[0230] Referring to FIG. 13, in an embodiment, a first antenna system 850 may include a first antenna radiator ① formed as at least part of a first side member 810, a ground G1 electrically connected to the first antenna radiator ①, and a first transmission line 851 between a first feed unit F1 and the first antenna radiator ①. In an embodiment, a second antenna system 860 may include a second antenna radiator ② formed as at least part of the first side member 810, a ground G2 electrically connected to the second antenna radiator ②, and a second transmission line 861 between a second feed unit F2 and the second antenna radiator ②. According to various embodiments, the first antenna system 850 may include the first antenna system 550 of FIG. 5, the first antenna system 650 of FIG. 6A, or the first antenna system 750 of FIG. 7A. According to various embodiments, the second antenna system 860 may include the second antenna system 560 of FIG. 5, the second antenna system 660 of FIG. 6A, or the second antenna system 760 of FIG. 7A.

[0231] According to an embodiment, the electronic device 800 may electrically connect a sixth conductive unit D6 (e.g., the sixth conductive unit B6 of FIG. 6A, or the sixth conductive unit C6 of FIG. 7A) to the ground G5 (e.g., the ground G51 of FIG. 6A, or the ground G5 of FIG. 7A) through a second lumped element 894 (e.g., the second lumped element 683 of FIG. 6A, or the second lumped element 794 of FIG. 7A), by controlling a frequency adjustment circuit 891 (e.g., the frequency adjustment circuit 691 of FIG. 6A, or the frequency adjustment circuit 791 of FIG. 7A) in the folded state. In the folded state, the electronic device 800 may electrically separate the sixth conductive unit D6 (e.g., the sixth conductive unit B6 of FIG. 6A, or the sixth conductive unit C6 of FIG. 7A) and the ground G5, by controlling the frequency adjustment circuit 891.

[0232] Referring to FIGS. 13 and 14, 1401 may indicate the antenna radiation performance on the frequency distribution of the second antenna system 860 if the sixth conductive unit D6 and the ground D5 are electrically separated by the frequency adjustment circuit 891 in the folded state of the electronic device 800. 1402 may indicate the antenna radiation performance of the second antenna system 660 if the sixth conductive unit D6 and the ground D5 are electrically connected through the second lumped element 894 by the frequency adjustment circuit 891 in the folded state of the electronic device 800.

[0233] According to an embodiment, if the sixth conductive unit D6 is electrically connected to the ground G5 through the second lumped element 894, the second lumped element 894 may act the loading of the electrical element (e.g., the inductance loading) on the second antenna system 860. Hence, the antenna radiation performance on the frequency distribution of the second antenna system 860 may differ if the sixth conductive unit D6 is electrically connected or is not electrically connected to the ground G5. According to an embodiment, in at least some frequency band, the antenna radiation performance for the second antenna system 860 may be higher if the sixth conductive unit D6 is electrically connected to the ground G5 through the second lumped element 894 (see 1401 and 1402).

[0234] FIG. 15A illustrates a circuit of an electronic device in an unfolded state according to an embodiment. FIG. 15B illustrates a frequency adjustment circuit according to an embodiment. FIG. 15C illustrates a circuit of the electronic device in a folded state according to an embodiment. FIG. 16 is a graph showing antenna radiation performance on a frequency distribution of a first antenna system in an electronic device of a folded state according to an embodiment.

[0235] Referring to FIGS. 15A and 15B, in an embodiment, an electronic device 1500 (e.g., the electronic device 500 of FIG. 5) may include at least one of a first side member 1510 (e.g., the first side member 510 of FIG. 5), a second side member 1520 (e.g., the second side member 620 of FIG. 5), a hinge structure 1530 (e.g., the hinge structure 530 of FIG. 5), at least one wireless communication circuit 1580 (e.g., the at least one wireless communication circuit 580 of FIG. 5), a processor 1590 (e.g., the processor 590 of FIG. 5), a first frequency adjustment circuit 1591 or a fifth frequency adjustment circuit 1592.

[0236] According to an embodiment, a first antenna radiator ① (e.g., the first antenna radiator ① of FIG. 5, the first antenna radiator ① of FIG. 6A, or the first antenna radiator ① of FIG. 7A) may be formed as at least part of a second conductive unit E2 (e.g., the second conductive unit A2 of FIG. 5, the second conductive unit B2 of FIG. 6A, or the second conductive unit C2 of FIG. 7A) included in the first side member 1510. A second antenna radiator ② (e.g., the second antenna radiator ② of FIG. 5, the second antenna radiator ② of FIG. 6A, or the second antenna radiator ② of FIG. 7A) may be formed as at least part of a third conductive unit E3 (e.g., the third conductive unit A3 of FIG. 5, the third conductive unit B3 of FIG. 6A, or the third conductive unit C3 of FIG. 7A) included in the first side member 1510. The first antenna radiator ① may be electrically connected to the at least one wireless communication circuit 1580 at at least one feeding point FP1 (e.g., the first feeding point FP1 of FIG. 6A or 7A), and may be electrically connected to the ground G1 (e.g., the ground G1 of FIG. 6A or 7A) at at least one grounding point GP1 (e.g., the first grounding point GP1 of FIG. 6A or 7B). The second antenna radiator ② may be electrically connected to the at least one wireless communication circuit 1580 at at least one feeding point FP2 (e.g., the second feeding point FP2 of FIG. 6A or 7A), and may be electrically connected to the ground G2 (e.g., the ground G2 of FIG. 6A or 7A) at at least one grounding point GP2 (e.g., the second grounding point GP2 of FIG. 6A or 7A).

[0237] According to an embodiment, the electronic device 1500 may include a first antenna system 1550 (e.g., the first antenna system 550 of FIG. 5, the first antenna system 650 of FIG. 6A, or the first antenna system 750 of FIG. 7A), or a second antenna system 1560 (e.g., the second antenna system 560 of FIG. 5, the second antenna system 660 of FIG. 6A, or the second antenna system 760 of FIG. 7A) electrically connected to the at least one wireless communication circuit 1580. For example, the first antenna system 1550 may include the first antenna radiator ①, the ground G1 electrically connected to the first antenna radiator ①, a first transmission line 1551 (e.g., the first transmission line 551 of FIG. 5, the first transmission line 651 of FIG. 6A, or the first transmission line 751 of FIG. 7A) between the first antenna radiator ① and the at least one wireless communication circuit 1580, or a tuner 1552 (e.g., the tuner 652 of

FIG. 6A, or the tuner 752 of FIG. 7A) connected to the first transmission line 1551. For example, the second antenna system 1560 may include the second antenna radiator ②, the ground G2 electrically connected to the second antenna radiator ②, a second transmission line 1561 (e.g., the second transmission line 561 of FIG. 5, the second transmission line 661 of FIG. 6A, or the second transmission line 761 of FIG. 7A) between the second antenna radiator ② and the at least one wireless communication circuit 1580, or a switch 1562 (e.g., the switch 662 of FIG. 6A, or the switch 762 of FIG. 7A) connected to the second transmission line 1561. Since the first antenna system 1550 or the second antenna system 1560 is substantially the same as FIG. 5, 6A or 7A, its detailed descriptions are omitted.

[0238] According to an embodiment, the second side member 1520 may include a fifth conductive unit E5 (e.g., the fifth conductive unit A5 of FIG. 5A, the fifth conductive unit B5 of FIG. 6A, or the fifth conductive unit C5 of FIG. 7A) disposed close to the first antenna radiator ①, and a sixth conductive unit E6 (e.g., the sixth conductive unit A6 of FIG. 5A, the sixth conductive unit B6 of FIG. 6A, or the sixth conductive unit C6 of FIG. 7A) disposed close to the second antenna radiator ②, in the folded state of the electronic device 1500 (see FIG. 15C).

[0239] According to an embodiment, the first frequency adjustment circuit 1591 may include an electrical element (e.g., an inductor) having the component such as inductance, capacitance, or conductance acting on the first antenna system 1550 and/or the second antenna system 1560, in the folded state of the electronic device 1500 (see FIG. 15C). The first frequency adjustment circuit 1591 may include the first frequency adjustment circuit 591 of FIG. 5, the frequency adjustment circuit 691 of FIG. 6A or the frequency adjustment circuit 791 of FIG. 7A, and detailed descriptions thereof are omitted.

[0240] According to an embodiment, the fifth frequency adjustment circuit 1592 (e.g., the fifth frequency adjustment circuit 595 of FIG. 5) may be electrically connected with the fifth conductive unit E5 and a ground G9 (e.g., the ground of FIG. 5). The fifth conductive unit E5 may be electrically connected to the ground G9 through the fifth frequency adjustment circuit 1592.

[0241] Referring to FIGS. 15A, 15B, and 15C, in an embodiment, the fifth frequency adjustment circuit 1592 may include a third switch 1593 including a sixth terminal 1593a and a seventh terminal 1593b. The sixth terminal 1593a may be electrically connected to a sixth position (e.g., a sixth grounding point GP6) on the fifth conductive unit E5 between a third position (e.g., a third grounding point GP3) and a third insulating unit 1543 (e.g., the third insulating unit 543 of FIG. 5, and the third insulating unit 643 of FIG. 6A, or the third insulating unit 743 of FIG. 7A). The seventh terminal 1593b may be electrically connected to a ground G9 (e.g., the ground G9 of FIG. 5). For example, referring to FIG. 15C, the sixth grounding point GP6 may be positioned to overlap an area 1509 opposite to the feeding point FP1 of the first antenna radiator ① based on the grounding point GP1 of the first antenna radiator ①, when viewed in a third direction 1503 (e.g., the third direction 203 of FIG. 2).

[0242] According to an embodiment, the fifth frequency adjustment circuit 1592 may include at least one fourth lumped element (or, a fourth passive element) 1595 such as inductor disposed on a fourth electrical path 1594 electri-

cally connected between the sixth grounding point GP6 and the ground G9. According to an embodiment, the processor 1590 may control the third switch 1593 to electrically connect the sixth terminal 1593a and the seventh terminal 1593b, in the folded state of the electronic device 1500. If the sixth terminal 1593a and the seventh terminal 1593b are electrically connected, the fifth conductive unit E5 may be electrically connected to the ground G9 through the at least one fourth lumped element 1595.

[0243] According to an embodiment, the fifth frequency adjustment circuit 1592 may provide an electrical element (e.g., loading of the electrical device) such as inductance, capacitance, or conductance to the first antenna system 1550 and/or the second antenna system 1560 in the folded state of the electronic device 1500. For example, the fifth frequency adjustment circuit 1592 may move a resonance frequency of the first antenna system 1550 (or, the first antenna radiator ① and/or the second antenna system 1560 (or, the second antenna radiator ②) to a designated frequency, or move it as designated in the folded state of the electronic device 1500. According to various embodiments, the fifth frequency adjustment circuit 1592 may reduce the electrical influence of the second side member 1520 on the first antenna system 1550 and/or the second antenna system 1560 in the folded state of the electronic device 1500.

[0244] According to various embodiments, the fifth frequency adjustment circuit 1592 may be implemented to change various values of the electrical element under of the processor 1590 or the at least one wireless communication circuit 1580. For example, the fourth lumped element 1595 of the fifth frequency adjustment circuit 1592 may include an element such as a variable inductor, a variable capacitor.

[0245] Referring to FIGS. 15C and 16, 1601 indicates the antenna radiation performance on the frequency distribution for the first antenna system 1550, if the third switch 1593 of the fifth frequency adjustment circuit 1592 is turned off and the inductance of 0 nanohenry (nH) is applied to the first antenna system 1550 in the folded state of the electronic device 1500. 1602 indicates the antenna radiation performance on the frequency distribution for the first antenna system 1550, if the fifth frequency adjustment circuit 1592 applies the inductance of about 5.6 nH to the first antenna system 1550 in the folded state of the electronic device 1500. 1603 indicates the antenna radiation performance on the frequency distribution for the first antenna system 1550, if the fifth frequency adjustment circuit 1592 applies the inductance of about 1 nH to the first antenna system 1550 in the folded state of the electronic device 1500. Referring to 1601, 1602, and 1603, in the folded state of the electronic device 1500, the resonance frequency of the first antenna system 1550 (or, the first antenna radiator ①) may be moved in a corresponding frequency band (e.g., the LB) (see 1605), according to the inductance value applied by the fifth frequency adjustment circuit 1592 to the first antenna system 1550.

[0246] According to various embodiments (not shown), the electronic device 1500 may further include a third frequency adjustment circuit (e.g., the third frequency adjustment circuit 593 of FIG. 5) disposed in a first housing structure (e.g., the first housing structure 210 of FIG. 2) including the first side member 1510. The third frequency adjustment circuit may be electrically connected to the first antenna radiator ① and the ground (e.g., the ground G7 of FIG. 5). The third frequency adjustment circuit may be

disposed in the first housing structure 1510, to be involved in the resonance frequency movement of the first antenna system 1550 (or the first antenna radiator ①) in the folded state of the electronic device 1500 (e.g., see 1605 of FIG. 16). According to some embodiment, the fifth frequency adjustment circuit 1592 may be omitted in the embodiment which mounts the third frequency adjustment circuit in the first housing structure 210.

[0247] FIG. 17A illustrates a circuit of an electronic device in an unfolded state according to an embodiment. FIG. 17B illustrates a frequency adjustment circuit according to an embodiment. FIG. 17C illustrates a circuit of the electronic device in a folded state according to an embodiment. FIG. 18 is a graph showing antenna radiation performance on a frequency distribution of a first antenna system in an electronic device of a folded state according to an embodiment. FIG. 19 is a graph showing antenna radiation performance on a frequency distribution of a second antenna system in an electronic device of a folded state according to an embodiment.

[0248] Referring to FIGS. 17A, 17B, and 17C, in an embodiment, an electronic device 1700 (e.g., the electronic device 500 of FIG. 5) may include at least one of a first side member 1710 (e.g., the first side member 510 of FIG. 5), a second side member 1720 (e.g., the second side member 520 of FIG. 5), a hinge structure 1730 (e.g., the hinge structure 530 of FIG. 5), at least one wireless communication circuit 1780 (e.g., the at least one wireless communication circuit 580 of FIG. 5), a processor 1790 (e.g., the processor 590 of FIG. 5), a first frequency adjustment circuit 1791 or a third frequency adjustment 1792.

[0249] According to an embodiment, the first antenna radiator ① (e.g., the first antenna radiator ① of FIG. 5, the first antenna radiator ① of FIG. 6A, or the first antenna radiator ① of FIG. 7A) may be formed as at least part of a second conductive unit H2 (e.g., the second conductive unit A2 of FIG. 5, the second conductive unit B2 of FIG. 6A, or the second conductive unit C2 of FIG. 7A) included in the first side member 1710. The second antenna radiator ② (e.g., the second antenna radiator ② of FIG. 5, the second antenna radiator ② of FIG. 6A, or the second antenna radiator ② of FIG. 7A) may be formed as at least part of a third conductive unit H3 (e.g., the third conductive unit A3 of FIG. 5, the third conductive unit B3 of FIG. 6A, or the third conductive unit C3 of FIG. 7A) included in the first side member 1710. According to an embodiment, the second side member 1720 may include a fifth conductive unit H5 (e.g., the fifth conductive unit A5 of FIG. 5, the fifth conductive unit B5 of FIG. 6A, or the fifth conductive unit C5 of FIG. 7A) disposed close to the first antenna radiator ①, and a sixth conductive unit H6 (e.g., the sixth conductive unit A6 of FIG. 5, the sixth conductive unit B6 of FIG. 6A, or the sixth conductive unit C6 of FIG. 7A) disposed close to the second antenna radiator ②, in the folded state of the electronic device 1700 (see FIG. 17C).

[0250] According to an embodiment, the first antenna system 1750 (e.g., the first antenna system 550 of FIG. 5, the first antenna system 650 of FIG. 6A, or the first antenna system 750 of FIG. 7A) may include the first antenna radiator ①, the ground G1 electrically connected to the first antenna radiator ①, a first transmission line 1751 (e.g., the first transmission line 551 of FIG. 5, the first transmission line 651 of FIG. 6A, or the first transmission line 751 of FIG. 7A) between the first antenna radiator ① and the at least one

wireless communication circuit 1780, or a tuner 1752 (e.g., the tuner 652 of FIG. 6A, or the tuner 752 of FIG. 7A) connected to the first transmission line 1751. According to an embodiment, the second antenna system 1760 (e.g., the second antenna system 560 of FIG. 5, the second antenna system 660 of FIG. 6A, or the second antenna system 760 of FIG. 7A) may include the second antenna radiator 2, the ground G2 electrically connected to the second antenna radiator ②, a second transmission line 1761 (e.g., the second transmission line 561 of FIG. 5, the second transmission line 661 of FIG. 6A, or the second transmission line 761 of FIG. 7A) between the second antenna radiator ② and the at least one wireless communication circuit 1780, or a switch 1762 (e.g., the switch 662 of FIG. 6A, or the switch 762 of FIG. 7A) connected to the second transmission line 1761. Since the first antenna system 1750 or the second antenna system 1760 is substantially the same as FIG. 5, 6A, or 7A, detailed descriptions thereof are omitted.

[0251] According to an embodiment, the first frequency adjustment circuit 1791 may include an electrical element (e.g., an inductor) having the component such as inductance, capacitance or conductance acting on the first antenna system 1750 and/or the second antenna system 1760, in the folded state of the electronic device 1700 (see FIG. 17C). The first frequency adjustment circuit 1791 may include the first frequency adjustment circuit 591 of FIG. 5, the frequency adjustment circuit 691 of FIG. 6A, or the frequency adjustment circuit 791 of FIG. 7A, and detailed descriptions thereof are omitted.

[0252] According to an embodiment, the third frequency adjusting circuit 1792 (e.g., the third frequency adjusting circuit 593 of FIG. 5) may be electrically connected with the first antenna radiator ① and a ground G7 (e.g., the ground G7 of FIG. 5). The first antenna radiator ① may be electrically connected to the ground G7 through the third frequency adjustment circuit 1792.

[0253] Referring to FIGS. 17A, 17B, and 17C, in an embodiment, the third frequency adjustment circuit 1792 may include a second switch 1793 including a fourth terminal 1793a and a fifth terminal 1793b. The fourth terminal 1793a may be electrically connected to a fifth position (e.g., a fifth grounding point GP5) on the second conductive unit H2 between a first position (e.g., a first feeding point FP1) and the first insulating unit 1741 (e.g., the first insulating unit 541 of FIG. 5, the first insulating unit 641 of FIG. 6A, or first insulating unit 741 of FIG. 7A). The fifth terminal 1793b may be electrically connected to a ground G7 (e.g., the ground G7 of FIG. 5). For example, referring to FIG. 17C, the fifth grounding point GP5 may be positioned in an area 1709 opposite to the feeding point FP1 based on the first grounding point GP1.

[0254] According to an embodiment, the third frequency adjustment circuit 1792 may include at least one third lumped element (or a third passive element) 1795 such as an inductor disposed on a third electrical path 1794 electrically connected between the fifth grounding point GP5 and the ground G7. According to an embodiment, the processor 1590 may control the second switch 1793 to electrically connect the fourth terminal 1793a and the fifth terminal 1793b, in the folded state of the electronic device 1500. If the fourth terminal 1793a and the fifth terminal 1793b are electrically connected, the second conductive unit H2 may be electrically connected to the ground G7 through the at least one third lumped element 1795.

[0255] According to an embodiment, the third frequency adjustment circuit 1792 may include an electrical element having the component such as inductance, capacitance or conductance acting on the first antenna system 1750 and/or the second antenna system 1760 in the folded state of the electronic device 1700. For example, the third frequency adjustment circuit 1792 may move the resonance frequency of the first antenna system 1750 (or, the first antenna radiator ①) and/or the second antenna system 1760 (or, the second antenna radiator ②) to a designated frequency, or move it as designated in the folded state of the electronic device 1700. According to various embodiments, the third frequency adjustment circuit 1792 may reduce the electrical influence of the second side member 1720 on the first antenna system 1750 and/or the second antenna system 1760 in the folded state of the electronic device 1700.

[0256] According to various embodiments, the third frequency adjustment circuit 1792 may be implemented to change various values of the electrical element under control of the processor 1790 or the at least one wireless communication circuit 1780. For example, the third lumped element 1795 of the third frequency adjustment circuit 1792 may include an element such as a variable inductor, or a variable capacitor.

[0257] In an embodiment, referring to FIG. 17C, in the folded state of the electronic device 1700, the fifth conductive unit H5 may be electrically connected to the ground G5 through the first frequency adjusting circuit 1791, and the antenna radiator ① may be electrically connected to the ground G7 through the third frequency adjustment circuit 1792. Referring to FIG. 18, 1801 indicates the antenna radiation performance on the frequency distribution for the first antenna system 1750, if the third frequency adjustment circuit 1792 applies the capacitance of 0 picofarad (pF) to the first antenna system 1750, and the first frequency adjustment circuit 1791 applies the inductance of about 5.6 nH to the first antenna system 1750, in the folded state of the electronic device 1500. 1802 indicates the antenna radiation performance on the frequency distribution for the first antenna system 1750, if the third frequency adjustment circuit 1792 applies the capacitance of about 1 pF to the first antenna system 1750, and the first frequency adjustment circuit 1791 applies the inductance of about 10 nH to the first antenna system 1750, in the folded state of the electronic device 1700. Referring to 1801 and 1802, for example, the antenna radiation performance for the first antenna system 1750 in a corresponding frequency band 1804 (e.g., the LB) may vary depending on the inductance applied by the first frequency adjustment circuit 1791 to the first antenna system 1750. For example, referring to 1801 and 1802, the resonance frequency of the first antenna system 1750 (or, the first antenna radiator ①) may be moved to about 700 MHz by the capacitance of about 1 pF applied by the first frequency adjustment circuit 1791 to the first antenna system 1750.

[0258] In an embodiment, referring to FIG. 17C, in the folded state of the electronic device 1700, the sixth conductive unit H6 may be electrically connected to the ground G5 through the first frequency adjusting circuit 1791, and the antenna radiator ① may be electrically connected to the ground G7 through the third frequency adjustment circuit 1792. Referring to FIG. 19, 1901 indicates the antenna radiation performance on the frequency distribution for the second antenna system 1760, if the first frequency adjustment circuit 1791 applies the inductance of about 1 nH to the

second antenna system 1760 and the third frequency adjustment circuit 1792 applies the inductance of 0 nH to the second antenna system 1760, in the folded state of the electronic device 1700. 1902 indicates the antenna radiation performance on the frequency distribution for the second antenna system 1760, if the first frequency adjustment circuit 1791 applies the inductance of about 1 nH to the second antenna system 1760 and the third frequency adjustment circuit 1792 applies the inductance of about 1 nH to the second antenna system 1760, in the folded state of the electronic device 1700. Referring to 1901 and 1902, for example, the antenna radiation performance for the second antenna system 1760 in a corresponding frequency band 1904 (e.g., the LB) may vary depending on the inductance applied by the third frequency adjustment circuit 1792 to the second antenna system 1760.

[0259] FIG. 20 illustrates a circuit of an electronic device in a folded state according to an embodiment. FIG. 21 illustrates an antenna extension structure according to an embodiment. FIG. 22 illustrates an antenna extension structure according to various embodiments. FIG. 23 is a graph showing antenna radiation performance of a first antenna system according to whether an antenna extension structure is applied or not according to an embodiment.

[0260] Referring to FIG. 20, in an embodiment, an electronic device 2000 (e.g., the electronic device 500 of FIG. 5) may include at least one of a first side member 2010 (e.g., the first side member 510 of FIG. 5), a second side member 2020 (e.g., the second side member 520 of FIG. 5), at least one wireless communication circuit 2080 (e.g., the at least one wireless communication circuit 580 of FIG. 5), a processor 2090 (e.g., the processor 590 of FIG. 5), a frequency adjustment circuit 2091, a first radiator extension structure 2070 or a second radiator extension structure 2080.

[0261] According to an embodiment, a first antenna radiator ① (e.g., the first antenna radiator ① of FIG. 5, the first antenna radiator ① of FIG. 6A, or the first antenna radiator ① of FIG. 7A) may be formed as at least part of a second conductive unit J2 (e.g., the second conductive unit A2 of FIG. 5, the second conductive unit B2 of FIG. 6A, or the second conductive unit C2 of FIG. 7A) included in the first side member 2010. A second antenna radiator ② (e.g., the second antenna radiator ② of FIG. 5, the second antenna radiator ② of FIG. 6A, or the second antenna radiator ② of FIG. 7A) may be formed as at least part of a third conductive unit J3 (e.g., the third conductive unit A3 of FIG. 5, the third conductive unit B3 of FIG. 6A, or the third conductive unit C3 of FIG. 7A) included in the first side member 2010. A fifth conductive unit J5 (e.g., the fifth conductive unit A5 of FIG. 5, the fifth conductive unit B5 of FIG. 6A, or the fifth conductive unit C5 of FIG. 7A) of the second side member 1720 may be disposed close to the first antenna radiator ① in the folded state of the electronic device 2000. A sixth conductive unit J6 (e.g., the sixth conductive unit B6 of FIG. 5, the sixth conductive unit B6 of FIG. 6A, or the sixth conductive unit C6 of FIG. 7A) of the second side member 1720 may be disposed close to the second antenna radiator ② in the folded state of the electronic device 2000. According to an embodiment, a first antenna system 2050 (e.g., the first antenna system 550 of FIG. 5, the first antenna system 650 of FIG. 6A, the first antenna system 750 of FIG. 7A, or the first antenna system 1550 of FIG. 15C) may include at least one the first antenna radiator ①, a ground G1 electrically connected to the first

antenna radiator ①, a first transmission line 2051 (e.g., the first transmission line 1551 of FIG. 15C) between the first antenna radiator ① and the at least one wireless communication 2080, a tuner 2052 (e.g., the tuner 1552 of FIG. 15C) connected to the first transmission line 2051, the first radiator extension structure 2070 or the second radiator extension structure 2080. According to an embodiment, a second antenna system 1760 (e.g., the second antenna system 560 of FIG. 5, the second antenna system 660 of FIG. 6A, the second antenna system 760 of FIG. 7A, or the second antenna system 1560 of FIG. 15B) may include at least one of the second antenna radiator ②, a ground G2 electrically connected to the second antenna radiator ②, a second transmission line 2061 (e.g., the second transmission line 1561 of FIG. 15C) between the second antenna radiator ② and the at least one wireless communication 2080, or a switch 2062 (e.g., the switch 1562 of FIG. 15C) connected to the second transmission line 2061.

[0262] According to an embodiment, in the folded state of the electronic device 2000, the frequency adjustment circuit 2091 may include an electrical device (e.g., an inductor) having a component such as inductance, capacitance or conductance acting on the first antenna system 2050 and/or the second antenna system 2060. The frequency adjustment circuit 2091 may include the first frequency adjustment circuit 591 of FIG. 5, the frequency adjustment circuit 691 of FIG. 6A or the frequency adjustment circuit 791 of FIG. 7A, and detailed descriptions thereof are omitted.

[0263] According to an embodiment, the first radiator extension structure 2070 may include a conductive pattern electrically connected to the first antenna radiator ① and the ground G1. According to an embodiment, the second radiator extension structure 2080 may include a conductive pattern electrically connected to the first antenna radiator ①. The first antenna radiator ① receives power from the at least one wireless communication circuit 2080, and a current may be transmitted to the first antenna radiator ①, the first radiator extension structure 2070 and the second radiator extension structure 2080 and flow into the ground G1. A width of the electronic device 2000 in the folded state (e.g., see FIG. 3) may be designed restrictively in consideration of portability, and accordingly the first antenna radiator ① may have a physical length with which it is difficult to achieve the resonance frequency in a corresponding frequency band (e.g., the LB). According to an embodiment, the first radiator extension structure 2070 or the second radiator extension structure 2080 may operate as a radiator together with the first antenna radiator ①, and extend an electrical length for the first antenna radiator ①. The electrical length may refer to an electrical wavelength unit length of a line or an element against a wavelength of a frequency to use in the electronic device 2000. The electrical length may refer to a value produced by dividing a physical length by the wavelength (2). According to some embodiment, the electrical length (or a phase length) may refer to a length of a conductive pattern (or an electrical conductor) for a phase shift exhibited by transmission through a conductor at a specific frequency. According to various embodiments, the electrical length may vary depending on the shape the length, width (or breadth), area (or extent) or shape of the conductive pattern which may be formed by the first radiator extension structure 2070 or the second radiator extension structure 2080.

[0264] According to an embodiment, the first radiator extension structure 2070 and/or the second radiator extension structure 2080 may extend the electrical length of the first antenna radiator ①, thus generating the resonance frequency for the corresponding frequency band (e.g., the LB).

[0265] According to various embodiments, at least one of the first radiator extension structure 2070 or the second radiator extension structure 2080 may be omitted.

[0266] Referring to FIG. 21, in an embodiment, a first housing structure 2110 (e.g., the first housing structure 210 of FIG. 2) may include a first side member 2111 (e.g., the first side member 2010 of FIG. 20) and/or a first internal structure 2117. The first side member 2111 may include a plurality of conductive units (e.g., a first conductive unit K1, a second conductive unit K2, and a third conductive unit K7) which are physically or electrically separated. For example, the first conductive unit K1 may be the first conductive unit A1 of FIG. 5, the second conductive unit K2 may be the second conductive unit A2 of FIG. 5 or the second conductive unit J2 of FIG. 20, and the third conductive unit K3 may be the third conductive unit A3 of FIG. 5 or the third conductive unit J3 of FIG. 20. A first insulating unit 2171 (e.g., the first insulating unit 541 of FIG. 5) may be disposed between the first conductive unit K1 and the second conductive unit K2. A second insulating unit 2172 (e.g., the second insulating unit 542 of FIG. 5) may be disposed between the second conductive unit K2 and the third conductive unit K3. According to an embodiment, the first insulating unit 2171 and/or the second insulating unit 2172 may extend from the first internal structure 2117 which is nonconductive and disposed inside the first housing structure 2110. The first internal structure 2117 may be coupled with the first side member 2111, and thus the first, second and third conductive units K1, K2, and K3 may be physically separated by the first internal structure 2117. For example, the first internal structure 2117 may be formed to be coupled with the first side member 2111 by the insert injection.

[0267] According to an embodiment, at least part of the second conductive unit K2 may be utilized as the first antenna radiator ① of FIG. 20, and at least part of the third conductive unit K3 may be utilized as the second antenna radiator ② of FIG. 20.

[0268] According to an embodiment, the first housing structure 2110 may include a first radiator extension structure 2170 (e.g., the first radiator extension structure 2070 of FIG. 20) or a second radiator extension structure 2180 (e.g., the second radiator extension structure 2080 of FIG. 20) electrically connected to the second conductive unit K2 (e.g., the first antenna radiator ① of FIG. 20). The first radiator extension structure 2170 or the second radiator extension structure 2080 may include a conductive pattern which is at least in part fixed on the first internal structure 2117. According to some embodiment, the first radiator extension structure 2170 or the second radiator extension structure 2180 may be at least in part disposed inside the first internal structure 2117.

[0269] According to an embodiment, the first radiator extension structure 2170 or the second radiator extension structure 2180 may be implemented with laser direct structuring (LDS). For example, the first radiator extension structure 2170 or the second radiator extension structure 2180 may be formed by designing a pattern on the first

internal structure 2117 (e.g., a structure formed of a resin such as polycarbonate) using laser, and plating a conductive material such as copper or nickel thereon.

[0270] According to an embodiment, the second conductive unit K2 may include a first terminal 2101 or a second terminal 2102 protruding and extending from the second conductive unit K2 into the inner space of the first housing structure 2110. The first terminal 2101 may be electrically connected with the first radiator extension structure 2170. The second terminal 2102 may be electrically connected with the second radiator extension structure 2180. For example, a conductive material such as solder may electrically interconnect the first terminal 2101 and the first radiator extension structure 2170, and/or the second terminal 2102 and the second radiator extension structure 2180.

[0271] According to some embodiment, the first radiator extension structure 2170 or the second radiator extension structure 2180 may be implemented as a flexible PCB (FPCB). According to various embodiments, the first radiator extension structure 2170 or the second radiator extension structure 2180 may be implemented as various conductive paths such as a conductive pattern or a cable provided on a PCB (e.g., the first substrate 451 of FIG. 4).

[0272] According to an embodiment, the first radiator extension structure 2170 and/or the second radiator extension structure 2180 may extend the electrical length of the first antenna radiator ① including at least part of the second conductive unit (K2), thus forming the resonance frequency for the selected or designated frequency band (e.g., the LB).

[0273] Referring to FIG. 22, in an embodiment, a first housing structure 2210 (e.g., the first housing structure 210 of FIG. 2) may include a first side member 2211 (e.g., the first side member 2010 of FIG. 20) and a first PCB 2251 (e.g., the first substrate 451 of FIG. 4). The first side member 2211 may include a plurality of conductive units (e.g., a first conductive unit M1, a second conductive unit M2, and a third conductive unit M3) which are physically or electrically separated. For example, the first conductive unit M1 may be the first conductive unit A1 of FIG. 5, the second conductive unit M2 may be the second conductive unit A2 of FIG. 5 or the second conductive unit J2 of FIG. 20, and the third conductive unit M3 may be the third conductive unit A3 of FIG. 5 or the third conductive unit J3 of FIG. 20. A first insulating unit 2271 (e.g., the first insulating unit 541 of FIG. 5) may be disposed between the first conductive unit M1 and the second conductive unit M2. A second insulating unit 2272 (e.g., the second insulating unit 542 of FIG. 5) may be disposed between the second conductive unit M2 and the third conductive unit M3.

[0274] According to an embodiment, at least part of the second conductive unit M2 may be utilized as the first antenna radiator ① of FIG. 20, and at least part of the third conductive unit M3 may be utilized as the second antenna radiator ② of FIG. 20.

[0275] According to an embodiment, the first housing structure 2210 may include a first radiator extension structure 2270 (e.g., the first radiator extension structure 2070 of FIG. 20) or a second radiator extension structure 2280 (e.g., the second radiator extension structure 2080 of FIG. 20) electrically connected to the second conductive unit M2 (e.g., the first antenna radiator ① of FIG. 20). According to an embodiment, the first radiator extension structure 2270 or

the second radiator extension structure **2280** may include a conductive pattern or a conductive path disposed on the first PCB **2251**.

[0276] According to an embodiment, the second conductive unit **M2** may include a first terminal **2201** or a second terminal **2202** protruding and extending from the second conductive unit **M2** into an inner space of the first housing structure **2210**. The first terminal **2201** may be electrically connected to the first radiator extension structure **2270**. The second terminal **2202** may be electrically connected to the second radiator extension structure **2280**.

[0277] According to various embodiments (not shown), a flexible conductive member such as a C clip, a pogo-pin, a spring, a conductive poron and rubber or a conductive tape may be disposed between the first terminal **2201** and the first radiator extension structure **2270**, or between the second terminal **2202** and the second radiator extension structure **2280**. The first terminal **2201** may elastically contact the first radiator extension structure **2270** by the flexible conductive member. The second terminal **2202** may elastically contact the second radiator extension structure **2280** by the flexible conductive member.

[0278] According to an embodiment, the first radiator extension structure **2270** and/or the second radiator extension structure **2280** may extend the electrical length of the first antenna radiator **①** including at least part of the second conductive unit **M2**, thus forming the resonance frequency for the selected or designated frequency band (e.g., the LB).

[0279] Referring to FIG. 23, **2301** indicates the antenna radiation performance on the frequency distribution with respect to the first antenna radiator if the radiator extension structure is not applied. **2302** indicates the antenna radiation performance on the frequency distribution with respect to the first antenna radiator (e.g., the first antenna radiator **①** of FIG. 20), if the radiator extension structure (e.g., the first or second radiator extension structure **2070** or **2080** of FIG. 20, the first or second radiator extension structure **2170** or **2180** of FIG. 21, or the first or second radiator extension structure **2270** or **2280** of FIG. 22) is applied. Referring to **2301** and **2302**, the resonance frequency of the first antenna radiator is moved in the frequency band such as the LB if the radiator extension structure is applied.

[0280] FIG. 24 illustrates a circuit of an electronic device in a folded state according to an embodiment. FIG. 25 is a graph showing antenna radiation performance on a frequency distribution of an antenna system according to a frequency adjustment circuit while an electronic device of FIG. 24 is folded according to an embodiment.

[0281] Referring to FIG. 24, in an embodiment, an electronic device **2400** (e.g., the electronic device **500** of FIG. 5) may include at least one of a first side member **2410** (e.g., the first side member **510** of FIG. 5), a second side member **2420** (e.g., the second side member **520** of FIG. 5), at least one wireless communication circuit **2480** (e.g., the at least one wireless communication circuit **580** of FIG. 5), a processor **2490** (e.g., the processor **590** of FIG. 5) or a frequency adjustment circuit **2491**.

[0282] According to an embodiment, a third antenna radiator **③** (e.g., the third antenna radiator **③** of FIG. 5) may be formed as at least part of an eighth conductive unit **N8** (e.g., the eighth conductive unit **A8** of FIG. 5) included in the first side member **2410**. A fourth antenna radiator **④** (e.g., the fourth antenna radiator **④** of FIG. 5) may be formed as at least part of a third conductive unit **N3** (e.g., the

third conductive unit **A3** of FIG. 5) included in the first side member **2410**. A tenth conductive unit **N10** (e.g., the tenth conductive unit **A10** of FIG. 5) of the second side member **2420** may be disposed close to the third antenna radiator **③** in the folded state of the electronic device **2400**. A sixth conductive unit **N6** (e.g., the sixth conductive unit **A6** of FIG. 5) of the second side member **2420** may be disposed close to the fourth antenna radiator **④** in the folded state of the electronic device **2400**.

[0283] According to an embodiment, the electronic device **2400** may include a third antenna system **2470** or a fourth antenna system **2480** electrically connected to the at least one wireless communication circuit **2480**. The third antenna system **2470** (e.g., the third antenna system **570** of FIG. 5) may include at least one of the third antenna radiator **③**, a ground **G3** (e.g., the ground **G3** of FIG. 5) electrically connected to the third antenna radiator **③**, a third transmission line **2471** (e.g., the third transmission line **571** of FIG. 5) between the third antenna radiator **③** and the wireless communication circuit **2480**, or a tuner **2452** (e.g., the tuner **652** of FIG. 6A) connected to the third transmission line **2471**. The fourth antenna system **2480** (e.g., the fourth antenna system **580** of FIG. 5) may include at least one of the fourth antenna radiator **④**, a ground **G4** (e.g., the ground **G4** of FIG. 5) electrically connected to the fourth antenna radiator **④**, a fourth transmission line **2481** (e.g., the fourth transmission line **581** of FIG. 5) between the fourth antenna radiator **④** and the wireless communication circuit **2480**, or a switch **2462** (e.g., the switch **662** of FIG. 6A) connected to the fourth transmission line **2481**.

[0284] According to an embodiment, the frequency adjustment circuit **2491** (e.g., the second frequency adjustment circuit **592** of FIG. 5) may include, in the folded state of the electronic device **2400**, an electrical element (e.g., an inductor) having the component such as inductance, capacitance or conductance acting on the third antenna system **2470** and/or the fourth antenna system **2480**. The frequency adjustment circuit **2491** may include the frequency adjustment circuit **691** of FIG. 6A or the frequency adjustment circuit **791** of FIG. 7A, and detailed descriptions thereof are omitted.

[0285] Referring to FIGS. 24 and 25, **2501** indicates the antenna radiation performance on the frequency distribution of the third antenna system **2470**, if the tenth conductive unit **N10** (e.g., the tenth conductive unit **A10** of FIG. 5) controls the frequency adjustment circuit **2491** not to electrically connect the ground **G6** (e.g., the ground **G6** of FIG. 5) in the folded state of the electronic device **2400**. **2502** indicates the antenna radiation performance on the frequency distribution of the third antenna system **2470**, if the tenth conductive unit **N10** controls the frequency adjustment circuit **2491** to electrically connect the ground **G6** through at least one lumped element (e.g., an inductor) in the folded state of the electronic device **2400**. If the tenth conductive unit **N10** is electrically connected to the ground **G6** through the at least one lumped element, the component such as inductance, capacitance or conductance of the at least one lumped element may act on the third antenna system **2470**. Referring to **2501** and **2502**, if the tenth conductive unit **N10** is electrically connected to the ground **G6** through the at least one lumped element, the antenna radiation performance for the third antenna system **2470** may be improved in at least some frequency band.

[0286] FIG. 26 is a block diagram of an electronic device including a frequency adjustment circuit according to an embodiment. FIG. 27 is a table for explaining control on a frequency adjustment circuit of FIG. 26 according to an embodiment.

[0287] Referring to FIG. 26, an electronic device 2600 (e.g., the electronic device 101 of FIG. 1, the electronic device 200 of FIG. 2, the electronic device 500 of FIG. 5, the electronic device 600 of FIG. 6A, the electronic device 700 of FIG. 7A, the electronic device 1500 of FIG. 15A, the electronic device 1700 of FIG. 17A, the electronic device 2000 of FIG. 20, or the electronic device 2400 of FIG. 24) may include at least one of a first conductive element 2611 and a second conductive element 2612 included a first side member 2610 (e.g., the first side member 510 of FIG. 5), a third conductive element 2621 and a fourth conductive element 2622 included in a second side member 2620, a first ground 2631, a second ground 2632, a wireless communication circuit 2640, a first frequency adjustment circuit 2710, a second frequency adjustment circuit 2720, a third frequency adjustment circuit 2730, a processor 2660, a memory 2670, or at least one sensor 2680.

[0288] According to an embodiment, at least part of the first conductive element 2611 may be electrically connected to the wireless communication circuit 2640 to operate as a first antenna radiator (e.g., the first antenna radiator ① of FIG. 5). According to an embodiment, at least part of the second conductive element 2612 may be electrically connected to the wireless communication circuit 2640 to operate as a second antenna radiator (e.g., the second antenna radiator ② of FIG. 5).

[0289] According to an embodiment, the third conductive element 2621 may be disposed close to the first conductive element 2611, in the folded state of the electronic device 2600 (e.g., see FIG. 3). According to an embodiment, the fourth conductive element 2622 may be disposed close to the second conductive element 2612, in the folded state of the electronic device 2600.

[0290] According to various embodiments, the first conductive element 2611 may be the second conductive unit A2 of FIG. 5, the second conductive unit B2 of FIG. 6A, the second conductive unit C2 of FIG. 7A, the second conductive unit E2 of FIG. 15A, the second conductive unit H2 of FIG. 17A, the second conductive unit J2 of FIG. 20, or the eighth conductive unit N8 of FIG. 24. According to an embodiment, the second conductive element 2612 may be the third conductive unit A3 of FIG. 5, the third conductive unit B3 of FIG. 6A, the third conductive unit C3 of FIG. 7A, the third conductive unit E3 of FIG. 15A, the third conductive unit H3 of FIG. 17A, the second conductive unit J3 of FIG. 20, or the third conductive unit N3 of FIG. 24.

[0291] According to an embodiment, the third conductive element 2621 may be the fifth conductive unit A5 of FIG. 5, the fifth conductive unit B5 of FIG. 6A, the fifth conductive unit C5 of FIG. 7A, the fifth conductive unit E5 of FIG. 15A, the fifth conductive unit H5 of FIG. 17A, the fifth conductive unit J5 of FIG. 20, or the tenth conductive unit N10 of FIG. 24. According to an embodiment, the fourth conductive element 2622 may be the sixth conductive unit A6 of FIG. 5, the sixth conductive unit B6 of FIG. 6A, the sixth conductive unit C6 of FIG. 7A, the sixth conductive unit E6 of FIG. 15A, the sixth conductive unit H6 of FIG. 17A, the sixth conductive unit J6 of FIG. 20, or the sixth conductive unit N6 of FIG. 24.

[0292] According to an embodiment, at least part of the first ground 2631 may be a ground G1, G2, G3, G4, G7 or G8 included in a first ground layer or a first ground plane (e.g., the first PCB 501 of FIG. 5) accommodated in the first housing structure 2610. According to an embodiment, at least part of the second ground 2632 may be a ground G5, G6, G9 or G10 included in a second ground layer or a second ground plane (e.g., the second PCB 502 of FIG. 5) accommodated in the second housing structure 2620.

[0293] According to an embodiment, the first ground 2631 may be electrically connected to the first conductive element 2611 through at least one fifth electrical path 2705. According to an embodiment, the first ground 2631 may be electrically connected to the second conductive element 2612 through at least one sixth electrical path 2706. The first ground 2631 may serve as an antenna ground for an antenna radiator (e.g., the first conductive element 2611 and/or the second conductive element 2612). According to various embodiments, the first ground 2631 and the second ground 2632 may be electrically connected. For example, referring to FIG. 5, the electronic device 500 may include a connection member such as an FPCB which electrically interconnects the first printed circuit board 501 and the second printed circuit board 502, and the connection member may be disposed across the hinge structure 530. The first ground 2631 and the second ground 2632 may be electrically connected through at least part of the connection member.

[0294] The wireless communication circuit 2640 may establish communication between, for example, the electronic device 2500 and an external electronic device (e.g., the electronic device 102 or 104 or the server 108 of FIG. 1). For example, the wireless communication circuit 2640 may be connected to a network (e.g., the first network 198 or the second network 199 of FIG. 1) through wireless communication to communicate with the external electronic device.

[0295] According to an embodiment, the wireless communication circuit 2640 may be electrically connected to the first conductive element 2611 through the seventh electrical path 2707, and the first conductive element 2611 may be utilized as a first antenna radiator (e.g., the first antenna radiator ①, or the third antenna radiator ③ of FIG. 5). The wireless communication circuit 2640 may transmit or receive a signal of a first frequency band, through the first antenna radiator including at least part of the first conductive element 2611.

[0296] According to an embodiment, the wireless communication circuit 2640 may be electrically connected to the second conductive element 2612 through the eighth electrical path 2708, and the second conductive element 2612 may be utilized as a second antenna radiator (e.g., the second antenna radiator ②, or the fourth antenna radiator ④ of FIG. 5). The wireless communication circuit 2640 may transmit or receive a signal of a second frequency band, through the second antenna radiator including at least part of the second conductive element 2612.

[0297] According to various embodiments, the electronic device 2600 may include a fourth switch 2801 (e.g., the switch 662 of FIG. 6A) connected to the eighth electrical path 2708 between the wireless communication circuit 2640 and the second conductive element 2612. The second conductive element 2612 may be electrically connected to the wireless communication circuit 2640 if the fourth switch

2681 is turned on, and may be electrically separated from the wireless communication circuit **2640** if the fourth switch **2681** is turned off.

[0298] According to an embodiment, the wireless communication circuit **2640** may include at least one of a cellular module, a WiFi module, a Bluetooth module, a GNSS module (e.g., a global positioning system (GPS) module, a Glonass module, a Beidou module, or a Galileo module), a near field communication (NFC) module or an RF module.

[0299] The cellular module may include, for example, a voice call, a video call, a text service, or an Internet service through a communication network. According to an embodiment, the cellular module may distinguish and authenticate the electronic device **2600** in the communication network using a subscriber identification module (e.g., the subscriber identification module **196** of FIG. 1). According to an embodiment, the cellular module may perform at least some function of functions included in the processor **2660**. According to an embodiment, the cellular module may include a communication processor (CP).

[0300] The WiFi module, the Bluetooth module, the GNSS module or the NFC module each may include, for example, a processor for processing data transmitted and received through a corresponding module. According to some embodiment, at least some (e.g., two or more) of the cellular module, the WiFi module, the Bluetooth module, the GNSS module or the NFC module may be included in one integrated chip (IC) or IC package.

[0301] The RF module may transmit and receive, for example, a communication signal (e.g., an RF signal). The RF module may include, for example, at least one of a transceiver, a power amp module (PAM), a frequency filter or a low noise amplifier (LNA). According to another embodiment, at least one of the cellular module, the WiFi module, the Bluetooth module, the GNSS module or the NFC module may transmit and receive an RF signal through a separate RF module.

[0302] According to an embodiment, the wireless communication circuit **2640** (e.g., the wireless communication module **192** of FIG. 1) may support communication of various types (e.g., the first network **198** or the second network **199** of FIG. 1) by utilizing the first conductive element **2611** and/or the second conductive element **2612** as the antenna radiator. The wireless communication circuit **2640** may transmit and/or receive the RF signal through the first conductive element **2611** and/or the second conductive element **2612**. The wireless communication circuit **2640** may include various RF components (e.g., an RFIC, a front end module (FEM)) between the antenna radiator (e.g., the first conductive element **2611** or the second conductive element **2612**) and the processor **2660**.

[0303] For example, in the radio signal reception, the wireless communication circuit **2640** (e.g., an RFIC) may receive a radio signal from the first conductive element **2611** and/or the second conductive element **2612**, convert the received radio signal into a baseband signal, and transmit the converted baseband signal to the processor **2660**. The processor **2660** may process the received baseband signal, and control a human/mechanical interface of the electronic device **2600** corresponding to the received baseband signal. For example, in the radio signal transmission, the processor **2660** may generate and output a baseband signal to the wireless communication circuit **2640**. The wireless communication circuit **2640** may receive a baseband signal from the

processor **2660**, convert the received baseband signal into a radio signal, and transmit it over the air through the first conductive element **2611** and/or the second conductive element **2612**.

[0304] For example, the wireless communication circuit **2640** (e.g., an FEM) may include a transmission/reception device for controlling the radio signal. The wireless communication circuit **2640** may connect at least one antenna and the RFIC and separate transmission and reception signals. For example, the FEM may serve as filtering and amplification, and may include a receiving stage front end module which embeds a filter for filtering the received signal, and a transmitting stage front end module which embeds a power amplifier module (PAM) for amplifying the transmit signal.

[0305] According to various embodiments, the wireless communication circuit **2640** may support at least one communication scheme of single input multiple output (SIMO), multiple input single output (MISO), diversity or multiple input multiple output (MIMO) using the first conductive element **2611** and/or the second conductive element **2612**.

[0306] According to various embodiments, the wireless communication circuit **2640** may process a transmit signal or a receive signal in at least one designated frequency band by using the first conductive element **2611** and/or the second conductive element **2612**. For example, the designated frequency band may include at least one of the LB (about 600~1 GHz), the MB (about 1~2.2 GHz), the HB (about 2.2~2.7 GHz) or the UHB (about 2.7~3.6 GHz).

[0307] According to an embodiment, the processor **2660** may be electrically connected with the wireless communication circuit **2640**, the fourth switch **2801**, the first frequency adjustment circuit **2710**, the second frequency adjustment circuit **2720**, the third frequency adjustment circuit **2730**, the memory **2670** and/or the at least one sensor **2680**. The processor **2660** may control the fourth switch **2801**, the first frequency adjustment circuit **2710**, the second frequency adjustment circuit **2720**, or the third frequency adjustment circuit **2730** according to instructions stored in the memory **2670**.

[0308] The memory **2670** may store, for example, instructions for the processor **2660** to control the fourth switch **2801**, the first frequency adjustment circuit **2710**, the second frequency adjustment circuit **2720**, or the frequency adjustment circuit **2730**. According to an embodiment, the memory **2670** may store at least one of a state detection instruction **2671** or a switch control instruction **2672**.

[0309] According to an embodiment, the state detection instruction **2671** may include routines for the processor **2660** to detect the unfolded state (e.g., see FIG. 2) or the folded state (e.g., see FIG. 3) of the electronic device **2600** using the at least one sensor **2680**.

[0310] According to an embodiment, the at least one sensor **2680** may include a proximity sensor (e.g., the proximity sensor disposed in the component mounting area **214** of the first housing structure **210**, in FIG. 2). For example, if the electronic device **2600** is in the folded state (e.g., see FIG. 3), the proximity sensor disposed in the first housing structure **2610** may output an electrical signal relating to the proximity of the second housing structure **2620**.

[0311] According to another embodiment, the at least one sensor **2680** may include a hall IC. For example, the first housing structure **2610** may include the hall IC, and the

second housing structure **2620** may include a magnet. If the electronic device **2600** is in the folded state (e.g., see FIG. 3), the hall IC disposed in the first housing structure **2610** and the magnet disposed in the second housing structure **2620** may be aligned, and the hall IC may recognize the magnet and output an electrical signal.

[0312] According to another embodiment, the at least one sensor **2680** may include an image sensor (e.g., the front camera disposed in the component mounting area **214** of the first housing structure **210**, in FIG. 2). For example, the image sensor disposed in the first housing structure **2610** may acquire image data related to the second housing structure **2620**, and the processor **2660** may determine the unfolded state or the folded state of the electronic device **2600** based on the image data.

[0313] According to various embodiments, the unfolded state or the folded state of the electronic device **2600** may be detected, using other various sensors.

[0314] According to some embodiment, the first housing structure **2610** may include a first contact, and the second housing structure **2620** may include a second contact. If the electronic device **2600** is in the folded state (e.g., see FIG. 3), the first contact disposed on the first housing structure **2610** and the second contact disposed on the second housing structure **2620** may physically contact to conduct, and the processor **2660** may recognize the folded state based on the conduction.

[0315] According to an embodiment, the switch control instruction **2672** may include routines causing the processor **2660** to control the fourth switch **2801** based on the unfolded state or the folded state of the electronic device **2600**. For example, the second conductive element **2612** may be selectively connected to the wireless communication circuit **2640** electrically based on the folded state or the unfolded state. For example, the processor **2660** may turn on the fourth switch **2801** in the unfolded state of the electronic device **2600**, and turn off the fourth switch **2801** in the folded state of the electronic device **2600**.

[0316] According to an embodiment, the switch control instruction **2672** may include routines causing the processor **2660** to control the fourth switch **2801** based on the communication mode. For example, the processor **690** may turn on the fourth switch **2681** in the first communication mode, and the wireless communication circuit **2640** may transmit or receive the signal of the designated frequency related to the first communication mode through the second conductive element **2612**. For example, the processor **2660** may turn off the fourth switch **2801** in the second communication mode which utilizes other frequency than the first communication mode.

[0317] According to an embodiment, the first frequency adjustment circuit **2710** may include a first electrical path **2701** (e.g., the first electrical path **671** of FIG. 6A, or the first electrical path **781** of FIG. 7A) electrically connected between the third conductive element **2621** and the second ground **2632**. The first frequency adjustment circuit **2710** may include an A switch **2711** (e.g., the first switch **672** of FIG. 6A) connected to the first electrical path **2701**. The first frequency adjustment circuit **2710** may include a first lumped element **2712** (e.g., the first lumped element **673** of FIG. 6A, or the first lumped element **793** of FIG. 7A) connected to the first electrical path **2701**. If the A switch **2711** is turned on, the third conductive element **2621** may be electrically connected to the second ground **2632** through

the first lumped element **2712**. If the A switch **2711** is turned off, the third conductive element **2621** may be electrically separated from the second ground **2632**.

[0318] According to an embodiment, the first frequency adjustment circuit **2710** may include a second electrical path **2702** (e.g., the second electrical path **681** of FIG. 6A, or the second electrical path **782** of FIG. 7A) electrically connected between the third conductive element **2621** and the second ground **2632**. The first frequency adjustment circuit **2710** may include a B switch **2713** (e.g., the second switch **682** of FIG. 6A) connected to the second electrical path **2702**. The first frequency adjustment circuit **2710** may include a second lumped element **2714** (e.g., the second lumped element **683** of FIG. 6A, or the second lumped element **794** of FIG. 7A) connected to the second electrical path **2702**. If the B switch **2713** is turned on, the fourth conductive element **2622** may be electrically connected to the second ground **2632** through the second lumped element **2714**. If the B switch **2713** is turned off, the fourth conductive element **2622** may be electrically separated from the second ground **2632**.

[0319] According to various embodiments, the A switch **2711** and the B switch **2713** may be implemented as the first switch **792** of FIG. 7A.

[0320] According to an embodiment, the switch control instruction **2672** stored in the memory **2670** may include routines causing the processor **2660** to control the A switch **2711** or the B switch **2713** according to the communication mode. The communication mode may include, for example, the first communication mode utilizing the LB (about 600~1 GHz), the second communication mode utilizing the MB (about 1~2.2 GHz), the third communication mode utilizing the HB (about 2.2~2.7 GHz), or the fourth communication mode utilizing the UHB (about 2.7~3.6 GHz). According to an embodiment, the processor **2660** turns on or off the first switch **2651** or the second switch **2652**, according to the first communication mode, the second communication mode, the third communication mode, or the fourth communication mode.

[0321] According to an embodiment, the switch control instruction **2672** stored in the memory **2670** may include routines causing the processor **2660** to control the A switch **2711** or the B switch **2713** based on the unfolded state or the folded state of the electronic device **2600**.

[0322] According to an embodiment, in the folded state of the electronic device **2600** (see FIG. 3), the first frequency adjustment circuit **2710** may include the first lumped element **2712** and/or the second lumped element **2714** having the component such as inductance, capacitance or conductance, thus reducing the influence of the second side member **2620** on the antenna radiation performance of the first conductive element **2611** and/or the second conductive element **2612**.

[0323] For example, the processor **2660** may turn on the A switch **2711** in the folded state of the electronic device **2600** (see FIG. 3), and the third conductive element **2621** may be electrically connected to the second ground **2632** through the first lumped element **2712**. The first lumped element **2712** may, in the folded state of the electronic device **2600**, reduce the influence of the third conductive element **2621** on the antenna radiation performance of the first conductive element **2611** and/or the antenna radiation performance of the second conductive element **2612**. In the folded state of the electronic device **2600**, at least part of the second side member **2620** may be close to the first conductive element

2611 and/or the second conductive element **2612** to generate capacitance (or capacitance component) (e.g., parasitic capacitance), and a parasitic resonance frequency may be generated by the capacitance. According to an embodiment, the first lumped element **2712** may adjust the parasitic resonance frequency not to include the parasitic resonance frequency generated in the folded state of the electronic device **2600** in a parasitic resonance frequency of the first conductive element **2611** and/or the second conductive element **2612**. The parasitic resonance frequency generated in the folded state of the electronic device **2600** may be moved out of the resonance frequency band of the first conductive element **2611** and/or the second conductive element **2612** by the first lumped element **2712**, thus reducing degradation of the antenna radiation performance due to the parasitic resonance frequency.

[0324] According to an embodiment, at least part of the third conductive element **2621** electrically connected to the first lumped element **2712** may operate as an additional antenna radiator for transmitting or receiving the signal having the selected or designated frequency, together with the first conductive element **2611** and/or the second conductive element **2612**, in the folded state of the electronic device **2600** (see FIG. 3). For example, at least part of the third conductive element **2621** may operate as the antenna radiator electromagnetically coupled with the first conductive element **2611** and/or the second conductive element **2612** in the folded state of the electronic device **2600**. According to an embodiment, the first lumped element **2712** may have inductance (e.g., inductance component) for reducing capacitance (e.g., parasitic capacitance) which may occur if the third conductive element **2621** is close to the first conductive element **2611**. For example, the first lumped element **2611** electrically connected to the second ground **2632** may be electrically connected to the third conductive element **2621** to operate as an inductor. According to an embodiment, the first lumped element **2712** may adjust impedance of at least part of the third conductive element **2621** to match impedance of a transmission line with respect to the first conductive element **2611** and/or the second conductive element **2612**. For example, the inductance of the first lumped element **2712** may adjust the impedance of at least part of the third conductive element **2621**.

[0325] For example, the processor **2660** may turn on the B switch **2713** in the folded state of the electronic device **2600** (see FIG. 3), and the fourth conductive element **2622** may be electrically connected to the second ground **2632** through the second lumped element **2714**. The second lumped element **2714** may, in the folded state of the electronic device **2600**, reduce the influence of the fourth conductive element **2622** on the antenna radiation performance of the first conductive element **2611** and/or the antenna radiation performance of the second conductive element **2612**. In the folded state of the electronic device **2600**, at least part of the second side member **2620** may be close to the first conductive element **2611** and/or the second conductive element **2612** to generate capacitance (or capacitance component) (e.g., parasitic capacitance), and a parasitic resonance frequency may be generated due to the capacitance. According to an embodiment, the second lumped element **2714** may adjust the parasitic resonance frequency not to include the parasitic resonance frequency generated in the folded state of the electronic device **2600** in the parasitic resonance frequency of the first conductive element **2611** and/or the

second conductive element **2612**. The parasitic resonance frequency generated in the folded state of the electronic device **2600** may be moved out of the resonance frequency band of the first conductive element **2611** and/or the second conductive element **2612** by the second lumped element **2714**, thus reducing degradation of the antenna radiation performance due to the parasitic resonance frequency.

[0326] According to an embodiment, at least part of the fourth conductive element **2622** electrically connected to the second lumped element **2714** may operate as an additional antenna radiator for transmitting or receiving the signal having the selected or designated frequency, together with the first conductive element **2611** and/or the second conductive element **2612** in the folded state of the electronic device **2600** (see FIG. 3). For example, at least part of the fourth conductive element **2622** may operate as the antenna radiator electromagnetically coupled with the first conductive element **2611** and/or the second conductive element **2612** in the folded state of the electronic device **2600**. According to an embodiment, the second lumped element **2714** may have inductance (or inductance component) for reducing capacitance (e.g., parasitic capacitance) which may occur if at least part of the fourth conductive element **2622** is close to the second conductive element **2612**. For example, the second lumped element **2714** electrically connected to the second ground **2632** may be electrically connected to the fourth conductive element **2622** to operate as an inductor. According to an embodiment, the second lumped element **2714** may adjust impedance of at least part of the fourth conductive element **2622** to match the impedance of the transmission line with respect to the first conductive element **2611** and/or the second conductive element **2612**. For example, the inductance of the second lumped element **2714** may adjust the impedance of at least part of the fourth conductive element **2622**.

[0327] According to an embodiment, in the folded state of the electronic device **2600** (see FIG. 3), the frequency adjustment circuit **2710** may function to reduce the degradation of the antenna radiating performance due to the parasitic resonance frequency by moving the parasitic resonance frequency generated by at least part of the second side member **620** which is close to the first conductive element **2611** and/or the second conductive element **2612**, and function to operate at least part of the second side member **2620** as the additional antenna radiator coupled with the first conductive element **2611** and/or the second conductive element **2612** by adjusting the impedance.

[0328] According to various embodiments, the first frequency adjustment circuit **2710** may move the resonance frequency of the first conductive element **2611** and/or the resonance frequency of the second conductive element **2612** to a designated frequency, or move it as designated in the folded state of the electronic device **200** (see FIG. 3), according to on or off of the A switch **2711** or the B switch **2713**.

[0329] For example, in the folded state of the electronic device **2600** (see FIG. 3), if the A switch **2711** is turned on under the control of the processor **2660**, the first lumped element **2712** may be electrically connected to the third conductive element **2621**, thus reducing the resonance frequency of the parasitic component which may occur in the frequency band related to the first conductive element **2611** if the third conductive element **2621** is close to the first conductive element **2611**, and/or the resonance frequency of

the parasitic component which may occur in the frequency band related to the second conductive element 2612 if the fourth conductive element 2622 is close to the second conductive element 2612. In the folded state of the electronic device 2600, if the A switch 2711 is turned off under the control of the processor 2660, the third conductive element 2621 may be electrically separated from the second ground 2632. If the third conductive element 2621 is electrically separated from the second ground 2632, the impedance (or the condition of the parasitic resonance) may be changed and thus reduce the antenna radiation performance degradation of the antenna system utilizing the first conductive element 2611 and/or the second conductive element 2612 due to the parasitic resonance.

[0330] For example, in the folded state of the electronic device 2600 (see FIG. 3), if the B switch 2713 is turned on by the control of the processor 2660, the second lumped element 2714 may be electrically connected with the fourth conductive element 2622, and thus reduce the resonance frequency of the parasitic component which may occur in the frequency band related to the second conductive element 2612 if the fourth conductive element 2622 is close to the second conductive element 2612, and/or the resonance frequency of the parasitic component which may occur in the frequency band related to the first conductive element 2611 if the third conductive element 2621 is close to the first conductive element 2611. In the folded state of the electronic device 2600, if the B switch 2713 is turned off under the control of the processor 2660, the fourth conductive element 2622 may be electrically separated from the second ground 2632. If the fourth conductive element 2622 is electrically separated from the second ground 2632, the impedance (or the condition of the parasitic resonance) may be changed and thus reduce the antenna radiation performance degradation of the antenna system utilizing the first conductive element 2611 and/or the second conductive element 2612 due to the parasitic resonance.

[0331] Referring to FIGS. 26 and 27, in an embodiment, if executing the first communication mode related to the LB in the folded state, the electronic device 2600 may transmit or receive a signal related to the first communication mode through the first conductive element 2611 utilized as the first antenna radiator. If executing the first communication mode related to the LB in the folded state of the electronic device 2600, the processor 2660 may turn on the A switch 2711 and turn off the B switch 2713.

[0332] Referring to FIGS. 26 and 27, in an embodiment, if executing the second communication mode related to the MB in the folded state, the electronic device 2600 may transmit or receive a signal related to the second communication mode through the first conductive element 2611 utilized as the first antenna radiator. If executing the second communication mode related to the MB in the folded state of the electronic device 2600, the processor 2660 may turn on the A switch 2711 and turn off the B switch 2713.

[0333] Referring to FIGS. 26 and 27, in an embodiment, if executing the second communication mode for the MB in the folded state, the electronic device 2600 may transmit or receive a signal related to the second communication mode through the second conductive element 2612 utilized as the first antenna radiator. If executing the second communication mode related to the MB in the folded state of the electronic device 2600, the processor 2660 may turn on both the A switch 2711 and the B switch 2713.

[0334] Referring to FIGS. 26 and 27, in an embodiment, if executing the third communication mode for the HB in the folded state, the electronic device 2600 may transmit or receive a signal related to the third communication mode through the second conductive element 2612 utilized as the second antenna radiator. If executing the third communication mode related to the HB in the folded state of the electronic device 2600, the processor 2660 may turn on the A switch 2711 and turn off the B switch 2713.

[0335] According to an embodiment, the second frequency adjustment circuit 2720 (e.g., the third frequency adjustment circuit 593 of FIG. 5, or the third frequency adjustment circuit 1792 of FIG. 17A) may include a third electrical path 2703 (e.g., the third electrical path 1794 of FIG. 17B) electrically connected between the first conductive element 2611 and the first ground 2631 implemented at least in part as the first antenna radiator. According to an embodiment, the second frequency adjustment circuit 2720 may include a second switch 2721 (e.g., the second switch 1793 of FIG. 17B) connected to the third electrical path 2703. The second frequency adjustment circuit 2720 may include a third lumped element 2722 (e.g., the third lumped element 1795 of FIG. 17B) connected to the third electrical path 2703.

[0336] According to an embodiment, the switch control instruction 2672 stored in the memory 2670 may include routines causing the processor 2660 to turn on or off the second switch 2721 of the second frequency adjustment circuit 2720 according to the communication mode (e.g., the first communication mode utilizing the LB, the second communication mode utilizing the MB, the third communication mode utilizing the HB, or the fourth communication mode utilizing the UHB). According to an embodiment, the switch control instruction 2672 stored in the memory 2670 may include routines causing the processor 2660 to control the second switch 2721 of the second frequency adjustment circuit 2720 based on the unfolded state or the folded state of the electronic device 2600.

[0337] According to an embodiment, in the folded state of the electronic device 2600, the second switch 2721 may be turned on, and the first conductive element 2611 may be electrically connected to the first ground 2631 through the third lumped element 2722. According to an embodiment, the third lumped element 2722 may apply the component such as inductance, capacitance or conductance to the antenna system utilizing the first conductive element 2611 and/or the second conductive element 2612, in the folded state of the electronic device 2600.

[0338] According to an embodiment, the third lumped element 2722 may move the resonance frequency of the first conductive element 2611 and/or the second conductive element 2612 to a designated frequency, or move it as designated in the folded state of the electronic device 2600. According to various embodiments, the third lumped element 2722 may reduce the electrical influence of the second side member 2620 on the antenna system utilizing the first conductive element 2611 and/or the second conductive element 2612 in the folded state of the electronic device 2600.

[0339] According to an embodiment, the third frequency adjustment circuit 2730 (e.g., the fifth frequency adjustment circuit 595 of FIG. 5, or the fifth frequency adjustment circuit 1592 of FIG. 17A) may include a fourth electrical path 2704 (e.g., the fourth electrical path 1594 of FIG. 15B)

electrically connected between the third conductive element **2621** and the second ground **2632**. According to an embodiment, the third frequency adjustment circuit **2730** may include a third switch **2731** (e.g., the third switch **1593** of FIG. **15B**) connected to the fourth electrical path **2704**. The third frequency adjustment circuit **2730** may include a fourth lumped element **2732** (e.g., the third lumped element **1595** of FIG. **15B**) connected to the fourth electrical path **2704**.

[0340] According to an embodiment, the switch control instruction **2672** stored in the memory **2670** may include routines causing the processor **2660** to turn on or off the third switch **2731** of the third frequency adjustment circuit **2730** according to the communication mode (e.g., the first communication mode utilizing the LB, the second communication mode utilizing the MB, the third communication mode utilizing the HB, or the fourth communication mode utilizing the UHB). According to an embodiment, the switch control instruction **2672** stored in the memory **2670** may include routines causing the processor **2660** to control the third switch **2731** of the third frequency adjustment circuit **2730** based on the unfolded state or the folded state of the electronic device **2600**.

[0341] According to an embodiment, in the folded state of the electronic device **2600**, the third switch **2731** is turned on, and the third conductive element **2621** may be electrically connected to the second ground **2632** through the fourth lumped element **2732**. According to an embodiment, the fourth lumped element **2732** may apply the component such as inductance, capacitance or conductance to the antenna system utilizing the first conductive element **2611** and/or the second conductive element **2612**, in the folded state of the electronic device **2600**.

[0342] According to an embodiment, the fourth lumped element **2723** may move the resonance frequency of the first conductive element **2611** and/or the second conductive element **2612** to a designated frequency, or move it as designated in the folded state of the electronic device **2600**. According to various embodiments, the third lumped element **2722** may reduce the electrical influence of the second side member **2620** on the antenna system utilizing the first conductive element **2611** and/or the second conductive element **2612** in the folded state of the electronic device **2600**.

[0343] According to various embodiments, at least one of the second frequency adjustment circuit **2720** and the third frequency adjustment circuit **2730** may be omitted.

[0344] According to various embodiments, based on the technical ideas of various embodiments of the present disclosure, an electronic device of other various rotation types (e.g., a slide type, a swivel type, etc.) in which at least two housings are rotatably operated by a connection unit may be implemented.

[0345] According to an embodiment of the present disclosure, an electronic device (e.g., the electronic device **700** of FIG. **7A**) may include a foldable housing (e.g., the foldable housing **200** of FIG. **2**). The foldable housing may include a hinge structure (e.g., the hinge structure **730** of FIG. **7A**), and a first housing structure (e.g., the first housing structure **210** of FIG. **2**) connected to the hinge structure, and including a first surface (e.g., the first surface **2001** of FIG. **2**) facing a first direction (e.g., the first direction **201** of FIG. **2**), a second surface (e.g., the second surface **2002** of FIG. **2**) facing a second direction (e.g., the second direction **202** of FIG. **2**) opposite to the first direction, and a first side member

(e.g., the first side member **710** of FIG. **7A**) at least partially surrounding a space between the first surface and the second surface, and including a first conductive unit (e.g., the first conductive unit **C1** of FIG. **7A**), a first insulating unit (e.g., the first insulating unit **741** of FIG. **7A**), a second conductive unit (e.g., the second conductive unit **C2** of FIG. **7A**), a second insulating unit (e.g., the second insulating unit **742** of FIG. **7A**) and a third conductive unit (e.g., the third conductive unit **C3** of FIG. **7A**) sequentially disposed from the hinge structure. The foldable housing may include a second housing structure (e.g., the second housing structure **220** of FIG. **2**) connected to the hinge structure, including a third surface (e.g., the third surface **2003** of FIG. **2**) facing a third direction (e.g., the third direction **203** of FIG. **2**), a fourth surface (e.g., the fourth surface **2004** of FIG. **2**) facing a fourth direction (e.g., the fourth direction **204** of FIG. **2**) opposite to the third direction, and a second side member (e.g., the first side member **710** of FIG. **7A**) at least partially surrounding a space between the third surface and the fourth surface, and including a fourth conductive unit (e.g., the fourth conductive unit **C4** of FIG. **7A**), a third insulating unit (e.g., the third insulating unit **743** of FIG. **7A**), a fifth conductive unit (e.g., the fifth conductive unit **C5** of FIG. **7A**), a fourth insulating unit (e.g., the fourth insulating unit **744** of FIG. **7A**) and a sixth conductive unit (e.g., the sixth conductive unit **C6** of FIG. **7A**) sequentially disposed from the hinge structure, and folding with the first housing structure based on the hinge structure. The first surface may face the third surface in a folded state, the third direction may be the same as the first direction in an unfolded state, the first insulating unit and the third insulating unit may substantially align in the folded state, and the second insulating unit and the fourth insulating unit may substantially align. The electronic device may include a flexible display (e.g., the display **300** of FIG. **2**) extending from the first surface to the third surface. The electronic device may include a first PCB (e.g., the first PCB **501** of FIG. **5**) disposed between the first surface and the second surface, and including at least one first ground layer (e.g., the grounds **G1**, **G2**, **G3**, **G4**, **G7**, and **G8** of FIG. **5**). The electronic device may include a second PCB (e.g., the second PCB **502** of FIG. **5**) disposed between the third surface and the fourth surface, and including at least one second ground layer (e.g., the grounds **G5**, **G9**, **G6**, and **G10** of FIG. **5**). The electronic device may include at least one wireless communication circuit (e.g., the wireless communication circuit **780** of FIG. **7A**) disposed on the first PCB, and configured to transmit and/or receive a first signal of a first frequency band by electrically connecting a first position (e.g., the first feeding point **FP1** of FIG. **7A**) closer to the second insulating unit than the first insulating unit, of the second conductive unit, and a second signal of a second frequency band by electrically connecting a second position (e.g., the second feeding point **FP2** of FIG. **7A**) close the second insulating unit of the third conductive unit. The electronic device may include at least one first switch (e.g., the first switch **792** of FIG. **7A** or **7B**) including a first terminal (e.g., the first terminal **792a** of FIG. **7A**) electrically connected to the second ground layer, a second terminal (e.g., the second terminal **792b** of FIG. **7B**) electrically connected to a third position (e.g., the third grounding point **GP3** of FIG. **7A** or **7B**) of the fifth conductive unit closer to the fourth insulating unit than the third insulating unit, and a third terminal (e.g., the third terminal **792c** of FIG. **7B**)

electrically connected to a fourth position (e.g., the fourth grounding point GP4 of FIG. 7A or 7B) of the sixth conductive unit close to the fourth insulating unit.

[0346] According to an embodiment of the present disclosure, the electronic device (e.g., the electronic device 700 of FIG. 7A) may further include at least one first passive element (e.g., the first lumped element 793 of FIG. 7A or 7B) on a first electrical path (e.g., the first electric path 781 of FIG. 7A or 7B) electrically connected between the third position and the second ground layer.

[0347] According to an embodiment of the present disclosure, the electronic device (e.g., the electronic device 700 of FIG. 7A) may further include at least one second passive element (e.g., the second lumped element 794 of FIG. 7A or 7B) on a second electrical path (e.g., the second electric path 782 of FIG. 7A or 7B) electrically connected between the fourth position and the second ground layer.

[0348] According to an embodiment of the present disclosure, the at least one first passive element (e.g., the first lumped element 793 of FIG. 7A), or the at least one second passive element (e.g., the second lumped element 794 of FIG. 7A) may include an inductor.

[0349] According to an embodiment of the present disclosure, the electronic device (e.g., the electronic device 1700 of FIG. 17A) may further include a second switch (e.g., the second switch 1793 of FIG. 17B) including a fourth terminal (e.g., the fourth terminal 1793a of FIG. 17B) and a fifth terminal (e.g., the fifth terminal 1793b of FIG. 17B). The fourth terminal may be electrically connected to a fifth position (e.g., the fifth grounding point GP5 of FIG. 17A or 17B) between the first position of the second conductive unit and the first insulating unit. The fifth terminal may be electrically connected to the first ground layer (e.g., the ground G7 of FIG. 17A or 17B).

[0350] According to an embodiment of the present disclosure, the electronic device (e.g., the electronic device 1700 of FIG. 17A) may further include at least one third passive element (e.g., the third lumped element 1795 of FIG. 17B) on a third electrical path (e.g., the third electrical path 1794 of FIG. 17A or 17B) electrically connected between the fifth position and the first ground layer.

[0351] According to an embodiment of the present disclosure, the electronic device (e.g., the electronic device 1500 of FIG. 15A) may further include a third switch (e.g., the third switch 1593 of FIG. 15B) including a sixth terminal (e.g., the sixth terminal 1593a of FIG. 15B) and a seventh terminal (e.g., the seventh terminal 1593b of FIG. 15B). The sixth terminal may be electrically connected to a sixth position (e.g., the sixth grounding point GP6 of FIG. 15A or 15B) between the third position of the fifth conductive unit and the third insulating unit. The seventh terminal may be electrically connected to the second ground layer (e.g., the ground G9 of FIG. 15A or 15B).

[0352] According to an embodiment of the present disclosure, the electronic device (e.g., the electronic device 1500 of FIG. 15A) may further include at least one fourth passive element (e.g., the fourth lumped element 1595 of FIG. 15B) on a fourth electrical path (e.g., the fourth electrical path 1594 of FIG. 15A or 15B) electrically connected between the sixth position and the second ground layer.

[0353] According to an embodiment of the present disclosure, the electronic device (e.g., the electronic device 700 of FIG. 7A) may further include a fifth electric path (e.g., the fifth electric path 2705 of FIG. 26) electrically connected

between a seventh position (e.g., the first ground point GP1 of FIG. 7A) of the second conductive unit between the first insulating unit and the first position and the first ground.

[0354] According to an embodiment of the present disclosure, in the folded state, when viewed from above the second surface (e.g., the second surface 2003 of FIG. 2), the fourth conductive unit (e.g., the fourth conductive unit of FIG. 7A) may overlap at least in part the first conductive unit (e.g., the first conductive unit C1 of FIG. 7A), the fifth conductive unit (e.g., the fifth conductive unit C5 of FIG. 7A) may overlap at least in part the second conductive unit (e.g., the second conductive unit C2 of FIG. 7A), and the sixth conductive unit (e.g., the sixth conductive unit C6 of FIG. 7A) may at least in part overlap the third conductive unit (e.g., the third conductive unit C3 of FIG. 7A).

[0355] According to an embodiment of the present disclosure, the first side member (e.g., the first side member 510 of FIG. 5) may form a first side surface (e.g., the first side surface 511 of FIG. 5) and a second side surface (e.g., the second side surface 521 of FIG. 5) facing opposite directions, and a third side surface (e.g., the third side surface 531 of FIG. 5) interconnecting the first side surface and the second side surface and perpendicular to the first side surface. The third conductive unit (e.g., the third conductive unit C3 of FIG. 5) may form the third side surface, part of the first side surface and part of the second side surface. The first conductive unit (e.g., the first conductive unit C1 of FIG. 5) and the second conductive unit (e.g., the second conductive unit C2 of FIG. 5) may form part of the first side surface or part of the second side surface.

[0356] According to an embodiment of the present disclosure, the electronic device (e.g., the electronic device 700 of FIG. 7A) may further include a sixth electrical path (e.g., the sixth electrical path 2706 of FIG. 26) electrically connected between an eighth position (e.g., the second grounding point GP2 of FIG. 7A), of the third conductive unit, of a portion forming the third side surface and the first ground layer (e.g., the ground G2 of FIG. 7A).

[0357] According to an embodiment of the present disclosure, the electronic device (e.g., the electronic device 700 of FIG. 7A) may further include at least one tuner (e.g., the tuner 752 of FIG. 7A) on a seventh electrical path (e.g., the seventh electrical path 2707 of FIG. 26) electrically connected between the first position and the at least one wireless communication circuit.

[0358] According to an embodiment of the present disclosure, the electronic device (e.g., the electronic device 700 of FIG. 7A) may further include a fourth switch (e.g., the switch 762 of FIG. 7A, or the switch 2681 of FIG. 26) on an eighth electrical path (e.g., the eighth electrical path 2708 of FIG. 26) electrically connected between the second position and the at least one wireless communication circuit.

[0359] According to an embodiment of the present disclosure, the fourth switch (e.g., the switch 762 of FIG. 7A) may be controlled to electrically separate the second position and the at least one wireless communication circuit in the first frequency band, and to electrically connect the second position and the at least one wireless communication circuit in the second frequency band.

[0360] According to an embodiment of the present disclosure, the electronic device (e.g., the electronic device 700 of FIG. 7A) may further include a processor (e.g., the processor 790 of FIG. 7A, or the processor 2660 of FIG. 26) electrically connected to the flexible display, the at least one

wireless communication circuit, and the first switch, and a memory (e.g., the memory 2670 of FIG. 26) operatively connected with the processor. The memory may store instructions, when executed, causing the processor to control the first switch (e.g., the first switch 792 of FIG. 7A) to electrically connect the first terminal to at least one of the second terminal and the third terminal according to the first frequency band or the second frequency band, in the folded state.

[0361] According to an embodiment of the present disclosure, the electronic device (e.g., the electronic device 2000 of FIG. 20) may further include at least one conductive pattern (e.g., the first radiator extension structure 2070 or the second radiator extension structure 2080 of FIG. 20, the first radiator extension structure 2170 or the second radiator extension structure 2180 of FIG. 21, or the first radiator extension structure 2270 or the second radiator extension structure 2280 of FIG. 22) disposed in a first housing structure (e.g., the first housing structure 2110 of FIG. 21, or the first housing structure 2210 of FIG. 22), and electrically connected to the second conductive unit.

[0362] According to an embodiment of the present disclosure, the at least one conductive pattern (e.g., the first radiator extension structure 2170 or the second radiator extension structure 2180 of FIG. 21) may be formed in a nonconductive structure (e.g., the first internal structure 2117 of FIG. 21) disposed in the first housing structure. According to various embodiments of the present disclosure, the at least one conductive pattern (e.g., the first radiator extension structure 2270 or the second radiator extension structure 2280 of FIG. 22) may be formed on the first PCB (e.g., the first PCB 2251 of FIG. 22).

[0363] According to an embodiment of the present disclosure, the at least one conductive pattern (e.g., the first radiator extension structure 2070 of FIG. 20) may be electrically connected to the first ground layer (e.g., the ground G1 of FIG. 20).

[0364] According to various embodiments of the present disclosure, an electronic device (e.g., the electronic device 700 of FIG. 7A) may include a foldable housing (e.g., the foldable housing 200 of FIG. 2). The foldable housing may include a hinge structure (e.g., the hinge structure 730 of FIG. 7A), and a first housing structure (e.g., the first housing structure 210 of FIG. 2) connected to the hinge structure, and including a first surface (e.g., the first surface 2001 of FIG. 2) facing a first direction (e.g., the first direction 201 of FIG. 2), a second surface (e.g., the second surface 2002 of FIG. 2) facing a second direction (e.g., the second direction 202 of FIG. 2) which is opposite to the first direction, and a first side member (e.g., the first side member 710 of FIG. 7A) which surrounds at least in part a space between the first surface and the second surface, and includes a first conductive unit (e.g., the first conductive unit C1 of FIG. 7A), a first insulating unit (e.g., the first insulating unit 741 of FIG. 7A), a second conductive unit (e.g., the second conductive unit C2 of FIG. 7A), a second insulating unit (e.g., the second insulating unit 742 of FIG. 7A) and a third conductive unit (e.g., the third conductive unit C3 of FIG. 7A) sequentially disposed from the hinge structure. The foldable housing may include a second housing structure (e.g., the second housing structure 220 of FIG. 2) connected to the hinge structure, including a third surface (e.g., the third surface 2003 of FIG. 2) facing a third direction (e.g., the third direction 203 of FIG. 2), a fourth surface (e.g., the fourth surface 2004 of

FIG. 2) facing a fourth direction (e.g., the fourth direction 204 of FIG. 2) which is opposite to the third direction, and a second side member (e.g., the first second side member 720 of FIG. 7A) which surrounds at least in part a space between the third surface and the fourth surface, and including a fourth conductive unit (e.g., the fourth conductive unit C4 of FIG. 7A), a third insulating unit (e.g., the third insulating unit 743 of FIG. 7A), a fifth conductive unit (e.g., the fifth conductive unit C5 of FIG. 7A), a fourth insulating unit (e.g., the fourth insulating unit 744 of FIG. 7A) and a sixth conductive unit (e.g., the sixth conductive unit C6 of FIG. 7A) sequentially disposed from the hinge structure, and folding with the first housing structure based on the hinge structure. The first surface may face the third surface in a folded state, the third direction may be the same as the first direction in an unfolded state, the first insulating unit and the third insulating unit may substantially align in the folded state, and the second insulating unit and the fourth insulating unit may substantially align. The electronic device may include a flexible display (e.g., the display 300 of FIG. 2) extended from the first surface to the third surface. The electronic device may include a first PCB (e.g., the first PCB 501 of FIG. 5) disposed between the first surface and the second surface, and including at least one first ground layer (e.g., the grounds G1, G2, G3, G4, G7, and G8 of FIG. 5). The electronic device may include a second PCB (e.g., the second PCB 502 of FIG. 5) disposed between the third surface and the fourth surface, and including at least one second ground layer (e.g., the grounds G5, G9, G6, and G10 of FIG. 5). The electronic device may include at least one wireless communication circuit (e.g., the wireless communication circuit 780 of FIG. 7A) disposed on the first PCB, and configured to transmit and/or receive a first signal of a first frequency band by electrically connecting to a first position (e.g., the first feeding point FP1 of FIG. 7A) closer to the second insulating unit than the first insulating unit, of the second conductive unit, and a second signal of a second frequency band by electrically connecting to a second position (e.g., the second feeding point FP2 of FIG. 7A) close to the second insulating unit of the third conductive unit. The electronic device may include at least one first switch (e.g., the first switch 792 of FIG. 7A or FIG. 7B) including a first terminal (e.g., the first terminal 792a of FIG. 7A) electrically connected to the second ground layer, a second terminal (e.g., the second terminal 792b of FIG. 7B) electrically connected to a third position (e.g., the third ground point GP3 of FIG. 7A or 7B) of the fifth conductive unit closer to the fourth insulating unit than the third insulating unit, and a third terminal (e.g., the third terminal 792c of FIG. 7B) electrically connected to a fourth position (e.g., the fourth grounding point GP4 of FIG. 7A or 7B) of the sixth conductive unit close to the fourth insulating unit. The electronic device may include at least one first passive element (e.g., the first lumped element of FIG. 7A or 7B) on a first electrical path (e.g., the first electrical path 781 of FIG. 7A or 7B) electrically connected between the third position and the second ground layer. The electronic device may include at least one second passive element (e.g., the second lumped element 794 of FIG. 7A or 7B) on a second electrical path (e.g., the second electrical path 782 of FIG. 7A or 7B) electrically connected between the fourth position and the second ground layer. According to an embodiment of the present disclosure, the electronic device (e.g., the electronic device 700 of FIG. 7A) may further include a processor

(e.g., the processor 790 of FIG. 7A, or the processor 2660 of FIG. 26) electrically connected with the flexible display, the at least one wireless communication circuit, and the first switch, and a memory (e.g., the memory 2670 of FIG. 26) operatively connected with the processor. The memory may store instructions, when executed, causing the processor to control the first switch (e.g., the first switch 792 of FIG. 7A) to electrically connect the first terminal with at least one of the second terminal and the third terminal according to the first frequency band or the second frequency band, in the folded state.

[0365] The embodiments of the present disclosure disclosed in the specification and the drawings merely present specific examples to easily describe the technical details of the present disclosure and help to understand the embodiment of the present disclosure, and are not intended to limit the present disclosure. Thus, the scope of various embodiments of the present disclosure should be construed that all changes or modified forms derived from the technical aspects of various embodiments of the present disclosure in addition to the embodiments disclosed herein are included in the scope of various embodiments of the present disclosure.

What is claimed is:

1. An electronic device comprising:

a hinge structure;

a foldable housing including:

a first housing connected with the hinge structure and including a first side member forming a side surface of the first housing, the first side member including a first conductive portion, a first insulating portion and a second conductive portion sequentially disposed from the hinge structure, and

a second housing connected with the hinge structure and including a second side member forming a side surface of the second housing, the second side member including a third conductive portion, a second insulating portion and a fourth conductive portion sequentially disposed from the hinge structure, the second insulating portion configured to be substantially aligned with the first insulating portion when the foldable housing is folded, and the third conductive portion and the fourth conductive portion configured to be at least partially aligned with the first conductive portion and the second conductive portion, respectively, when the foldable housing is folded;

a flexible display accommodated in the first housing and the second housing;

a first printed circuit board accommodated in the first housing and including a first ground;

a second printed circuit board accommodated in the second housing and including a second ground;

a wireless communication circuit disposed on the first printed circuit board, and configured to transmit or receive a first signal corresponding to a first frequency band via the first conductive portion, and a second signal corresponding to a second frequency band via the second conductive portion; and

a switch accommodated in the second housing and including a first terminal and a second terminal electrically connected with the third conductive portion and the second ground, respectively.

2. The electronic device of claim 1, wherein the third conductive portion is configured to be electrically connected with the second ground via the switch when the foldable housing is folded.

3. The electronic device of claim 1, further comprising: another switch accommodated in the second housing and including a third terminal, a fourth terminal and a fifth terminal electrically connected with the third conductive portion, the fourth conductive portion and the second ground, respectively.

4. The electronic device of claim 3, wherein the third conductive portion is configured to be electrically connected with the second ground via the switch when the foldable housing is folded.

5. The electronic device of claim 4, wherein the fourth conductive portion is configured to be electrically connected with the second ground via the other switch when the foldable housing is folded.

6. The electronic device of claim 3, wherein, when the foldable housing is folded and the first signal, and not the second signal, is transmitted or received via the wireless communication circuit, the third conductive portion is configured to be electrically connected with the second ground via the switch.

7. The electronic device of claim 3, wherein, when the foldable housing is folded and the second signal, and not the first signal, is transmitted or received via the wireless communication circuit, the fourth conductive portion is configured to be electrically connected with the second ground via the other switch.

8. The electronic device of claim 1, further comprising: a tuner connected on an electrical path between the first conductive portion and the wireless communication circuit.

9. The electronic device of claim 1, further comprising: a third switch connected on an electrical path between the second conductive portion and the wireless communication circuit.

10. The electronic device of claim 9, wherein the third switch is configured to:

electrically disconnect the second conductive portion from the wireless communication circuit while the first signal is transmitted or received via the first conductive portion.

11. The electronic device of claim 10, wherein the third switch is configured to:

electrically connect the second conductive portion with the wireless communication circuit such that the second signal is transmitted or received via the second conductive portion.

12. The electronic device of claim 1, wherein the fourth conductive portion forms at least part of a corner area of the second side member.

13. The electronic device of claim 1, wherein the first conductive portion is electrically connected with the first ground.

14. The electronic device of claim 13, wherein the second conductive portion is electrically connected with the first ground.

15. The electronic device of claim 1, further comprising: a processor configured to control the switch to electrically connect the third conductive portion with the second ground when the foldable housing is in a folded state.

16. The electronic device of claim **1**, wherein the switch includes at least one passive element disposed on an electrical path between the first terminal and the second terminal.

17. The electronic device of claim **16**, wherein the passive element is a variable inductor configured to adjust an inductance value based on a control signal from the wireless communication circuit.

18. The electronic device of claim **1**, wherein the switch is configured to connect the third conductive portion to the second ground through a lumped element having an inductance.

19. The electronic device of claim **1**, wherein the third conductive portion is positioned to overlap with a feeding point of the first conductive portion when the foldable housing is in a folded state.

20. The electronic device of claim **1**, wherein the switch forms part of a frequency adjustment circuit configured to shift a resonant frequency of the first conductive portion when the foldable housing is in a folded state.

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